# LABOUR MARKET POWER, AUTOMATION AND WELFARE*

Aseem Patel†
*University of Essex*

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**Abstract**

How does labour market power shape the gains from labour-replacing technologies such as industrial automation and AI? I develop a quantitative general equilibrium model of oligopsonistic labour markets with endogenous task-based automation. I establish two results. (i) Relative to an economy without wage-setting power, large firms employ too few workers and over-automate. These distortions intensify when labour's comparative advantage rises less steeply across the tasks it still performs. (ii) The same comparative advantage schedule determines whether cheaper automation strengthens or weakens a firm's wage-setting power, shaping allocative efficiency and the distribution of automation gains. Consistent with the theory, French firms with greater labour market power hold more automation capital. A benchmark without endogenous automation understates the welfare cost of labour market power by 10–24%. Cheaper automation delivers almost all welfare gains to capital owners while increasing the welfare cost of labour market power by 3–4%.

**Keywords:** Labour market power, automation, task-based model, oligopsony, production function estimation.
**JEL codes:** J20, J42, O33, D24.

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†Email: aseem.patel@essex.ac.uk

# 1 Introduction

Automation can raise aggregate productivity and output while redistributing income from labour to capital and between workers in differently exposed occupations (Acemoglu and Restrepo, 2018, 2019, 2022b; Eden and Gaggl, 2018; Moll et al., 2022). Recent empirical research also documents substantial wage-setting power in labour markets, allowing firms to pay workers below their marginal revenue products (Card et al., 2018; Lamadon et al., 2022). Within local labour markets, larger firms tend to exercise more labour market power than smaller firms (Berger et al., 2022; Yeh et al., 2022).

In this paper, I connect these two strands of research by showing that automation and labour market power affect one another, with consequences for welfare. Firms' incentives to automate depend on the relative cost of automation capital and labour. Wage-setting power changes the marginal cost of labour and therefore influences which firms automate and how intensively they do so. Automation, in turn, can strengthen or weaken firms' wage-setting power through scale and displacement effects that change their relative size within local labour markets. This interaction shapes both the welfare cost of labour market power and the magnitude and distribution of the gains from automation.

To examine these mechanisms and quantify their welfare implications, I develop a quantitative general equilibrium model with oligopsonistic labour markets and endogenous automation choices. Firms compete for workers in local labour markets, where workers' preferences over wages and workplace amenities give employers wage-setting power. Production consists of tasks that firms can assign to labour or automation capital. Firms choose both whether to automate and, conditional on adoption, how many tasks to automate, with differences in productivity, amenities and installation costs generating variation in these choices. The equilibrium jointly determines the distributions of automation and labour market power across firms.

I establish three main results. First, more productive firms employ more workers, pay higher wages, exercise greater wage-setting power and automate more tasks. Their larger relative size within the local labour market makes the labour supply they face less elastic, widening the wedge between their wage and their marginal labour cost. Higher wages and this wider wedge

both raise their marginal labour cost, which leads them to assign more tasks to automation capital.

Whether more productive firms also earn higher profit margins depends on two opposing forces. Greater wage-setting power allows them to retain a larger share of the revenue generated by their workers, but greater automation reduces their reliance on labour and hence the scope for earning labour-market rents. More productive firms earn higher profit margins only when the first force outweighs the second.

Second, wage-setting power distorts both employment and automation choices. Near the benchmark without labour markdowns, reducing wage-setting power widens the wage premium of more productive firms, which exercise greater employer power, and draws workers towards them from their less productive rivals. This reallocation lowers marginal labour costs at more productive firms and raises them at their rivals, encouraging the former to return automated tasks to labour and the latter to automate more. Wage-setting power therefore leaves more productive firms with too few workers and too many automated tasks relative to the benchmark. These task distortions are larger when labour's comparative advantage schedule is relatively flat—meaning that labour's productivity relative to automation capital increases slowly across tasks—because a small change in relative costs then shifts more tasks between labour and automation capital.

Third, cheaper automation can strengthen or weaken the wage-setting power of more productive firms. It reduces production costs and encourages firms to expand—the scale effect—but also displaces workers from additional tasks. More productive firms benefit from a larger scale effect because they already automate more. Whether their relative size within their local labour market increases depends on how much displacement they experience relative to their rivals. Labour's comparative advantage schedule governs this balance by determining the fraction of each firm's remaining labour tasks that becomes worth automating. If the stronger displacement effect at more productive firms outweighs their larger scale effect, their relative size and wage-setting power decline; when the larger scale effect dominates, both increase.

Consequently, the interaction between automation and labour market power depends on both labour's comparative advantage across tasks and how strongly firms' wage-setting power

increases with their relative size.

To examine the model's prediction that firms with greater labour market power automate more, I study the relationship between wage-setting power and automation among French manufacturing firms over 2009–2019. I measure automation using stocks of automation capital constructed from firms' import histories. These stocks capture the extensive margin—whether firms hold automation capital—and the intensive margin—how much they hold among adopters.

I measure wage-setting power using two complementary approaches. First, I construct model-consistent wage markdowns from firms' employment shares within local labour markets and the model's labour supply elasticities. I compute these employment shares using administrative workforce data, which I link to firms' import histories. Second, I estimate wage markdowns using the production-function approach. This requires additional information from firms' balance sheets and a production survey that reports prices and quantities separately. These data allow me to address biases in production-function estimation arising from unobserved productivity and variation in output and input prices.

Both approaches reveal the same pattern: within local labour markets and years, firms with greater wage-setting power are more likely to hold automation capital and, among adopters, hold more automation capital relative to lagged value added. This evidence supports the model's prediction that automation and labour market power are positively related across firms.

I estimate the model by simulated method of moments using the 2019 cross-section of firms under three alternative comparative advantage schedules. These schedules differ in their *displacement hazards*—how readily firms replace their remaining labour tasks as automation becomes cheaper relative to labour—and in how these hazards vary with the share of tasks already automated. This allows me to assess how the quantitative implications depend on the task structure that governs the theoretical results.

To quantify how the interaction between automation and labour market power shapes welfare, I conduct two counterfactual experiments using the estimated model.

First, I measure the welfare cost of labour market power with and without automation. I remove labour markdowns in two economies: one in which firms choose which tasks to

automate, and another in which labour performs every task. With endogenous automation, removing markdowns raises welfare by 6.3–9.0 per cent of 2019 consumption. The welfare gain from removing labour market power is larger with endogenous automation: excluding automation understates this gain by 10–24 per cent across the three estimated economies. Automation introduces two opposing forces. By allowing firms to reassign tasks between workers and capital, it facilitates employment adjustment when markdowns are removed. However, automation also reduces labour's share of production costs, weakening firms' production response to changes in marginal labour costs.

Second, I examine how labour market power affects the magnitude of the gains from cheaper automation and their distribution between workers and firm owners. In each estimated economy, I lower the price of automation capital so that its stock in the oligopsonistic economy grows in line with the projected growth of France's industrial robot stock over 2019–2029. I apply the same price reduction to the corresponding economy without labour markdowns. In the oligopsonistic economy, cheaper automation raises social welfare by 15–29 per cent of 2019 consumption, with almost all gains accruing to firm owners. Workers' welfare gains are approximately three times as large without markdowns. Lower production costs encourage firms to expand, but displacement limits the increase in labour demand and wages, allowing much of the benefit to accrue as profit. Labour market power further limits workers' gains. Total welfare also rises more without markdowns, so cheaper automation increases the welfare cost of labour market power by approximately 3–4 per cent of its initial level.

**Related Literature.** This paper contributes to four strands of the literature. First, it brings endogenous task automation into quantitative models of labour market power. Task models explain how automation changes labour demand with firms taking wages as given (Acemoglu and Restrepo, 2018, 2019, 2020, 2022b), while models of imperfect labour market competition explain wage setting and employment allocation (Card et al., 2018; Lamadon et al., 2022; Berger et al., 2022; Deb et al., 2022, 2024; Azkarate-Askasua and Zerecero, 2025). My framework encompasses both competitive labour markets with endogenous automation and oligopsonistic labour markets without automation as special cases. I show that labour market power also distorts task allocation between labour and capital within firms, with the extent of misallocation

depending on labour's comparative advantage across tasks.

Second, this paper contributes to the recent literature on automation and labour market power.[1] Azar et al. (2024) show that monopsonistic firms face higher marginal labour costs than wage-taking firms and therefore have stronger incentives to automate. Rubens (2025) identifies an investment holdup mechanism through which employer bargaining power encourages technology adoption.[2] Kikuchi (2023) documents a positive relationship between labour market power and labour share among similarly sized establishments in Japan, which he rationalises through weaker automation incentives in a framework similar to mine.

Relative to these papers, I make four contributions. First, I characterise how automation can, in turn, strengthen or weaken firms' wage-setting power, depending on labour's comparative advantage across tasks.[3] Second, I establish sufficient conditions for a unique equilibrium in oligopsonistic labour markets with endogenous automation, and for *global* identification of the model's parameters from the joint distribution of wages, employment and revenue. Third, I develop a solution method that computes the equilibrium for any number of heterogeneous firms and local labour markets. Fourth, I quantify the consequences of ignoring the interaction between automation and labour market power when evaluating the welfare cost of employer power and the gains from cheaper automation.

Third, this paper contributes to the empirical literature studying the implications of automation for employment and firm performance (Acemoglu and Restrepo, 2020, 2022a; Aghion et al., 2026; Dauth et al., 2021; Koch et al., 2021; Stiebale et al., 2024; Bonfiglioli et al., 2024). Byambasuren (2026) examines its relationship with labour market power in Germany. I combine French customs records, balance sheets, workforce records and surveys reporting prices and quantities to address simultaneity, output price and input price biases in production function

[1] On the product-market side, Firooz et al. (2025) study the interaction between automation and heterogeneous markups using a CES technology over robots and labour. In their framework, product-market power distorts the allocation of production across firms but does not directly distort adopters' input mix at given input prices. Here, labour market power also distorts the allocation of tasks between labour and capital.

[2] Labour market power also affects the development of new technologies. Lehr (2026) shows that monopsony over inventors reduces innovation and growth by depressing R&D employment and distorting its allocation across firms.

[3] Automation can also affect employer power through wage bargaining. Leduc and Liu (2024) show that the option to automate strengthens firms' bargaining position. My mechanism operates through employment reallocation: automation changes firms' relative size in their local labour markets and hence the elasticity of the labour supply they face.

estimates of markdowns (Wong, 2026). I document a positive correlation between labour market power and automation at both the extensive margin (whether firms adopt) and the intensive margin (how intensively adopters use automation capital). This relationship also holds using model-consistent markdowns derived from employment shares and labour supply elasticities.

Finally, this paper contributes to the literature on automation's welfare consequences and policies addressing labour market power. Existing work examines redistribution across occupations (Eden and Gaggl, 2018; vom Lehn, 2020; Humlum, 2021; Jaimovich et al., 2021) and inequality through returns to capital (Moll et al., 2022), largely with firms taking wages as given. I examine how endogenous wage-setting power affects the distribution of automation's gains and the efficiency of labour allocation.

This interaction also connects the paper to work evaluating minimum wages (Berger et al., 2025; Hurst et al., 2025) and collective bargaining (Azkarate-Askasua and Zerecero, 2025). These policies can counteract employer power but also encourage automation, as Brynjolfsson et al. (2026) document for minimum wage increases. My framework provides a basis for studying policies that maximise welfare while accounting for both firms' task choices and the resulting changes in wage-setting power.

**Outline.** The paper is organised as follows. Section 2 sets out the model. Section 3 characterises the equilibrium in a duopsonistic labour market. Section 4 extends the model for the quantitative analysis. Section 5 describes the data, presents evidence on the mechanism and estimates the model. Section 6 performs counterfactual experiments, and Section 7 concludes.

The Online Appendix contains the proofs and the solution algorithm. The Supplemental Material contains derivations and additional information on the data and the estimation of the model.

# 2 Model

**Environment.** The economy has four classes of agent: workers, firm owners, a representative final-good producer, and a representative producer of automation capital. There is a mass

$L$ of workers, each supplying one unit of labour inelastically. Each intermediate-good firm has a distinct owner, who supplies no labour and consumes her profit. A single consumption good is produced competitively and serves as the numéraire; automation capital is produced competitively from the final good.

Intermediate firms compete monopolistically in the product market and oligopsonistically in local labour markets. In the product market, firms produce differentiated varieties that are aggregated with a constant elasticity of substitution, so each faces a downward-sloping demand curve and charges a constant markup $\mu$. In the labour market, firms are organised into many local markets, each containing $N$ firms that compete by posting wages à la Bertrand–Nash.

Production is task-based: a continuum of tasks can be performed by automation capital or by labour, and cost minimisation splits the task space between the two, so firms differ in how many tasks they automate.

## 2.1 Workers

Worker $\ell$ obtains indirect utility $u_{\ell ij} = w_i \, z_{\ell i} \, v_{\ell j}$ from working at firm $i$ in market $j$, where $w_i$ is the posted wage, $z_{\ell i}$ is an idiosyncratic preference for the firm, and $v_{\ell j}$ an idiosyncratic preference for the local market. The two shocks are independent Fréchet draws: $z_{\ell i}$ has scale $T_i$ (a firm amenity) and shape $\varsigma$, and $v_{\ell j}$ has unit scale and shape $\eta$. I assume $\varsigma > 1$ and $0 < \eta < \varsigma$, so that the expected within-market offer is finite and jobs across markets are more differentiated from a worker's perspective than jobs within a market. Lower values of $\varsigma$ and $\eta$ imply fatter Fréchet tails and hence greater dispersion in tastes.

The worker chooses in two steps: first a market, then a firm within it. Standard properties of the nested Fréchet structure (Supplemental Material S1.1) give the choice probabilities and the labour supply facing firm $i$,

$$l_i^s(w_i) \;=\; s_{i|j} \; s_j \; L, \qquad s_{i|j} = \frac{T_i \, w_i^{\varsigma}}{\Phi_j}, \qquad s_j = \frac{\Gamma^{\eta} \, \Phi_j^{\eta/\varsigma}}{\Phi}, \tag{1}$$

where $\Phi_j \equiv \sum_{i' \in j} T_{i'} \, w_{i'}^{\varsigma}$ and $\Phi \equiv \sum_{j'} \Gamma^{\eta} \Phi_{j'}^{\eta/\varsigma}$ are the market and economy wage indices, and $\Gamma \equiv \Gamma(1 - 1/\varsigma)$ is a constant. A firm that raises its wage draws workers both from rivals in its

own market and, through $\Phi_j$, from other markets. Internalising its effect on $\Phi_j$ while taking rivals' wages and $\Phi$ as given, the firm faces the wage elasticity of labour supply

$$\varepsilon_i \;\equiv\; \frac{d\log l_i^s}{d\log w_i} \;=\; \varsigma\left(1 - s_{i|j}\right) \;+\; \eta\, s_{i|j}, \tag{2}$$

a share-weighted average of the within- and across-market shapes. Because $\eta < \varsigma$, the elasticity is decreasing in the firm's within-market share, so a larger firm faces less elastic supply; in the limit $s_{i|j} \to 0$ it equals the constant $\varsigma$.

## 2.2 Producers

### 2.2.1 Final-good producer

A competitive representative producer aggregates the intermediate varieties, indexed by the finite set $\mathcal{I}$, into the final good with a constant-elasticity-of-substitution technology,

$$Y \;=\; \Big(\sum_{i\in\mathcal{I}} y_i^{(\rho-1)/\rho}\Big)^{\rho/(\rho-1)}, \qquad \rho > 1, \tag{3}$$

where $\rho$ is the elasticity of substitution across varieties. The final good is the numéraire, and cost minimisation gives each variety a downward-sloping inverse demand,

$$p_i \;=\; \Big(\frac{Y}{y_i}\Big)^{1/\rho}, \qquad P \;=\; \Big(\sum_i p_i^{1-\rho}\Big)^{1/(1-\rho)} = 1. \tag{4}$$

Revenue is therefore a power function of own output,

$$r_i \;=\; p_i\, y_i \;=\; Y^{1/\rho}\, y_i^{1/\mu}, \qquad \mu \equiv \frac{\rho}{\rho-1} \;>\; 1, \tag{5}$$

with $Y$ taken as given by each firm and $\mu$ the demand-side markup.

### 2.2.2 Automation-capital producer

Automation capital is produced in a competitive sector from the final good. Producing one unit of automation capital requires $c_k$ units of the final good. A representative producer chooses the aggregate supply of automation capital to maximise profit, $\max_K (p_k - c_k) K$, where $p_k$ is the price of a unit of automation capital. Constant marginal cost and competition imply

$$p_k = c_k, \tag{6}$$

with zero profit in equilibrium.

### 2.2.3 Intermediate-good producers

**Technology.** Each firm produces its variety from a measure $M$ of tasks, indexed by $\tau \in [0, M]$. Tasks are combined in a logarithmic Cobb–Douglas aggregate with returns to scale $\kappa_f \in (0, 1)$ and Hicks-neutral productivity $\phi_{ih}$, and each task is produced from automation capital and labour, which are perfect substitutes in efficiency units:

$$y_i = \phi_{ih} \left[ \exp\left( \frac{1}{M} \int_0^M \log\big(M\, y_i(\tau)\big)\, d\tau \right) \right]^{\kappa_f}, \qquad y_i(\tau) = \phi_k(\tau)\, k_i(\tau) + \phi_l(\tau)\, l_i(\tau), \tag{7}$$

where $k_i(\tau)$ and $l_i(\tau)$ are the capital and labour allocated to task $\tau$ and $\phi_k(\cdot)$, $\phi_l(\cdot)$ are task-level factor productivities.

**Assumption 1** (Task schedule)**.** *The task productivities $\phi_k, \phi_l$ are strictly positive on $(0, M)$, and labour's comparative advantage, $R(\tau) \equiv \phi_l(\tau)/\phi_k(\tau)$, is continuously differentiable with $R'(\tau) > 0$ on $(0, M)$, with $R(\tau) \to 0$ as $\tau \to 0$ and $R(\tau) \to \infty$ as $\tau \to M$.*

Under Assumption 1, capital's comparative advantage is strongest at the bottom of the task space and vanishes at the top. Online Appendix A.1 shows that this structure pins down the cost-minimising assignment of factors to tasks: at any factor prices, almost every task is served by a single factor, and the assignment is a threshold rule — there is a unique $x_i \in (0, M)$ such that tasks below $x_i$ use capital only and tasks above it use labour only. Because the schedule starts at zero, automation capital is cheaper than labour on the first tasks at any

positive factor prices, so every firm automates a positive measure of tasks; Section 4.2 relaxes this. The task-level allocation can therefore be written

$$y_i(\tau) = \begin{cases} \phi_k(\tau)\, k_i(\tau), & \tau \le x_i, \\ \phi_l(\tau)\, l_i(\tau), & \tau > x_i, \end{cases} \tag{8}$$

with the automation threshold $x_i$ treated as a choice variable whose optimal value is characterised below. I write $\tilde{x}_i \equiv x_i / M \in (0,1)$ for the automated share of tasks, the measure of automation intensity.

**The firm's problem.** The firm chooses its posted wage $w_i$ and the allocation of capital and labour across tasks, $\{k_i(\tau)\}$ and $\{l_i(\tau)\}$, to maximise profit, subject to hiring the workforce its wage generates through (1). Hence the task allocation takes the threshold form (8), so the problem reduces to choosing the wage, the threshold, and the factor quantities:

$$\max_{w_i,\, x_i,\, k_i(\cdot),\, l_i(\cdot)} \; Y^{1/\rho}\, y_i^{1/\mu} - c_k \int_0^{x_i} k_i(\tau)\, d\tau - w_i \int_{x_i}^{M} l_i(\tau)\, d\tau \quad \text{s.t.} \quad \int_{x_i}^{M} l_i(\tau)\, d\tau = l_i^{s}(w_i), \tag{9}$$

with $y_i$ from (7) under (8). Attaching multiplier $\xi_i \ge 0$ to the labour-supply constraint, the Lagrangian is

$$\mathcal{L}_i = Y^{1/\rho} y_i^{1/\mu} - c_k \int_0^{x_i} k_i(\tau)\, d\tau - w_i \int_{x_i}^{M} l_i(\tau)\, d\tau + \xi_i \Big[\, l_i^{s}(w_i) - \int_{x_i}^{M} l_i(\tau)\, d\tau \,\Big]. \tag{10}$$

**First-order conditions.** The conditions for capital and labour by task set the marginal revenue product of each input equal to its price,

$$\frac{\kappa_f\, r_i}{\mu M\, k_i(\tau)} = c_k, \qquad \frac{\kappa_f\, r_i}{\mu M\, l_i(\tau)} = w_i + \xi_i, \tag{11}$$

so capital and labour are uniform within their task ranges and the marginal labour cost in each task is $w_i + \xi_i$. The condition for an interior automation threshold equates the cost per

efficiency unit of the two factors at the marginal task,

$$\frac{c_k}{\phi_k(x_i)} = \frac{w_i + \xi_i}{\phi_l(x_i)}, \tag{12}$$

and the condition for the wage is

$$-l_i + \xi_i \, l_i^{s\prime}(w_i) = 0. \tag{13}$$

Since $l_i^{s\prime}(w_i) = \varepsilon_i \, l_i / w_i$ from (2), this implies

$$\xi_i = \frac{w_i}{\varepsilon_i}. \tag{14}$$

**Labour and capital demand.** Because capital is uniform across automated tasks and labour across non-automated tasks, the task inputs equal the firm's totals spread over the relevant measure,

$$k_i(\tau) = \frac{k_i}{x_i} \;\text{ for }\; \tau \le x_i, \qquad l_i(\tau) = \frac{l_i}{M - x_i} \;\text{ for }\; \tau > x_i, \tag{15}$$

with $k_i \equiv \int_0^{x_i} k_i(\tau)\, d\tau$ and $l_i \equiv \int_{x_i}^{M} l_i(\tau)\, d\tau$ total capital and employment. Integrating the task conditions (11) over each range, the factor demands are constant shares of revenue,

$$c_k \, k_i = \alpha_i \, r_i, \qquad (w_i + \xi_i) \, l_i = \left(\frac{\kappa_f}{\mu} - \alpha_i\right) r_i, \qquad \alpha_i \equiv \frac{\kappa_f}{\mu} \, \tilde{x}_i, \tag{16}$$

where $\alpha_i$ is the capital revenue share.

**Reduced-form production function.** Substituting the uniform allocations (15) into (7), output is Cobb–Douglas in total inputs,

$$y_i \;=\; \tilde{\phi}_i(\tilde{x}_i)^{\kappa_f} \, k_i^{\mu\alpha_i} \, l_i^{\kappa_f - \mu\alpha_i}, \tag{17}$$

where the composite productivity

$$\tilde{\phi}_i(\tilde{x}_i) \;\equiv\; \phi_{ih}^{1/\kappa_f} \, \bar{\phi}_k(\tilde{x}_i)^{\tilde{x}_i} \, \bar{\phi}_l(\tilde{x}_i)^{1-\tilde{x}_i} \, \tilde{x}_i^{-\tilde{x}_i} (1 - \tilde{x}_i)^{-(1-\tilde{x}_i)} \tag{18}$$

combines Hicks-neutral productivity $\phi_{ih}$ with the geometric-mean task productivities $\bar{\phi}_k(\tilde{x}_i)$ and $\bar{\phi}_l(\tilde{x}_i)$ over the automated and non-automated ranges.[4] Revenue then follows from (5):

$$r_i \;=\; Y^{1/\rho}\,\tilde{\phi}_i^{\kappa_f/\mu}\,k_i^{\alpha_i}\,l_i^{\kappa_f/\mu-\alpha_i}. \tag{19}$$

**Wages and markdowns.** With $\xi_i = w_i/\varepsilon_i$ from (14), the marginal labour cost is the posted wage plus the shadow value,

$$\omega_i \;\equiv\; w_i + \xi_i \;=\; \frac{w_i}{\delta_i}, \qquad \delta_i \equiv \frac{\varepsilon_i}{1+\varepsilon_i} \;\in\; (0,1), \tag{20}$$

so the posted wage is a fraction $\delta_i$ of the marginal labour cost: hiring one more worker costs the posted wage plus the raise paid to every worker already employed. Differentiating (19) with respect to $l_i$, holding capital and the threshold fixed, the marginal revenue product of labour is

$$\mathrm{MRPL}_i \;=\; \left(\frac{\kappa_f}{\mu}-\alpha_i\right)\frac{r_i}{l_i}, \tag{21}$$

and comparing with the labour demand (16) gives $\omega_i = \mathrm{MRPL}_i$: at the optimum the marginal revenue product of labour equals its marginal labour cost. The posted wage therefore marks the marginal revenue product down by the factor $\delta_i$,

$$w_i \;=\; \delta_i\,\mathrm{MRPL}_i \;=\; \delta_i\left(\frac{\kappa_f}{\mu}-\alpha_i\right)\frac{r_i}{l_i}. \tag{22}$$

The markdown depends on the equilibrium only through the within-market share, $\delta_i = \delta(s_{i|j})$, with

$$\delta(s) \equiv \frac{\varsigma(1-s)+\eta s}{1+\varsigma(1-s)+\eta s}, \tag{23}$$

strictly decreasing in $s$ since $\eta < \varsigma$: a firm with a larger within-market share faces less elastic supply and posts a wage further below its marginal labour cost. As $s_{i|j} \to 0$ the markdown approaches $\varsigma/(1+\varsigma)$, and as $s_{i|j} \to 1$ it approaches $\eta/(1+\eta)$.

[4] Formally, $\bar{\phi}_k(\tilde{x}) \equiv \exp\{\frac{1}{\tilde{x}}\int_0^{\tilde{x}} \log\phi_k(Mu)\,du\}$ and $\bar{\phi}_l(\tilde{x}) \equiv \exp\{\frac{1}{1-\tilde{x}}\int_{\tilde{x}}^{1} \log\phi_l(Mu)\,du\}$.

**Optimal automation.** Written with the marginal labour cost $\omega_i$, the condition (12) for the automation threshold is

$$R(x_i) \;=\; \frac{\omega_i}{c_k}. \tag{24}$$

Under Assumption 1 the schedule runs from zero to infinity, so this equation has a unique root for every positive ratio $\omega_i/c_k$, and the root is strictly increasing in that ratio. The automation rule is therefore

$$\tilde{x}_i \;=\; \frac{1}{M}\, R^{-1}\!\left(\frac{\omega_i}{c_k}\right) \;\in\; (0,1). \tag{25}$$

A firm automates every task at which automation capital is cheaper than labour per efficiency unit, and the automated share rises with the marginal labour cost relative to the price of automation capital along the task schedule.

For two firms with marginal labour costs $\omega_H > \omega_L$, monotonicity of the schedule gives $\tilde{x}_H > \tilde{x}_L$: the high-cost firm automates every task automated by the low-cost firm and the additional interval between the two thresholds. Figure 1 displays the rule: each firm automates the tasks below the intersection of its line $\omega_i/c_k$ with the task schedule.

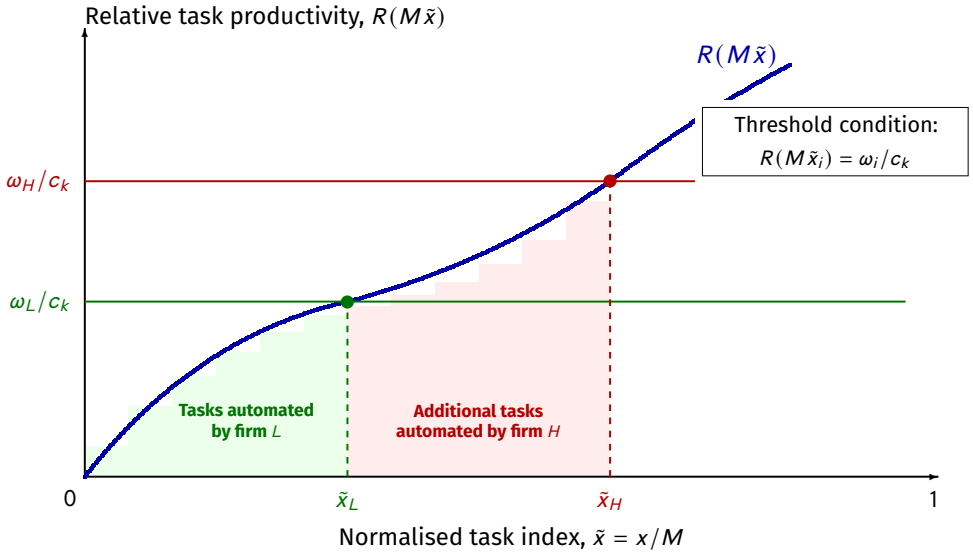
**Displacement hazard.** How far a change in the marginal labour cost moves the threshold depends on the task schedule. Differentiating (25),

$$\frac{d\log(1-\tilde{x}_i)}{d\log(\omega_i/c_k)} \;=\; -\,h(\tilde{x}_i), \qquad h(\tilde{x}) \;\equiv\; \frac{1}{(1-\tilde{x})\,\frac{d\log R(M\tilde{x})}{d\tilde{x}}}, \tag{26}$$

where $h$ is the displacement hazard: the fraction of the tasks labour still performs that switch to automation capital when the marginal labour cost rises by one per cent relative to the price of automation capital.[5] Where labour's comparative advantage rises steeply across tasks, the hazard is low and the same change in relative cost moves few tasks; where it rises slowly, the hazard is high and it moves many.

[5]The distribution is over log comparative advantage, $\log R(\tau)$. A firm at threshold $\tilde{x}$ has automated the fraction $\tilde{x}$ of tasks with the lowest values. The hazard $h(\tilde{x})$ is the density at the threshold divided by the mass of tasks still performed by labour. A constant hazard corresponds to an exponential distribution of log comparative advantage above its lower endpoint.

Figure 1: TASK SCHEDULE AND THE AUTOMATION THRESHOLD




*Notes:* The horizontal axis normalises the task index as $\tilde{x} = x/M \in [0, 1]$, so the increasing task schedule is $R(M\tilde{x})$, which starts at zero and rises without bound. Each firm's horizontal line is its marginal labour cost relative to the price of automation capital, $\omega_i/c_k$, and the firm automates the tasks below the intersection of this line with the task schedule. Green denotes firm $L$ and its automated interval; red denotes firm $H$ and the additional tasks it automates, with $\omega_L < \omega_H$ and hence $\tilde{x}_L < \tilde{x}_H$.

**Profit.** Since the wage bill values labour at $w_i = \delta_i \omega_i$ while the labour demand (16) values it at $\omega_i$, profit is

$$\tilde{\pi}_i = r_i - c_k k_i - w_i l_i = \Big[ \underbrace{\left(1 - \tfrac{\kappa_f}{\mu}\right)}_{\text{product-market rent}} + \underbrace{\left(1 - \delta_i\right) \tfrac{\kappa_f}{\mu} \left(1 - \tilde{x}_i\right)}_{\text{labour-market rent}} \Big] r_i. \tag{27}$$

The labour share of revenue is $w_i l_i / r_i = \delta_i \, (\kappa_f/\mu)(1 - \tilde{x}_i)$.

## 2.3 Aggregation and market clearing

**Factor markets.** Each worker supplies one unit of labour and the choice probabilities sum to one, so employment adds up to the labour force. Aggregate automation capital is the sum of

the firm-level demands (16):

$$\sum_{i\in\mathcal{I}} l_i = L, \qquad K \equiv \sum_{i\in\mathcal{I}} k_i = \frac{1}{c_k}\sum_{i\in\mathcal{I}} \alpha_i r_i. \tag{28}$$

**Aggregate output.** The goods market clears, $Y = \sum_i r_i$. Substituting the optimal capital and labour from (16) into revenue, with the threshold evaluated at its equilibrium value inside $\alpha_i$ and $\tilde{\phi}_i$, concentrates out both factors and leaves revenue a common power of aggregate output,

$$r_i = Y^{1/[\rho(1-\kappa_f/\mu)]}\,\Xi_i, \qquad \Xi_i \equiv \left[\tilde{\phi}_i^{\kappa_f/\mu}\left(\frac{\alpha_i}{c_k}\right)^{\alpha_i}\left(\frac{\kappa_f/\mu-\alpha_i}{\omega_i}\right)^{\kappa_f/\mu-\alpha_i}\right]^{1/(1-\kappa_f/\mu)}. \tag{29}$$

The market-clearing identity then inverts in closed form,

$$Y = \left(\sum_{i\in\mathcal{I}} \Xi_i\right)^{\rho(1-\kappa_f/\mu)/[(\rho-1)(1-\kappa_f)]}, \tag{30}$$

using $\rho(1-\kappa_f/\mu)-1 = (\rho-1)(1-\kappa_f) > 0$.

**Budget constraint.** Firm revenue is $r_i = w_i l_i + c_k k_i + \tilde{\pi}_i$; summing over firms and using goods-market clearing, national income equals factor payments plus profit,

$$Y = \mathcal{B} + c_k K + \Pi, \qquad \mathcal{B} \equiv \sum_i w_i l_i, \qquad \Pi \equiv \sum_i \tilde{\pi}_i. \tag{31}$$

Workers consume their wages and owners their profits, so aggregate consumption is output net of the resources used to produce and install automation capital,

$$C = \mathcal{B} + \Pi = Y - c_k K, \tag{32}$$

and $\sigma \equiv \mathcal{B}/C$ is the workers' share of consumption.

**Social welfare.** Workers' welfare is measured by $\mathcal{W} = \Phi^{1/\eta}$, where $\Phi$ is the aggregate wage index. Supplemental Material S1.1 derives a representative-household formulation, following Azkarate-Askasua and Zerecero (2025), that reproduces the labour supply system and has

maximised utility $\log \mathcal{W}$. This welfare index remains well defined when $\eta \leq 1$, even though mean utility in the Fréchet formulation is infinite.

To express workers' welfare in consumption-equivalent units, define $\mathcal{N} \equiv \mathcal{W}/\mathcal{B}$, where $\mathcal{B}$ is the aggregate wage bill. This employment index captures workers' utility from the allocation of employment, net of the wage-bill level, and is invariant to a proportional change in all wages. Relative to a reference allocation with index $\bar{\mathcal{N}}$, workers' consumption-equivalent welfare is $\mathcal{B}\mathcal{N}/\bar{\mathcal{N}}$.

Owners consume their profits and have linear utility, so their consumption-equivalent welfare equals aggregate profit, $\mathcal{W}^f = \Pi$. Social welfare adds the two groups' consumption equivalents with equal weights per unit of reference consumption:

$$\mathcal{S} = \frac{\sigma\, C\, \mathcal{N}}{\bar{\mathcal{N}}} + (1-\sigma)\, C, \tag{33}$$

where $C = \mathcal{B} + \Pi$ and $\sigma = \mathcal{B}/C$.[6]

## 2.4 Equilibrium

**Definition 1** (Equilibrium)**.** Given firm draws $(\phi_{ih}, T_i)$, an equilibrium consists of wages $w_i$, employment $l_i$, automated shares $\tilde{x}_i$, capital $k_i$, output $y_i$ and profit $\tilde{\pi}_i$ for each firm; market indices $\Phi_j$ and the aggregate index $\Phi$; and aggregate output $Y$, such that:

(i) workers choose employers optimally, generating employment $l_i = l_i^s(w_i)$ as in (1);

(ii) each firm chooses its wage, automation threshold and inputs to maximise profit, taking $Y$, $\Phi$ and rivals' wages as given. These choices solve (9) and satisfy the wage equation (22), the automation rule (25) and the factor demands (16);

(iii) the goods market clears, $Y = \sum_i r_i$.

**Proposition 1.** *Under regularity conditions, the equilibrium exists and is unique.*

The proof is in Online Appendix A.2. It also covers the extensions in Section 4.

[6] Supplemental Material S1.1 derives this measure.

Table 1: SPECIAL CASES

| Restriction | Model | Markdown | Automation |
|---|---|---|---|
| $\varsigma = \eta \to \infty$ | Competitive labour markets | $\delta_i = 1$ | $\tilde{x}_i > 0$ |
| $\phi_k(\tau) = 0$ for all $\tau$ | Oligopsonistic labour markets without automation | $\delta_i = \delta(s_{i\|j})$ | $\tilde{x}_i = 0$ |
| None | This paper | $\delta_i = \delta(s_{i\|j})$ | $\tilde{x}_i > 0$ |

*Notes:* $\varsigma$ and $\eta$ are the within- and between-market Fréchet shapes of Section 2.1, $\delta(\cdot)$ is the markdown schedule (23), $\phi_k$ is the task productivity of automation capital, and $\tilde{x}_i$ is the automated share of tasks.

## 2.5 Special cases

The model has two kinds of variation across firms: firms make different automation choices, and they set different markdowns. Each restriction in Table 1 removes one or both.

**Competitive labour markets.** When workers' tastes for firms and markets become identical, labour supply to each firm is perfectly elastic. Every markdown equals one and all firms pay the same wage. Since the automation choice depends only on the marginal labour cost relative to the price of automation capital, all firms then automate the same share of tasks. Automation is a choice, but neither markdowns nor automation differ across firms. This is the structure of task-based models with endogenous automation and competitive labour markets, as in Acemoglu and Restrepo (2018).

**Oligopsonistic labour markets without automation.** When automation capital is removed, so that labour performs every task, the model becomes a labour-only model of oligopsonistic labour markets. Markdowns depend on each firm's share of employment in its market and so differ across firms, but no firm automates. This is the structure of models of oligopsonistic labour markets, as in Berger et al. (2022) and Azkarate-Askasua and Zerecero (2025).

# 3 Characterisation

This section works with the *analytical economy*: one local labour market with two firms, a high-productivity firm $H$ and a low-productivity firm $L$, $\phi_{Hh} > \phi_{Lh}$, with a common amenity $T$. The single market employs the whole labour force.[7] Labour market power raises a firm's marginal labour cost and so its automation. Automation moves workers across firms and so moves labour market power. Proofs are in Online Appendix A.

## 3.1 Equilibrium ranking

**Proposition 2.** *In any equilibrium of the analytical economy, the more productive firm:*

(i) is larger: $s_H > s_L$, $l_H > l_L$, *and* $r_H > r_L$;

(ii) exercises more labour market power despite paying more: $\delta_H < \delta_L$, *with* $w_H > w_L$ *and hence* $\omega_H > \omega_L$;

(iii) automates more: $\tilde{x}_H > \tilde{x}_L$;

(iv) has a share of profit exceeding its share of revenue *if and only if*

$$(1 - \tilde{x}_H)(1 - \delta_H) \;>\; (1 - \tilde{x}_L)(1 - \delta_L). \tag{34}$$

The proof is in Online Appendix A.3.

Proposition 2 says that the more productive firm is larger, sets a lower markdown, and automates more. This ranking depends on the marginal labour cost, $\omega_i = w_i/\delta_i$. A larger firm posts a higher wage and sets a lower markdown. The lower markdown raises its marginal labour cost relative to the posted wage, $\omega_i/w_i = 1/\delta_i$, because hiring one more worker raises the wage of every worker the firm already employs. Both raise its marginal labour cost, and a higher marginal labour cost tips more tasks towards automation capital.

[7] Each firm still takes $\Phi$ as given when it sets its wage, so the labour supply elasticity it perceives is (2) and its markdown depends on $\eta$, even though the market employs every worker.

To see this, substitute $\omega_i = w_i/\delta_i$ from (20) into the automation rule (25):

$$\tilde{x}_i \;=\; \frac{1}{M}\, R^{-1}\!\Big(\frac{w_i}{\delta_i\, c_k}\Big). \tag{35}$$

Productivity affects automation only through size, since it cancels from the relative cost of the marginal task. Both components of the marginal labour cost rise with firm size. The more productive firm produces at lower marginal cost, meets more demand, and hires more workers. To attract them it posts a higher wage, $w_H > w_L$. Its larger employment share makes its labour supply less elastic, so it sets a lower markdown, $\delta_H < \delta_L$, by (23). Both push the firm's threshold further up the task schedule, so firm $H$ automates a larger share of tasks.

Part (iv) ranks profit per euro of revenue. By (27), each euro of revenue carries a product-market rent, $1-\kappa_f/\mu$, which is the same at both firms, and a labour-market rent, $(1-\delta_i)(\kappa_f/\mu)(1-\tilde{x}_i)$, which is the markdown applied to the share of the euro spent on labour tasks. The productive firm takes the larger markdown on each euro of labour cost, but it has automated more of its tasks, so fewer of its euros go to labour. It earns the higher margin only when its markdown advantage outweighs its smaller labour share, and the task schedule decides which.

A high hazard (26) moves many tasks off labour for a given difference in marginal labour costs, so the productive firm's labour share falls fast with its size and its margin falls below its rival's. Profit is then less concentrated than revenue. Section 6.2 returns to this when automation becomes cheaper.

## 3.2 From labour market power to automation

The benchmark is the equilibrium in which neither firm has wage-setting power: each firm pays its marginal revenue product, $w_i = \text{MRPL}_i$ and $\delta_i = 1$, while workers sort on posted wages as in Section 2.1, firms choose automation capital and tasks as in Section 2.2, and the goods market clears. I call it the *equilibrium without labour markdowns*. It keeps the product-market markup, so it is a counterfactual wage rule rather than the competitive limit of Table 1, and it isolates what labour market power does.

To move between the benchmark and the equilibrium of Definition 1, parametrise the wage

rule by $t \in [0,1]$,

$$w_i \;=\; \left\{1 - t\left[1 - \delta(s_{i|j})\right]\right\}\omega_i, \tag{36}$$

so that $t = 0$ is the equilibrium without labour markdowns and $t = 1$ is the wage-posting rule (22).

**Proposition 3.** *In the analytical economy, a marginal introduction of wage-setting power, $dt > 0$ at $t = 0$,*

(i) moves workers from the productive firm to its rival*: $ds_H/dt < 0$, and the wage premium $w_H/w_L$ falls;*

(ii) raises the marginal labour cost at the productive firm and lowers it at its rival*: $d\omega_H/dt > 0 > d\omega_L/dt$;*

(iii) shifts automation from the rival to the productive firm*: $d\tilde{x}_H/dt > 0 > d\tilde{x}_L/dt$, so the gap in automated shares widens;*

(iv) misallocates tasks in proportion to the displacement hazard*: the gap in the shares of tasks that labour performs widens by*

$$\frac{d}{dt}\Big[\log(1-\tilde{x}_L) - \log(1-\tilde{x}_H)\Big] \;=\; h(\tilde{x}_H)\,\frac{d\log\omega_H}{dt} \;+\; h(\tilde{x}_L)\left|\frac{d\log\omega_L}{dt}\right| \;>\; 0. \tag{37}$$

The proof is in Online Appendix A.5.

Starting from the equilibrium without labour markdowns, wage-setting power moves workers away from the productive firm towards its rival and shifts automation from the rival to the productive firm. The channel is the markdown gap. The productive firm sets the lower markdown, so wage-setting power lowers its posted wage relative to its rival's; workers sort on posted wages, and some of them move to $L$. The move raises labour's marginal product at $H$ and lowers it at $L$, and each firm's marginal labour cost equals its marginal revenue product of labour at the optimum. The automation rule (35) turns these changes in marginal labour costs into changes in automation: labour becomes more expensive at $H$, which therefore automates more tasks than it would without markdowns, and less expensive at $L$, which automates fewer.

Part (iii) says the gap in automated shares widens. Part (iv) says by how much. Equation (37)

shows that the gap in labour's task shares widens by each firm's change in marginal labour cost, multiplied by its displacement hazard (26), the fraction of its remaining labour tasks that a one per cent rise in that cost moves to automation. A flat task schedule has a high hazard and a steep one a low hazard, so the same change in costs misallocates more tasks where the schedule is flat. Since $H$ automates more, $\tilde{x}_H > \tilde{x}_L$, the two firms sit at different points of the schedule and face different hazards. For cost changes of similar size, if the hazard rises along the schedule then $h(\tilde{x}_H) > h(\tilde{x}_L)$ and most of the widening comes from $H$ automating too many tasks; if it falls, most comes from $L$ automating too few.

Labour market power, thus, misallocates workers across firms and, through them, tasks across firms.

## 3.3 From automation to labour market power

Cheaper automation moves workers across firms, and because markdowns depend on employment shares, it moves wage-setting power with them. Write $z \equiv -d \log c_k > 0$ for the size of a marginal fall in the price of automation capital. Which firm gains workers depends on whose labour demand shifts out by more. Let $g(\tilde{x}_i)$ denote firm $i$'s direct labour-demand response to the price fall, holding its marginal labour cost and aggregate demand fixed, and define

$$\Delta_\Upsilon \equiv g(\tilde{x}_H) - g(\tilde{x}_L). \tag{38}$$

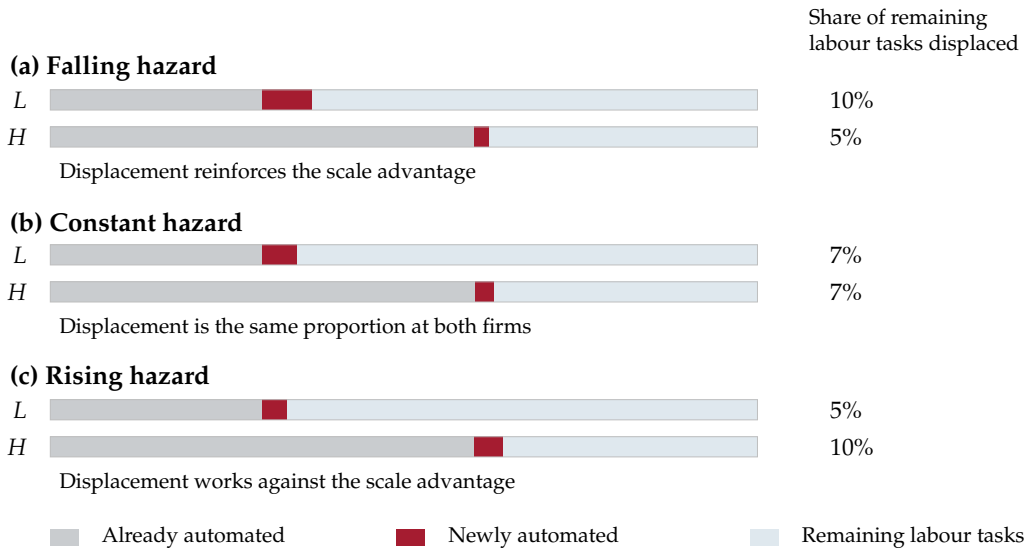
I call the incidence of cheaper automation *concentrating* when $\Delta_\Upsilon > 0$ and *equalising* when $\Delta_\Upsilon < 0$.

**Proposition 4.** *In the analytical economy, after a marginal fall in the price of automation capital, with $\Delta_\Upsilon$ from (38):*

(i) both firms automate more*: $d\tilde{x}_H > 0$ and $d\tilde{x}_L > 0$;*

(ii) workers move towards the firm whose labour demand shifts out by more*: $\text{sign}(ds_H) = \text{sign}(\Delta_\Upsilon)$, so firm $H$ gains employment share under concentrating incidence and loses it under equalising incidence, and shares do not change to first order if $\Delta_\Upsilon = 0$;*

Figure 2: DISPLACEMENT AND THE SHAPE OF THE TASK SCHEDULE




*Notes:* Each bar represents the same total measure of tasks, ordered by labour's comparative advantage. Firms $L$ and $H$ initially automate 30% and 60% of tasks, respectively. Grey denotes tasks already automated, red denotes tasks newly automated after the same small fall in the log automation price holding marginal labour costs fixed, and pale blue denotes tasks that remain with labour. The percentages beside the bars divide newly automated tasks by the tasks performed by labour before the price fall (red plus pale blue). Segment lengths and percentages are schematic first-order illustrations, not estimates or equilibrium responses.

(iii) wage-setting power follows the workers*: under concentrating incidence $\delta_H$ falls, $\delta_L$ rises, and the markdown gap $\delta_L - \delta_H$ widens; under equalising incidence these movements reverse.*

The proof is in Online Appendix A.6.

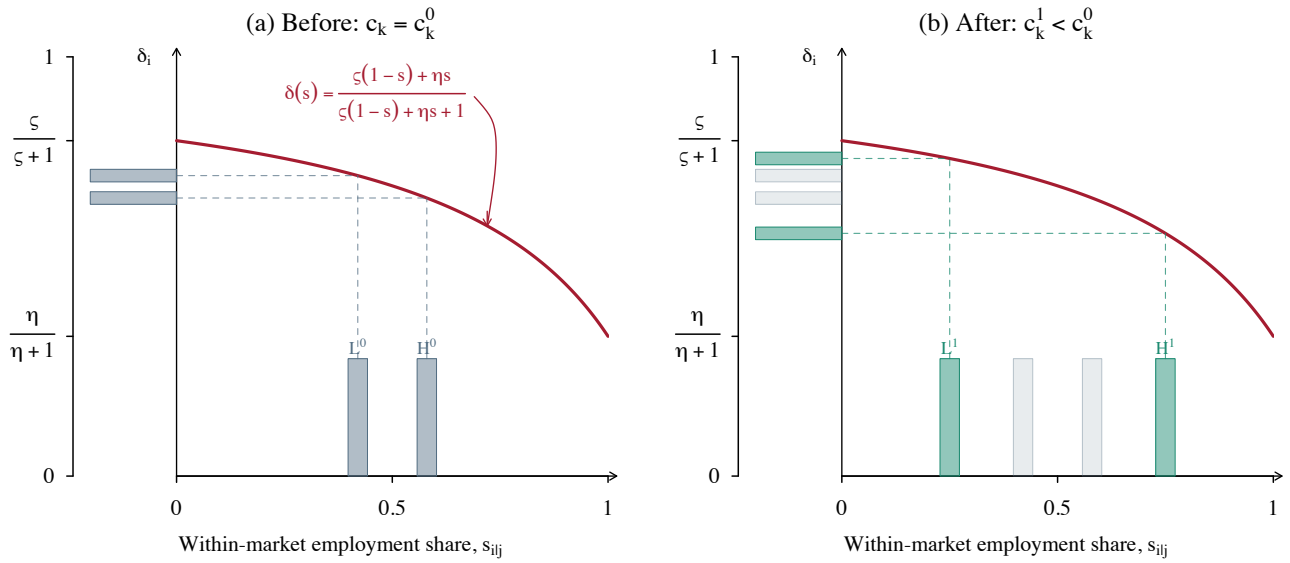
Cheaper automation raises each firm's marginal labour cost relative to the price of automation capital, so both automate more—part (i).

For part (ii), cheaper capital expands production but also replaces labour in additional tasks. Holding marginal labour costs and aggregate demand fixed, the direct labour-demand response is

$$g(\tilde{x}_i) \equiv -\left.\frac{\partial \log l_i^d}{\partial \log c_k}\right|_{\omega_i, Y} = \underbrace{\frac{\kappa_f}{\mu - \kappa_f}\tilde{x}_i}_{\text{scale effect}} - \underbrace{h(\tilde{x}_i)}_{\text{displacement effect}} . \tag{39}$$

The scale effect grows with the share of tasks already automated; the displacement hazard (26)

Figure 3: EMPLOYMENT REALLOCATION AND MARKDOWN INCIDENCE




*Notes:* Superscripts 0 and 1 denote allocations before and after a fall in the automation price, and $L$ and $H$ denote the less and more productive firms. The red curve is the markdown schedule $\delta(s)$. Vertical bars show the firms' within-market employment shares; horizontal bars to the left of the vertical axis show the implied markdown distribution. Panel (a) shows the initial allocation. Panel (b) shows concentrating incidence: firm $H$ gains employment share, firm $L$ loses it, and the markdown gap widens. Pale bars reproduce the initial allocation. The parameter values $\varsigma = 4$ and $\eta = 0.5$ are illustrative. Under equalising incidence, the employment-share movements reverse and the markdown gap narrows.

measures the fraction of remaining labour tasks that a marginal price fall moves to capital.

With total employment fixed, workers move towards the firm with the larger direct labour-demand response. The difference between firms is

$$\Delta_\Upsilon = \underbrace{\frac{\kappa_f}{\mu - \kappa_f}\left(\tilde{x}_H - \tilde{x}_L\right)}_{\text{scale advantage of } H} - \underbrace{\left[h(\tilde{x}_H) - h(\tilde{x}_L)\right]}_{\text{displacement disadvantage of } H}. \tag{40}$$

Because $H$ automates more, it has a larger scale effect. Whether it gains workers depends on whether displacement offsets this advantage. It can gain even when labour demand shifts inward at both firms, provided its demand contracts less.

The shape of the task schedule determines this balance. With a falling hazard, $H$ loses a smaller fraction of its remaining labour tasks, reinforcing its scale advantage. With a constant

hazard, displacement is proportionately equal at both firms, so the scale advantage again makes incidence concentrating. With a rising hazard, incidence is equalising only if $H$'s displacement disadvantage outweighs its scale advantage:

$$h(\tilde{x}_H) - h(\tilde{x}_L) > \frac{\kappa_f}{\mu - \kappa_f}(\tilde{x}_H - \tilde{x}_L).$$

Figure 2 illustrates the three cases.

Part (iii) follows because a firm's markdown falls as its employment share rises. Under concentrating incidence, $H$ gains workers and its markdown falls, while $L$'s rises, widening the markdown gap. Under equalising incidence, these movements reverse. Figure 3 illustrates the concentrating case.

Together, Propositions 3 and 4 establish the two-way relationship: wage-setting power changes the allocation of automation across firms, and cheaper automation reallocates workers and wage-setting power according to the shape of the task schedule.

# 4 Extensions for the quantitative model

The quantitative model adds heterogeneous installation costs, an extensive margin of automation, and physical capital. The first two features generate differences in automation across firms, including non-adoption. Physical capital aligns revenue in the model with value added in the data.

## 4.1 Heterogeneous installation costs

Installing automation equipment requires adaptation to firms' production processes, often through specialised integrators (Békés et al., 2025). I capture these costs with a firm-specific installation requirement $b_i > 0$, drawn from $G_b$. Installing one unit of automation capital requires $b_i$ units of the automation-capital sector's output, giving firm $i$ the unit cost

$$c_{k,i} = c_k \, b_i. \tag{41}$$

The baseline first-order conditions therefore apply with $c_k$ replaced by $c_{k,i}$. Automation depends on marginal labour cost relative to installation cost, $\omega_i/(c_k b_i)$. Firms with identical installation requirements retain the ranking of Proposition 2. A more productive firm can nevertheless automate less than its rival if its installation requirement is sufficiently higher.

## 4.2 The extensive margin

The baseline task schedule implies that every firm automates, whereas more than half of French manufacturing firms hold no automation capital. To allow non-adoption, I give the schedule a positive intercept, $R(0) > 0$. A firm adopts only when automation is cheaper than labour at the task best suited to capital:

$$\tilde{x}_i = \begin{cases} \dfrac{1}{M} R^{-1}\Big(\dfrac{\omega_i}{c_k b_i}\Big), & \omega_i > c_k\, b_i\, R(0), \\ 0, & \omega_i \leq c_k\, b_i\, R(0). \end{cases} \tag{42}$$

The adoption indicator is $d_i = 1\{\tilde{x}_i > 0\}$. Among adopters, automation intensity increases with $\omega_i/(c_k b_i)$.

**Proposition 5.** *Labour market power can induce adoption that would not occur in the equilibrium without labour markdowns. Consider the analytical economy with $R(0) > 0$ and a common installation requirement $b$. If the productive firm is at the adoption threshold in the equilibrium without labour markdowns, $\omega_H = c_k b R(0)$, neither firm adopts. A sufficiently small introduction of wage-setting power induces the productive firm to adopt.*

The proof is in Online Appendix A.7.

The productive firm adopts because wage-setting power raises its marginal labour cost, making automation cheaper than labour at the first tasks.

## 4.3 Physical capital

Firms combine the task composite $F_i$ with physical capital $k_i^p$:

$$y_i \;=\; \phi_{ih}\underbrace{\left[\exp\!\left(\frac{1}{M}\int_0^M \log\!\big(M\, y_i(\tau)\big)\, d\tau\right)\right]}_{\equiv F_i}^{\kappa_f}\big(k_i^p\big)^{\kappa_p}, \qquad \kappa \;=\; \kappa_f + \kappa_p \;\in\; (0,1). \tag{43}$$

Physical capital performs no tasks, so labour and automation capital remain the only factors that substitute within tasks.

A competitive producer supplies physical capital using the final good, with resource cost and supply price

$$G_p\big(K^p\big) \;=\; \frac{e}{1+e}\big(K^p\big)^{1+1/e}, \qquad p_p \;=\; \big(K^p\big)^{1/e}, \tag{44}$$

where $e > 0$ is the supply elasticity. A finite elasticity makes physical capital more expensive as its aggregate quantity rises, limiting output expansion when automation becomes cheaper.

Each firm spends the share $\kappa_p/\mu$ of revenue on physical capital. Summing over firms gives $p_p K^p = (\kappa_p/\mu)Y$, so the supply curve sets the price of physical capital as a function of aggregate output,

$$p_p \;=\; \left(\frac{\kappa_p}{\mu}\,Y\right)^{1/(1+e)}. \tag{45}$$

The capital producer's profit, $\Pi_p^U \equiv p_p K^p - G_p(K^p)$, accrues to firm owners in proportion to their profit shares at the welfare reference allocation and enters their welfare. Aggregate consumption is

$$C \;=\; Y - \sum_i c_{k,i} k_i - G_p\big(K^p\big).$$

Supplemental Material S1.2 and Online Appendix B provide the extended model's equilibrium conditions and solution algorithm.

Proposition 1 extends to this quantitative model provided that $\mu - 1 \geq \kappa_p/(1+e)$.

# 5 Quantification

This section takes the model to French manufacturing data. Section 5.1 describes the data and the measurement of automation. Section 5.2 documents that, within local labour markets, firms with more labour market power automate more, at both the extensive and the intensive margin. Section 5.3 sets out the parametric assumptions and normalisations, Section 5.4 the externally fixed parameters, and Section 5.5 the simulated method of moments. Section 5.6 reports the estimates under three task schedules and the fit of the model.

## 5.1 Data

**Sources.** The data are a firm-year panel of French manufacturing firms from 2009 to 2019, built from four administrative sources that share the firm identifier. Balance sheets (FARE) give sales, value added, physical capital, intermediate inputs, employment, labour costs, location, and industry. The production survey (EAP) reports revenue and quantities by product for every firm with at least 20 employees or turnover above €5 million and for a representative sample of smaller firms; because it reports prices and quantities separately, it yields firm-level output prices. Workforce records (DADS) report, for every job, the worker's wage, hours, age, occupation, and commuting zone, which I aggregate to the firm and year. Customs records report each firm's imports by product and year from 1996 to 2019, so every firm-year in the panel has a complete history of automation imports.

**Sample construction.** I merge the balance sheets with the production survey and keep firms present in both, which limits the sample to firms that meet the survey's reporting thresholds, then add the workforce records and the Customs history. I drop five two-digit industries with very few observations or whose firms may be intermediaries for automation equipment, observations with missing prices, quantities, inputs, employment, or location, and firms with fewer than five employees on average, and I keep each firm's longest run of consecutive years, provided it lasts at least three years. Monetary variables are in 2015 euros, winsorised at the first and ninety-ninth percentiles. The result is a panel of firms in eighteen manufacturing industries.

I estimate production functions on the twelve of these industries in which automation capital is pervasive, which I call the production-function sample. A local labour market is a five-digit industry by commuting zone, employment is the firm's number of workers, and the within-market share $s_{i|j}$ is its share of the market's employment. Supplemental Material S3.1 provides additional details.

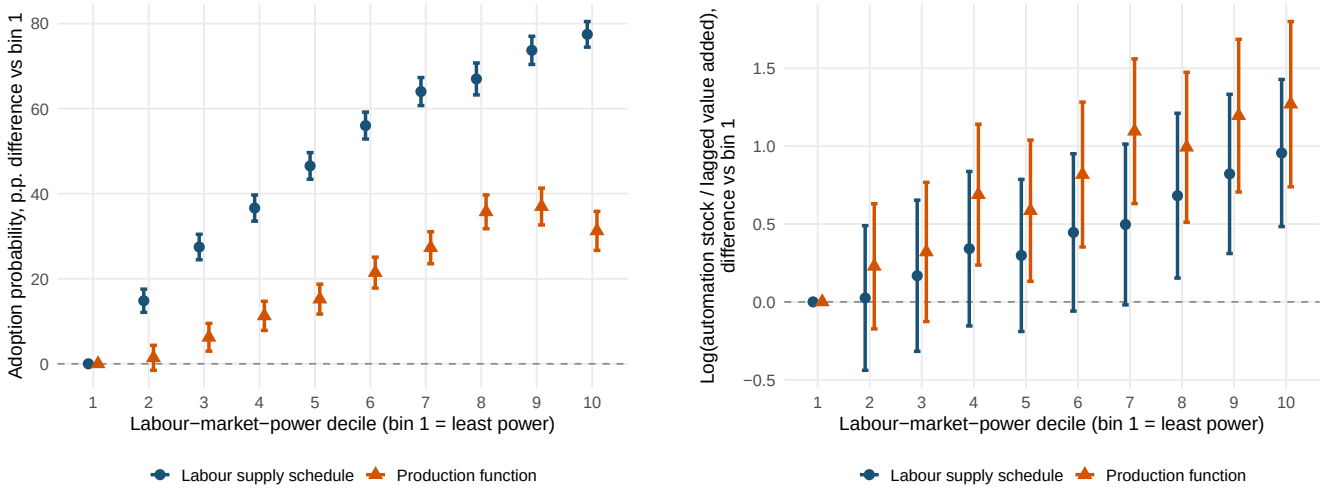
**Automation.** Automation is measured from imports. An import is automation capital when its product code is on the list of industrial robots, numerically controlled machines, automatic machine tools, and related equipment of Acemoglu and Restrepo (2022a). Let $e_{it}$ denote firm $i$'s imports of automation capital in year $t$, in 2015 euros. The automation stock accumulates these imports over an eight-year straight-line service life,

$$K^A_{it} = \sum_{s=0}^{7} \left(1 - \frac{s}{8}\right) e_{i,t-s}, \tag{46}$$

so each purchase enters the stock at its full value, loses one eighth of that value each year, and leaves after eight years. A firm is an adopter in year $t$ when $K^A_{it} > 0$, and its automation intensity is $K^A_{it}/\mathrm{VA}_{i,t-1}$, the stock per euro of lagged value added. The model has no intermediate inputs, so value added is the data counterpart of a firm's revenue in the model. Supplemental Material S3.1.2 gives the product list.

**Sample statistics.** The panel has 21,918 firms, 15,189 of them in 2019. The production-function sample has 16,525 firms and 134,762 firm-years and accounts for 86 per cent of the panel's sales. The production-function markdowns are estimated on the production-function sample; the model-consistent markdowns, the market structure and the estimation moments use the whole panel. Automation is common but lumpy: in 2019, 44 per cent of the panel's firms hold a positive stock; in a given year about three firm-years in ten in the production-function sample record automation imports; and the median firm, with 25 workers, imports none. Supplemental Material S3.1 reports the sample tables.

Figure 4: AUTOMATION AND LABOUR MARKET POWER




(a) Extensive margin

(b) Intensive margin

*Notes:* Estimates of $\beta_b$ from regression (47), relative to bin 1, the highest markdowns; bin 10 has the most labour market power. Navy circles use the model-consistent markdown $\delta(s_{i|j})$ on the whole panel; orange triangles use the production-function markdown of Supplemental Material S3 on the production-function sample. Panel (a): probability of holding a positive automation stock, in percentage points. Panel (b): log automation stock over lagged value added, in log points, adopters only. Standard errors are clustered by firm; the bands are 95 per cent confidence intervals. Panel (a) uses 92,709 firm-years (14,266 firms) for the model-consistent series and 70,784 (10,431 firms) for the production-function series; panel (b) uses 34,767 and 29,360 adopter firm-years.

## 5.2 Labour market power and automation in the data

**Labour market power.** Markdowns are measured using the production-function approach and in a model-consistent manner. Under the production-function approach, following Yeh et al. (2022), the markdown is the labour share of revenue times the markup over the output elasticity of labour, and I estimate the output elasticities by two-digit industry. The estimation corrects for unobserved productivity with a control function, for unobserved output prices with the firm-level prices from the production survey, and for unobserved input prices with a second control function and sector-by-location fixed effects. The median markdown is 0.68, and markdowns fall with firm size (Supplemental Material S3.3). The model-consistent markdown is $\delta(s_{i|j})$, computed from each firm's within-market employment share by (23), with the labour-supply elasticities $(\varsigma, \eta)$ of Table 2.

**Automation adoption and intensity.** I sort firm-year observations into deciles of the markdown, from the least labour market power in bin 1, the highest markdowns, to the most in bin 10, and estimate for each margin

$$y_{it} \;=\; \sum_{b=2}^{10} \beta_b \, 1\{b_{it} = b\} \;+\; \gamma_{j(i)t} \;+\; \varepsilon_{it}, \tag{47}$$

where $\gamma_{j(i)t}$ is a local-labour-market by year fixed effect, so each $\beta_b$ compares firms with competitors in the same market and year. For the extensive margin the outcome is the adoption indicator. For the intensive margin the sample is restricted to adopters and the outcome is the log of the automation stock divided by lagged value added.[8]

Figure 4 reports the estimates. With both markdowns, the probability of holding automation capital and the automation intensity of adopters rise with labour market power within local labour markets. By the model-consistent markdown, firms in the top decile are 78 percentage points more likely to hold automation capital than firms in the bottom decile of the same market and year, and adopters in the top decile hold 0.96 log points more automation capital per euro of lagged value added, about 2.6 times as much. The production-function markdown gives the same positive relation on both margins: firms in the top decile are 31 percentage points more likely to hold automation capital, and adopters in the top decile hold about 3.5 times as much per euro of lagged value added.[9]

## 5.3 Parametric assumptions and normalisations

**Task schedule.** The estimation adopts the power–exponential family

$$\phi_l(\tau) = \zeta\Big(\frac{\tau}{M}\Big)^{a} \exp\!\Big(\beta_R \frac{\tau}{M}\Big), \qquad \phi_k(\tau) = \zeta\Big(\frac{\tau}{M}\Big)^{a}\Big(1 - \frac{\tau}{M}\Big)^{a}. \tag{48}$$

[8] Value added enters with a lag because contemporaneous value added is itself an outcome that automation affects. The bin boundaries are constructed once on the pooled sample and held fixed across the two margins.

[9] The estimated production function does not include the automation stock, so its labour elasticity responds to automation only through the mix of capital, labour and intermediate inputs. If automation lowers a firm's labour elasticity by more, the lower labour share is attributed to a lower markdown, which would steepen the production-function gradient. The model-consistent markdown does not use the production function and is unaffected.

Here $a \in (0,1)$, $\beta_R > -a$, and $\zeta > 0$. The task productivities imply

$$R(\tau) = \frac{\phi_l(\tau)}{\phi_k(\tau)} = \exp\!\left(\beta_R \frac{\tau}{M}\right)\left(1 - \frac{\tau}{M}\right)^{-a}. \tag{49}$$

The restriction $\beta_R > -a$ makes $R$ strictly increasing, with $R(M) = \infty$. Because labour and capital productivity share the scale $\zeta$, $R(0) = 1 > 0$: the family is the positive-intercept schedule of Section 4.2, under which a firm automates nothing when $\omega_i \leq c_k b_i$. The common factor $(\tau/M)^a$ in (48) gives capital productivity a finite comparative advantage at the bottom of the task space and permits a no-automation corner. The exponential term controls how rapidly labour's comparative advantage increases across tasks. A higher $\beta_R$ makes this increase steeper, while leaving $R(0) = 1$, and hence the adoption cutoff, unchanged.

When $\beta_R = 0$, the specification reduces to the power specification used by Guerreiro et al. (2022). In this family the displacement hazard is $h(\tilde{x}) = [a + \beta_R(1 - \tilde{x})]^{-1}$. Under the power specification the hazard is constant: a given fall in the automation price displaces the same share of every adopter's remaining labour tasks, so the task schedule cannot generate equalising incidence. The sign of $\beta_R$ sets the shape. A negative value gives the falling hazard of Figure 2, a positive value the rising one.

**Distributional assumptions.** Three firm-level primitives are drawn i.i.d. across firms. Hicks-neutral productivity is lognormal, $\log \phi_{ih} \sim N(\mu_\phi, \sigma_\phi^2)$, with dispersion $\sigma_\phi$, and amenities are Pareto, $T_i \sim \text{Pareto}(x_{\min}, \iota)$, with scale $x_{\min}$ and shape $\iota$. Productivity and amenity are dependent through one parameter: with $F_N$ the standard normal distribution function, the two draws are generated from a pair of standard normals with correlation $\mathcal{R}_{\phi T}$,

$$\begin{pmatrix} z_{\phi,i} \\ z_{T,i} \end{pmatrix} \sim N\left(0, \begin{pmatrix} 1 & \mathcal{R}_{\phi T} \\ \mathcal{R}_{\phi T} & 1 \end{pmatrix}\right), \quad \phi_{ih} = \exp(\mu_\phi + \sigma_\phi\, z_{\phi,i}), \quad T_i = x_{\min}\left[1 - F_N(z_{T,i})\right]^{-1/\iota}. \tag{50}$$

Since $1 - F_N(z_{T,i})$ is uniform on $(0,1)$, $T_i$ has the Pareto marginal for every $\mathcal{R}_{\phi T}$, and $\phi_{ih}$ the lognormal one: (50) is a Gaussian copula. The installation requirement has distribution $G_b$, with $\log b_i \sim N(\mu_b, \upsilon^2)$ and dispersion $\upsilon$, independent of $(\phi_{ih}, T_i)$.

**Normalisations.** Every targeted moment is dimensionless, so the levels and units of several objects are not identified. Table 2 lists the parameters set by normalisation, and Supplemental Material S2 explains each one.

## 5.4 Externally fixed parameters

The labour-supply, product-demand, and technology parameters are fixed outside the SMM estimation. Both labour-supply shapes are taken from Azkarate-Askasua and Zerecero's (2025) estimates for French manufacturing. I set the between-market Fréchet shape to $\eta = 0.33$ and the common within-market shape to $\varsigma = 3.455$, the upper end of their sector-specific estimates.

On the product-market side, I set the demand elasticity to $\rho = 4.333$, so that the CES markup is $\mu = \rho/(\rho - 1) = 1.3$, the lower end of the 1.3–1.6 range documented across OECD manufacturing by Calligaris et al. (2018).

Total returns to scale are $\kappa = 0.972$ and the physical-capital elasticity is $\kappa_p = 0.338$, both from Azkarate-Askasua and Zerecero's (2025) estimates for French manufacturing. The task-composite elasticity is the residual $\kappa_f = \kappa - \kappa_p = 0.634$, and the net-revenue curvature is $\tilde{\mu} = \mu - \kappa_p = 0.962$. The supply elasticity of physical capital is $e = 1.5$.

The market structure is taken from the data: $J$ and $\{N_j\}$ are the observed local labour markets of the panel in 2019 and their firm counts, and $L$ is their total employment.

Figure S3.1 in Supplemental Material S3.1.4 shows the market structure the model inherits. Local labour markets are highly concentrated in counts: 71 per cent of five-digit-industry-by-commuting-zone markets contain a single firm. Economic activity is spread considerably more evenly, because single-firm markets are disproportionately small: they hold 44 per cent of firms and around 48 per cent of employment and sales, while markets with six or more firms account for only 2 per cent of markets but 14 per cent of firms and 11 per cent of employment.[10]

**Proposition 6.** *Under the maintained identification assumptions, the parameters*

$$(\sigma_\phi, \iota, \mathcal{R}_{\phi T}, \upsilon, c_k, a, \beta_R)$$

[10]The coarser three-digit market definition gives larger markets: 59 per cent of markets contain a single firm, holding 27 per cent of firms and roughly 31 per cent of employment and sales.

*are globally identified from the joint population distribution of wages, employment and revenue, together with the observed market structure.*

The proof is in Online Appendix A.8.

## 5.5 Simulated method of moments

Proposition 6 establishes identification from the full population distribution of observables. To assess how the quantitative implications depend on the task technology, I estimate the model separately under falling, constant and rising displacement hazards, re-estimating firm heterogeneity and installation costs against the same empirical targets. The baseline sets $a = 0.5$; Section 6.3 examines sensitivity to $a = 0.1$ and $a = 0.9$. The three baseline schedules set $\beta_R = -0.25$, 0 and 1, respectively. Conditional on each schedule, I estimate $\theta = (\sigma_\phi, \iota, \mathcal{R}_{\phi T}, \upsilon, c_k)$ by simulated method of moments:

$$\hat{\theta}(a, \beta_R) = \arg\min_{\theta} \big[\mathbf{m}^{\text{data}} - \mathbf{m}(\theta; a, \beta_R)\big]' \mathbf{W} \big[\mathbf{m}^{\text{data}} - \mathbf{m}(\theta; a, \beta_R)\big], \tag{51}$$

where $\mathbf{m}^{\text{data}}$ contains the five empirical moments, $\mathbf{m}(\theta; a, \beta_R)$ their simulated counterparts, and $\mathbf{W}$ is the weighting matrix. The same moments and weighting matrix are used for every task schedule. The model moments are computed using the equilibrium algorithm in Online Appendix B. Table 3 reports the five targeted moments and their simulated counterparts.

**Measurement.** The empirical moments use the 2019 cross-section. Wages are deflated wage bills per worker, and marginal labour costs are $\omega_i = w_i/\delta_i$ with the model-consistent markdowns of Section 5.2. Supplemental Material S4.2 gives sample restrictions and moment formulas.

**Firm heterogeneity.** The first block disciplines productivity dispersion, $\sigma_\phi$, amenity dispersion, $\iota$, and the correlation between the two, $\mathcal{R}_{\phi T}$. More productive firms are larger and pay higher wages, whereas firms with better amenities attract more workers at lower wages. The within-market variances of log wages (M1) and log employment (M2) and their covariance (M3) separate the two sources: productivity dispersion raises all three, amenity dispersion raises both

variances and lowers the covariance. The correlation is identified because a productive firm that also offers good amenities reaches a large size with a smaller wage premium: the higher the correlation, the smaller the dispersion of wages relative to the dispersion of employment (Supplemental Material S4.1).

**Adoption selection.** The second block disciplines the dispersion of installation requirements, $\upsilon$, given the task schedule. A firm adopts when $\omega_i > c_k b_i$. If every firm faced the same installation requirement, adoption would follow a single cutoff in the marginal labour cost: every firm above it would adopt, none below. Heterogeneous requirements blur that cutoff. A firm with a low marginal labour cost adopts because installation is easy for it. A firm with a higher marginal labour cost stays out because installation is hard. On all eligible firms I estimate the linear probability model

$$d_i^{\text{data}} = \gamma_j + \beta_4 \log \omega_i^{\text{data}} + v_i, \tag{52}$$

where $\gamma_j$ is a local-labour-market fixed effect, and target the slope $\hat{\beta}_4$ (M4). The slope measures how adoption rises with the marginal labour cost within a market. The more dispersed the installation requirements, the flatter it is.

**Adoption rate.** The third block disciplines the common capital cost, $c_k$, given the task schedule and the installation dispersion $\upsilon$. A lower capital cost moves more firms across the adoption threshold. I therefore target the aggregate adoption rate (M5), the share of adopters among all eligible 2019 firms.

**Weighting.** I estimate the covariance matrix of the empirical moments, $\hat{\mathbf{C}}$, using 1,000 non-parametric bootstrap replications clustered by local labour market. The weighting matrix $\mathbf{W}$ is diagonal, with inverse bootstrap variances as weights, so the objective is the sum of squared discrepancies standardised by each moment's standard error. The full covariance matrix is used for inference.

Table 2: MODEL PARAMETERS

| | | | | | |
|---|---|---|---|---|---|
| *Externally fixed* | | | *Set by normalisation* | | |
| $\rho$ | Demand elasticity | 4.333 | $M$ | Task measure | 1 |
| $\kappa$ | Total returns to scale | 0.972 | $x_{\min}$ | Amenity scale | $e^{-1/\iota}$ |
| $\kappa_p$ | Physical-capital elasticity | 0.338 | $A_p$ | Cost scale of $K^p$ | 1 |
| $\kappa_f$ | Task-composite elasticity | 0.634 | $\zeta$ | Task-schedule level | 1 |
| $e$ | Supply elasticity of $K^p$ | 1.5 | $R(0)$ | Comparative-advantage level | 1 |
| $\varsigma$ | Within-market shape | 3.455 | $\mu_\phi$ | Productivity location | $-\sigma_\phi^2/2$ |
| $\eta$ | Between-market shape | 0.330 | $\mu_b$ | Installation location | $-\upsilon^2/2$ |
| | | | $L$ | Worker mass | Sample |

| | | Task schedule | | |
|---|---|---|---|---|
| | | Falling hazard | Constant hazard | Rising hazard |
| *Fixed schedule parameters* | | | | |
| $a$ | Task-schedule curvature | 0.5 | 0.5 | 0.5 |
| $\beta_R$ | Task-schedule shape | −0.25 | 0 | 1 |
| *Estimated* | | | | |
| $\sigma_\phi$ | Productivity dispersion | 0.577 (0.010) | 0.567 (0.009) | 0.546 (0.009) |
| $\iota$ | Amenity dispersion | 0.755 (0.020) | 0.755 (0.020) | 0.754 (0.020) |
| $\mathcal{R}_{\phi T}$ | Productivity–amenity correlation | 0.363 (0.029) | 0.375 (0.028) | 0.403 (0.027) |
| $\upsilon$ | Installation dispersion | 0.513 (0.034) | 0.563 (0.039) | 0.706 (0.090) |
| $c_k$ | Automation capital price | 0.957 (0.030) | 1.075 (0.041) | 1.694 (0.169) |

*Notes:* The demand elasticity implies a markup of $\mu = 1.3$; $\kappa_f = \kappa - \kappa_p$ and $\tilde{\mu} = \mu - \kappa_p = 0.962$. The normalisations set $\mathbb{E}[\log T_i] = 0$, $\mathbb{E}[\phi_{ih}] = 1$, and $\mathbb{E}[b_i] = 1$; the copula (50) leaves the marginals and the normalisations unchanged. With $R(0) = 1$ and $\mathbb{E}[b_i] = 1$, a firm adopts when $\omega_i > c_k b_i$, and $c_k$ is the average unit cost of installed automation capital; $A_p$ is the scale of the physical-capital cost function (Supplemental Material S2), and $L$ equals employment in the panel in 2019. The three schedules share $a = 0.5$ and differ in $\beta_R$; standard errors from (53) in parentheses.

## 5.6 Estimation results and model fit

Table 2 collects the model's parameters in three panels: those fixed externally (Section 5.4), those set by normalisation (Section 5.3), and those estimated by SMM under each of the three task schedules. Under each schedule the Jacobian of the five moments with respect to the five parameters has full rank, so the estimates are locally identified (Supplemental Material S4.1).

**Standard errors.** The estimated covariance matrix of $\hat{\theta}$ is

$$\hat{\mathbf{V}} = \left(\hat{\mathbf{D}}'\mathbf{W}\hat{\mathbf{D}}\right)^{-1}\hat{\mathbf{D}}'\mathbf{W}\left(\hat{\mathbf{C}} + \hat{\mathbf{C}}_{\text{sim}}\right)\mathbf{W}\hat{\mathbf{D}}\left(\hat{\mathbf{D}}'\mathbf{W}\hat{\mathbf{D}}\right)^{-1}, \tag{53}$$

Table 3: TARGETED MOMENTS

| | Empirical target | Data | Model | | |
|---|---|---|---|---|---|
| | | | Falling | Constant | Rising |
| *Block 1: firm heterogeneity* | | | | | |
| (M1) | Within-market variance of log wages | 0.048 (0.003) | 0.048 | 0.048 | 0.048 |
| (M2) | Within-market variance of log employment | 0.765 (0.020) | 0.765 | 0.765 | 0.765 |
| (M3) | Within-market covariance of log wages and log employment | 0.020 (0.003) | 0.020 | 0.020 | 0.020 |
| *Block 2: adoption selection* | | | | | |
| (M4) | Within-market slope of automation adoption on the marginal labour cost | 0.438 (0.021) | 0.438 | 0.438 | 0.438 |
| *Block 3: adoption rate* | | | | | |
| (M5) | Aggregate automation-adoption rate | 0.437 (0.006) | 0.437 | 0.437 | 0.437 |

*Notes:* The Data column reports the empirical moments computed on the 2019 cross-section of the panel, with a local labour market defined as a five-digit industry $\times$ commuting zone and an adopter defined by a positive eight-year automation stock. Standard errors, in parentheses, come from a nonparametric cluster bootstrap that draws local labour markets with replacement over 1,000 replications. The Model columns report the simulated counterparts at the estimated parameters of Table 2 under each task schedule.

where $\hat{\mathbf{D}}$ is the Jacobian of the five model moments with respect to the five estimated parameters, evaluated at $\hat{\theta}$, $\mathbf{W}$ is the weighting matrix, and $\hat{\mathbf{C}}$ is the bootstrap covariance matrix of the empirical moments. The simulation covariance $\hat{\mathbf{C}}_{\text{sim}}$, estimated from 100 independent sets of firm draws at $\hat{\theta}$, accounts for computing the model moments from a single simulated economy. The standard errors in Table 2 are the square roots of the diagonal elements of $\hat{\mathbf{V}}$.

**Model fit on targeted moments.** With five parameters and five moments the model is exactly identified, and it matches every targeted moment (Table 3).

# 6 Counterfactual Experiments

I conduct two experiments. The first measures how accounting for automation changes the welfare gain from removing labour markdowns. The second examines how cheaper automation changes the welfare cost of labour market power. I first report results for the three baseline

task schedules, with $a = 0.5$.

Section 6.3 examines sensitivity to the task schedule and the definition of local labour markets. Both welfare-cost findings survive every alternative task schedule and market definition examined: ignoring automation understates the welfare cost of labour market power, and cheaper automation increases that cost. Supplemental Material S5 reports the estimates, model fit and counterfactual results for each specification.

**Environment.** Each experiment holds fixed the estimated parameters, the firm draws, the market structure and total employment. The price of physical capital moves with aggregate output through (45). The product-market markup is kept, so the equilibrium without labour markdowns is not the first best. Welfare is measured by (33), with the reference fixed at the estimated 2019 economy.

## 6.1 The welfare cost of labour market power with and without automation

I remove labour markdowns in two economies. In the first, firms choose their automated shares before and after the reform. In the second, labour performs every task. Removing markdowns sets $\delta_i = 1$ at every firm, so wages equal marginal revenue products. The welfare cost of labour market power is the gain from this reform: $G$ with endogenous automation and $G^0$ without automation, both expressed as percentages of 2019 consumption.

Table 4 shows that removing markdowns raises social welfare by 6.3–9.0 per cent of 2019 consumption with endogenous automation, compared with 4.8–8.1 per cent without automation. Excluding automation therefore understates the welfare gain by 0.9–1.5 per cent of 2019 consumption, or 10–24 per cent of the gain with automation.

Proposition 3 identifies a reason why automation matters for this comparison. In the analytical economy, introducing wage-setting power shifts workers away from the productive firm and induces it to automate more tasks, while its rival employs more workers and automates fewer. Labour market power therefore distorts both employment and task assignment. Removing markdowns allows firms to adjust both margins: where marginal labour costs fall, firms return automated tasks to workers; where they rise, firms substitute automation capital

Table 4: THE WELFARE COST OF LABOUR MARKET POWER

| | Falling hazard | | Constant hazard | | Rising hazard | |
|---|---|---|---|---|---|---|
| | Endogenous automation | No automation | Endogenous automation | No automation | Endogenous automation | No automation |
| **Panel A. Oligopsonistic economy, levels with the 2019 economy** = 1 | | | | | | |
| Consumption $C$ | 1 | 0.513 | 1 | 0.563 | 1 | 0.719 |
| Wage bill $\mathcal{B}$, share of 2019 $C$ | 0.130 | 0.124 | 0.144 | 0.136 | 0.186 | 0.174 |
| Workers' welfare $\widetilde{\mathcal{W}}$ | 1 | 0.902 | 1 | 0.905 | 1 | 0.926 |
| Owners' welfare $\mathcal{W}^f$ | 1 | 0.447 | 1 | 0.499 | 1 | 0.669 |
| Social welfare $\mathcal{S}$ | 1 | 0.506 | 1 | 0.558 | 1 | 0.717 |
| **Panel B. Equilibrium without labour markdowns, levels with the 2019 economy** = 1 | | | | | | |
| Consumption $C$ | 0.994 | 0.507 | 0.993 | 0.556 | 0.990 | 0.710 |
| Wage bill $\mathcal{B}$, share of 2019 $C$ | 0.309 | 0.293 | 0.343 | 0.321 | 0.439 | 0.410 |
| Workers' welfare $\widetilde{\mathcal{W}}$ | 2.904 | 2.615 | 2.903 | 2.623 | 2.899 | 2.682 |
| Owners' welfare $\mathcal{W}^f$ | 0.787 | 0.246 | 0.760 | 0.274 | 0.677 | 0.368 |
| Social welfare $\mathcal{S}$ | 1.062 | 0.554 | 1.070 | 0.614 | 1.090 | 0.798 |
| Welfare cost $G$, $G^0$ (% of 2019 $C$) | 6.25 | 4.77 | 6.95 | 5.58 | 9.00 | 8.11 |
| Difference $G - G^0$ (% of 2019 $C$) | 1.47 | | 1.37 | | 0.89 | |
| Difference (% of $G$) | 23.6 | | 19.7 | | 9.9 | |

*Notes:* Each pair of columns is the economy estimated under the stated task schedule, with firms choosing their automated shares (endogenous automation) or every automated share held at zero (no automation). Levels are relative to the estimated 2019 economy, with the welfare reference fixed at that economy. The wage bill is expressed as a share of 2019 consumption, so that the two economies are compared in the same unit. Workers' and owners' welfare are each normalised by their respective 2019 levels. Workers' welfare is their consumption equivalent, $\widetilde{\mathcal{W}} \equiv \sigma C \mathcal{N}/\tilde{\mathcal{N}}$, the first term of (33). Social welfare is measured by (33). The welfare cost is the change in social welfare from Panel A to Panel B, and the difference is $G - G^0$.

for labour. These adjustments allow employment to change without an equivalent change in output. Without automation, firms accommodate employment changes entirely through production and conventional capital.

Task adjustment does not, however, guarantee a larger welfare gain from removing markdowns. Automation also reduces labour's share of production costs, so a given proportional fall in marginal labour cost induces less expansion in production. The quantitative comparison incorporates both forces, together with changes in wages, capital prices and employment across markets. Across all three estimated task schedules, the welfare gain is larger when firms can automate.

Consumption nevertheless falls after markdowns are removed in both economies. The

reform raises relative wages most in single-firm markets, where firms have greater wage-setting power.[11] These firms have lower marginal revenue products of labour in the estimated economies, so the resulting movement of workers across markets outweighs the output gains from reallocation towards more productive firms within markets. Workers value job amenities as well as consumption, however, and their gains from changing jobs outweigh the aggregate consumption loss. Task reassignment provides a channel through which production can accommodate this movement towards preferred jobs, even when aggregate consumption falls.

## 6.2 Cheaper automation and the welfare cost of labour market power

I next ask whether a fall in the price of automation capital raises the welfare cost of labour market power.[12] The welfare cost at automation price $c_k$ is the gap between social welfare without and with labour markdowns,

$$G(c_k) \equiv 100\,\frac{\mathcal{S}^{\mathrm{B}}(c_k) - \mathcal{S}(c_k)}{\bar{C}}, \tag{54}$$

where the superscript B marks the equilibrium without labour markdowns and $\bar{C}$ is 2019 consumption. At the 2019 price, $G(c_k)$ is the gain $G$ of Table 4. I lower the price in both economies and compare $G(c_k)$ before and after. The change in $G(c_k)$ equals the gain from cheaper automation without labour markdowns less the gain in the oligopsonistic economy.

Table 5 reports the results. Because the cut is set separately for each schedule to deliver the same growth in automation capital, differences across columns reflect both the shape of the task schedule and the size of the cut. Cheaper automation raises social welfare by 15 to 29 per cent of 2019 consumption in the oligopsonistic economy. The price reduction lowers the cost of performing tasks already automated and makes automation profitable for additional tasks. Firms expand production, and consumption rises by 15 to 29 per cent: the economy produces more consumption goods with the same workforce.

[11] At the five-digit-industry-by-commuting-zone level, single-firm markets account for 71 per cent of markets and around 48 per cent of employment.

[12] The size of the fall is set so that automation capital in the oligopsonistic economy grows by 80 per cent, the projected growth of France's stock of industrial robots from 2019 to 2029 (Supplemental Material S4.3). Each estimated economy therefore has its own cut, between 12.5 and 14.8 per cent.

Table 5: CHEAPER AUTOMATION AND THE WELFARE COST OF LABOUR MARKET POWER

| | Falling hazard | Constant hazard | Rising hazard |
|---|---|---|---|
| Automation-price reduction (%) | 12.5 | 12.9 | 14.8 |
| Increase in consumption, oligopsonistic economy (%) | 28.95 | 25.61 | 15.49 |
| **Welfare cost of labour market power (% of 2019 $C$)** | | | |
| At the 2019 price | 6.25 | 6.95 | 9.00 |
| At the lower price | 6.52 | 7.25 | 9.28 |
| Increase (% of 2019 $C$) | 0.27 | 0.30 | 0.28 |
| Increase (% of the cost at the 2019 price) | 4.4 | 4.3 | 3.1 |
| **Gain from cheaper automation (% of 2019 $C$)** | | | |
| Oligopsonistic economy: social welfare $\mathcal{S}$ | 29.03 | 25.68 | 15.49 |
| Workers' welfare $\widetilde{\mathcal{W}}$ | 0.37 | 0.46 | 0.60 |
| Owners' welfare $\mathcal{W}^f$ | 28.66 | 25.22 | 14.89 |
| No labour markdowns: social welfare $\mathcal{S}$ | 29.30 | 25.98 | 15.77 |
| Workers' welfare $\widetilde{\mathcal{W}}$ | 1.09 | 1.34 | 1.76 |
| Owners' welfare $\mathcal{W}^f$ | 28.21 | 24.64 | 14.01 |

*Notes:* Each gain from cheaper automation compares an economy at the lower price with the same economy at the 2019 price. Workers' and owners' gains are changes in $\widetilde{\mathcal{W}}$ and $\mathcal{W}^f$, in per cent of 2019 consumption, and sum to the social welfare gain. Welfare uses the 2019 economy as reference.

Owners receive almost all of the welfare gain. Workers gain 0.4 to 0.6 per cent of 2019 consumption, while owners gain 15 to 29 per cent. Cheaper automation expands production, but it also replaces workers in additional tasks, limiting the increase in labour demand and wages. Firms can therefore produce substantially more without a comparable increase in their wage bills, and much of the benefit accrues as profit. Labour market power further limits workers' gains: without markdowns, their welfare gain is about three times as large.

Cheaper automation also raises the welfare cost of labour market power by 0.27 to 0.30 per cent of 2019 consumption, or 3 to 4 per cent of its initial level. Proposition 4 identifies how cheaper automation can change the underlying distortion. Firms that already automate more benefit from a larger reduction in production costs, giving them a stronger incentive to expand. If this scale advantage outweighs their additional displacement of labour, they attract workers and acquire greater wage-setting power.

This mechanism links the allocation of automation gains to the welfare cost of labour market

power. When expansion concentrates employment at firms that already have greater wage-setting power, those firms pay a smaller fraction of their marginal revenue product to workers, limiting how much workers benefit from cheaper automation. The quantitative experiment incorporates this response alongside changes in production, capital and task assignment. Across all three task schedules, welfare rises more without markdowns, so cheaper automation increases the welfare gain available from removing labour market power.

### 6.3 Sensitivity analysis

**Task schedule.** Re-estimating the model at $a = 0.1$ and $a = 0.9$, under each hazard shape, preserves both findings. Across the baseline and alternative schedules, ignoring automation understates the gain from removing markdowns by approximately 4–48 per cent, while cheaper automation increases the welfare cost by 1.0–5.2 per cent. The baseline ranges of 10–24 and 3.1–4.4 per cent are therefore conditional on $a = 0.5$.

**Market definition.** Defining local labour markets as three-digit industries within commuting zones produces similar results. Ignoring automation understates the gain from removing markdowns by 8–20 per cent, and cheaper automation increases the welfare cost by 2.8–4.0 per cent.

## 7 Conclusion

In this paper, I connect two strands of the literature, one focused on automation and another on the wage-setting power of firms, and show that their interaction has consequences for welfare. I develop a quantitative general equilibrium model of oligopsonistic labour markets in which firms' automation choices and wage-setting power are jointly determined. I establish sufficient conditions for a unique equilibrium and for global identification of the model's parameters. I present empirical evidence from French manufacturing supporting the model's prediction that firms with greater wage-setting power are more likely to hold automation capital and use it more intensively.

The quantitative analysis shows that this interaction matters for welfare. Excluding automation understates the gains from removing labour market power by 10–24 per cent. Cheaper automation raises social welfare, delivers most of its gains to firm owners and increases the welfare cost of labour market power by 3–4 per cent.

These findings have implications for policies addressing labour market power. Minimum wages (Berger et al., 2025) and collective bargaining (Azkarate-Askasua and Zerecero, 2025) can counteract employer power while encouraging automation, so their evaluation must account for firms' technology choices. The framework developed here provides a basis both for evaluating these policies and for understanding how firms' labour market power shapes the welfare consequences of technological advances such as AI (Althoff and Reichardt, 2026).

# References


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# ONLINE APPENDIX

# LABOUR MARKET POWER, AUTOMATION AND WELFARE

Aseem Patel
*University of Essex*

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This Online Appendix is organised as follows. Appendix A proves the propositions of the main text. Appendix B gives the algorithm that solves the quantitative model. Equations, propositions, sections, tables and figures without a letter prefix refer to the main paper; those with an S prefix refer to the Supplemental Material.

# A Proofs

## A.1 Assumptions and lemmas

This section states a maintained assumption and two lemmas used throughout the proofs.

**Maintained assumption.**

(A1) Each firm takes $Y$ and $\Phi$ as given, as in Definition 1(ii), including in the comparative statics, in which $Y$ and $\Phi$ move but no firm internalises its own effect on them. In the analytical economy the single market employs every worker, so each firm prices its labour supply with the between-market shape $\eta$ as in (2) without internalising that the market's share of the labour force is one.

The lemmas are stated for the baseline model of Section 2, in which the comparative-advantage schedule starts at zero and every firm automates some, but not all, tasks. The final paragraph of this section extends them to a positive intercept and to the adoption margin of Section 4.2.

**Notation.** Write $\tilde{R}(\tilde{x}) \equiv R(M\tilde{x})$ for the schedule in normalised units. In terms of its slope, the displacement hazard (26) and the direct labour-demand shift (39) are

$$\psi(\tilde{x}) \equiv \frac{d \log \tilde{R}(\tilde{x})}{d\tilde{x}} > 0, \qquad h(\tilde{x}) = \frac{1}{(1-\tilde{x})\,\psi(\tilde{x})}, \qquad g(\tilde{x}) = \frac{\kappa_f\,\tilde{x}}{\mu - \kappa_f} - h(\tilde{x}).$$

For firm $i$, $h_i \equiv h(\tilde{x}_i)$ and $g_i \equiv g(\tilde{x}_i)$. Since $\varepsilon'(s) = -(\varsigma - \eta) < 0$ because $\eta < \varsigma$, the markdown map (23) is strictly decreasing,

$$\delta'(s) = -\frac{\varsigma - \eta}{\big(1 + \varepsilon(s)\big)^2} < 0. \tag{A.1}$$

**Tasks.**

**Lemma 1** (Task assignment and the automation threshold)**.** *Let Assumption 1 hold and fix factor prices $c > 0$ for automation capital and $\omega > 0$ for labour.*

(i) *In any cost-minimising production plan, almost every task is served by a single factor, and the assignment takes a threshold form: there exists a unique* $x \in (0, M)$ *such that tasks* $\tau < x$ *use capital only and tasks* $\tau > x$ *use labour only, with* $x = R^{-1}(\omega/c)$.

(ii) *The automated share* $\tilde{x} = x/M \in (0,1)$ *depends on prices only through* $\omega/c$*, is invariant to Hicks-neutral productivity* $\phi_h$*, and is continuously differentiable and strictly increasing in* $\omega/c$*, with*

$$d\tilde{x} = (1-\tilde{x})\, h(\tilde{x})\, \big(d\log\omega - d\log c\big), \qquad d\log(1-\tilde{x}) = -\,h(\tilde{x})\, \big(d\log\omega - d\log c\big). \tag{A.2}$$

**Proof.** (i) Every task is essential under the logarithmic aggregate in (7), so a cost-minimising plan produces $y(\tau) > 0$ at almost every task, and minimises cost task by task: producing one unit of task output costs $c/\phi_k(\tau)$ with capital and $\omega/\phi_l(\tau)$ with labour, and since task output is linear in the two inputs, the minimal cost is $\min\{c/\phi_k(\tau),\, \omega/\phi_l(\tau)\}$, attained by using only the cheaper factor. Capital is strictly cheaper at task $\tau$ if and only if $c/\phi_k(\tau) < \omega/\phi_l(\tau)$, that is, $R(\tau) < \omega/c$. Under Assumption 1, $R$ is continuous and strictly increasing on $(0, M)$ with $R(\tau) \to 0$ as $\tau \to 0$ and $R(\tau) \to \infty$ as $\tau \to M$, so the set of tasks at which capital is strictly cheaper is the non-empty interval $[0, x)$ with $x = R^{-1}(\omega/c)$. Strict monotonicity of $R$ makes the threshold unique, and indifference between the factors occurs at most at the single task $\tau = x$, a set of measure zero. Because $R(\tau) \to \infty$ as $\tau \to M$, labour is strictly cheaper on a neighbourhood of $M$, so $x < M$.

(ii) Part (i) gives $\tilde{x} = M^{-1}R^{-1}(\omega/c)$, a function of $\omega/c$ alone, so $\phi_h$ does not enter, and $R \in C^1$ with $R' > 0$ gives continuous differentiability and strict monotonicity. Taking logs of $\tilde{R}(\tilde{x}) = \omega/c$ and differentiating, $\psi(\tilde{x})\, d\tilde{x} = d\log\omega - d\log c$, and $1/\psi(\tilde{x}) = (1-\tilde{x})\, h(\tilde{x})$ by the definition of the hazard. Dividing by $-(1-\tilde{x})$ gives the second expression in (A.2). □

The lemma is stated for arbitrary factor prices, so it applies at each firm's prices $(c_k, \omega_i)$, where the marginal labour cost $\omega_i = w_i + \xi_i$ includes the endogenous shadow value of the labour-supply constraint: whatever wage the firm posts, its cost-minimising task assignment has the threshold form (8) used in the main text.

**Costs and demands at given factor prices.** Throughout, the task productivities satisfy

$$\int_0^M \left(|\log \phi_k(\tau)| + |\log \phi_l(\tau)|\right) d\tau < \infty, \tag{A.3}$$

so the unit cost (A.5) below is finite.

**Lemma 2** (Costs, output, revenue, and labour demand at given factor prices)**.** *Fix factor prices* $(c_k, \omega)$ *and productivity* $\phi_h$*, and let* $\tilde{x} = \tilde{x}(\omega/c_k)$ *be the threshold of Lemma 1. Write* $\varrho \equiv (\mu - 1)/(\mu - \kappa_f) \in (0, 1)$ *and* $\beta \equiv \kappa_f \mu/(\mu - \kappa_f) > 0$ (local to this appendix)*.*

(i) Costs. *The minimised cost of producing output* $y$*, valuing labour at its marginal labour cost* $\omega$ *rather than at the posted wage, is*

$$TC(y;\, c_k, \omega, \phi_h) = \left(\frac{y}{\phi_h}\right)^{1/\kappa_f} q(c_k, \omega), \tag{A.4}$$

*where the unit cost of the task bundle is the geometric mean across tasks of the cheaper factor's cost per efficiency unit,*

$$\log q(c_k, \omega) \equiv \int_0^{\tilde{x}} \log \frac{c_k}{\phi_k(Mu)}\, du + \int_{\tilde{x}}^1 \log \frac{\omega}{\phi_l(Mu)}\, du. \tag{A.5}$$

*It has the Shephard properties*

$$\frac{\partial \log q}{\partial \log \omega} = 1 - \tilde{x} \in (0, 1), \qquad \frac{\partial \log q}{\partial \log c_k} = \tilde{x} \in (0, 1). \tag{A.6}$$

(ii) Output and revenue. *Given* $Y$*, the output that maximises* $Y^{1/\rho} y^{1/\mu} - TC(y)$ *is unique and log-linear in the primitives,*

$$\log y = \beta \left[\log \frac{\kappa_f}{\mu} + \frac{1}{\rho} \log Y + \frac{1}{\kappa_f} \log \phi_h - \log q(c_k, \omega)\right], \tag{A.7}$$

*with revenue* $\log r = \frac{1}{\rho}\log Y + \frac{1}{\mu}\log y$ *and elasticities*

$$\frac{\partial \log r}{\partial \log \phi_h} = \frac{1}{\mu - \kappa_f}, \quad \frac{\partial \log r}{\partial \log \omega} = -\frac{\kappa_f\,(1-\tilde{x})}{\mu - \kappa_f}, \quad \frac{\partial \log r}{\partial \log c_k} = -\frac{\kappa_f\,\tilde{x}}{\mu - \kappa_f}, \quad \frac{\partial \log r}{\partial \log Y} = \varrho. \tag{A.8}$$

(iii) Labour demand. *The labour demand* $l^d$*, defined by* $\omega\, l^d = (\kappa_f/\mu)(1-\tilde{x})\, r$ *as in* (16)*, satisfies*

$$\frac{\partial \log l^d}{\partial \log \phi_h} = \frac{1}{\mu - \kappa_f}, \quad \frac{\partial \log l^d}{\partial \log \omega} = -\mathcal{F}(\tilde{x}), \quad \frac{\partial \log l^d}{\partial \log c_k} = -g(\tilde{x}), \quad \frac{\partial \log l^d}{\partial \log Y} = \varrho, \tag{A.9}$$

*where*

$$\mathcal{F}(\tilde{x}) \;\equiv\; \underbrace{h(\tilde{x})}_{\text{task displacement}} \;+\; \underbrace{\frac{\kappa_f\,(1-\tilde{x})}{\mu - \kappa_f}}_{\text{scale}} \;+\; \underbrace{1}_{\text{factor intensity}} \;>\; 1, \tag{A.10}$$

$$\mathcal{F}(\tilde{x}) + g(\tilde{x}) \;=\; \frac{\mu}{\mu - \kappa_f} \;=\; \rho\,\varrho. \tag{A.11}$$

(iv) Equilibrium consistency. *Evaluated at* $(\omega_i, \phi_{ih})$ *the plan satisfies the threshold* (24)*, the factor demands* (16)*, and* $\mathrm{MRPL}_i = \omega_i$*; hence in equilibrium* $l_i = l^d(\omega_i;\phi_{ih})$*,* $r_i = r(\omega_i;\phi_{ih})$*,* $y_i = y(\omega_i;\phi_{ih})$*.*

(v) Global optimality. *Along the labour supply curve the firm perceives,* (1) *with rivals' wages,* $\Phi$*, and* $Y$ *held fixed per (A1), profit is strictly quasi-concave in the posted wage, so the first-order point is the unique global optimum.*

**Proof.** (i) Let $Q$ denote the bracketed task aggregate in (7), so that $y = \phi_h Q^{\kappa_f}$ and $Q = (y/\phi_h)^{1/\kappa_f}$. By Lemma 1(i), one unit of task output at $\tau$ costs the cheaper factor's cost per efficiency unit, $m(\tau)$, so the firm chooses $\{y(\tau)\}$ to solve

$$\min_{\{y(\tau)\}} \int_0^M m(\tau)\, y(\tau)\, d\tau \qquad \text{s.t.} \qquad \frac{1}{M}\int_0^M \log\big(M\, y(\tau)\big)\, d\tau = \log Q,$$
$$m(\tau) \equiv \min\Big\{\frac{c_k}{\phi_k(\tau)}, \frac{\omega}{\phi_l(\tau)}\Big\}.$$

The first-order condition $m(\tau)\, y(\tau) = \lambda/M$ equalises expenditure across tasks, with the multiplier $\lambda$ equal to total cost. Substituting $M y(\tau) = \lambda/m(\tau)$ into the constraint and solving for the multiplier,

$$\lambda = Q \exp\Big\{\int_0^1 \log m(Mu)\, du\Big\} = Q\, q(c_k, \omega),$$

with $q$ as in (A.5) because the minimum inside $m$ selects capital below the threshold and labour above it. This is (A.4). For (A.6), differentiate $\log q$: the direct price arguments contribute $(1-\tilde{x})\, d\log\omega + \tilde{x}\, d\log c_k$, and the movement of the limit $\tilde{x}$ contributes $[\log(c_k/\phi_k(M\tilde{x})) - \log(\omega/\phi_l(M\tilde{x}))]\, d\tilde{x} = 0$, because the two costs coincide at the threshold.

(ii) The objective $Y^{1/\rho} y^{1/\mu} - (y/\phi_h)^{1/\kappa_f} q(c_k, \omega)$ is strictly concave in $y$ (revenue strictly concave since $1/\mu < 1$; cost strictly convex since $1/\kappa_f > 1$). The first-order condition,

$$\frac{1}{\mu}\, Y^{1/\rho}\, y^{1/\mu - 1} = \frac{1}{\kappa_f}\, \phi_h^{-1/\kappa_f}\, y^{1/\kappa_f - 1}\, q(c_k, \omega),$$

collects to $y^{1/\kappa_f - 1/\mu} = (\kappa_f/\mu)\, Y^{1/\rho} \phi_h^{1/\kappa_f} q(c_k, \omega)^{-1}$ with $1/\kappa_f - 1/\mu = 1/\beta$, and taking logs gives (A.7). For the elasticities, $\log r = \frac{1}{\rho}\log Y + \frac{1}{\mu}\log y$, so each revenue elasticity is $1/\mu$ times the corresponding output elasticity plus the direct $\log Y$ term:

$$\frac{\partial \log r}{\partial \log \phi_h} = \frac{\beta}{\kappa_f \mu} = \frac{1}{\mu - \kappa_f}, \qquad \frac{\partial \log r}{\partial \log \omega} = -\frac{\beta}{\mu}\frac{\partial \log q}{\partial \log \omega} = -\frac{\kappa_f(1-\tilde{x})}{\mu - \kappa_f},$$
$$\frac{\partial \log r}{\partial \log c_k} = -\frac{\beta}{\mu}\frac{\partial \log q}{\partial \log c_k} = -\frac{\kappa_f \tilde{x}}{\mu - \kappa_f},$$

using both Shephard properties (A.6), and

$$\frac{\partial \log r}{\partial \log Y} = \frac{1}{\rho} + \frac{1}{\mu}\cdot\frac{\beta}{\rho} = \frac{1}{\rho}\left(1 + \frac{\kappa_f}{\mu - \kappa_f}\right) = \frac{\mu}{\rho\,(\mu - \kappa_f)} = \frac{\mu - 1}{\mu - \kappa_f} = \varrho,$$

the last equality because $\mu/\rho = 1/(\rho - 1) = \mu - 1$.

(iii) Taking logs of $\omega\, l^d = (\kappa_f/\mu)(1-\tilde{x})\, r$,

$$\log l^d = \log\frac{\kappa_f}{\mu} + \log(1-\tilde{x}) + \log r - \log\omega.$$

Differentiating with respect to $\log\phi_h$ and $\log Y$, only $r$ moves, giving $1/(\mu - \kappa_f)$ and $\varrho$. With

respect to $\log \omega$, using (A.2) for the threshold term and (A.8) for the revenue term,

$$\frac{\partial \log I^d}{\partial \log \omega} = -h(\tilde{x}) - \frac{\kappa_f (1-\tilde{x})}{\mu - \kappa_f} - 1 = -\mathcal{F}(\tilde{x}),$$

each term strictly negative. With respect to $\log c_k$, the threshold term is $+h(\tilde{x})$ by (A.2), so

$$\frac{\partial \log I^d}{\partial \log c_k} = h(\tilde{x}) - \frac{\kappa_f \, \tilde{x}}{\mu - \kappa_f} = -g(\tilde{x}),$$

which proves (39). The identity (A.11) follows because the displacement terms cancel: $\mathcal{F} + g = 1 + \kappa_f/(\mu - \kappa_f) = \mu/(\mu - \kappa_f) = \rho\varrho$.

(iv) The plan's threshold is that of Lemma 1, which is (24) at the firm's prices. At the optimal output, marginal cost equals marginal revenue $r/(\mu y)$, and since $TC$ is homogeneous of degree $1/\kappa_f$ in $y$, the minimised cost at these prices is $TC = \kappa_f \, y \, TC'(y) = (\kappa_f/\mu)\, r$; by part (i) expenditure is equal across tasks, so the capital bill is the share $\tilde{x}$ of it and the labour bill the share $1 - \tilde{x}$, which are (16). Equating marginal revenue to marginal cost delivers MRPL $= \omega$ at the chosen employment. Conversely the equilibrium allocation satisfies exactly these conditions at $(\omega_i, \phi_{ih})$; since the plan at given $(\omega, \phi_h)$ is unique (strict concavity in $y$; the unique threshold of Lemma 1), the two coincide.

(v) Parameterise the firm's strategy by its employment $I$: the perceived supply $I^s(w)$ is continuous and strictly increasing ($\varepsilon \geq \eta > 0$ everywhere), so $w^s(I)$ is well defined and strictly increasing. Write profit as $S(I) - C(I)$, where $S(I)$ is the maximised capital-net revenue at employment $I$ and $C(I) \equiv w^s(I)\, I$. By the envelope theorem, $S'(I) = \text{MRPL}(I)$; by part (iv) and strict monotonicity of $I^d$ in $\omega$ from (iii), $S'(I) = \omega^d(I)$, the inverse of $I^d(\cdot\,; \phi_h)$, strictly decreasing in $I$. On the cost side, $C'(I) = w^s(I)\big(1 + 1/\varepsilon(I)\big) = w^s(I)/\delta(I)$: along the supply curve the share, and hence $1/\delta$, is non-decreasing in $I$, and $w^s(I)$ is strictly increasing, so $C'(I)$ is strictly increasing. Therefore $S'(I) - C'(I)$ is strictly decreasing, so profit is strictly concave in $I$ and hence strictly quasi-concave in the posted wage, and the first-order point, MRPL $= w/\delta$, is the unique global maximiser. □

**The lemmas at the extensive margin.** With the positive intercept $R(0) > 0$ of Section 4.2, the lemmas carry over as follows.

- *Task assignment.* The argument of Lemma 1(i) goes through unchanged except at the bottom of the task space: capital is strictly cheaper on $[0, x)$ with $x = R^{-1}(\omega/c)$ when $\omega/c > R(0)$, and on no task when $\omega/c \leq R(0)$, in which case $x = 0$. At firm $i$'s prices $(c_{k,i}, \omega_i)$ this is the adoption rule (42).
- *The threshold.* The threshold is non-decreasing in $\omega/c$ and strictly increasing on the interior branch, where (A.2) applies as stated. Strictly below the cutoff, $\omega/c < R(0)$, it is locally constant, so $h$ and $g$ are replaced by zero there.
- *At the cutoff.* At $\omega/c = R(0)$ the threshold is zero but need not be locally constant, since any rise in $\omega/c$ moves it onto the interior branch; the proof of Proposition 5 never differentiates the threshold there.
- *Costs and demands.* In Lemma 2, the unit cost (A.4) at the corner is the all-labour cost, $\log q = \int_0^1 \log(\omega/\phi_l(Mu))\, du$. The Shephard properties (A.6) hold on both branches with $\tilde{x} = 0$ at the corner, since $1 - \tilde{x}$ is continuous across the cutoff, so the elasticities (A.9) and the identity (A.11) hold on both branches away from the cutoff. At the cutoff, $\tilde{x}$ and $q$, and hence revenue and labour demand, are continuous in $\omega$, so labour demand is strictly decreasing in $\omega$ across the cutoff as well.
- *Non-adopters.* A non-adopter has $k_i = 0$ and $\alpha_i = 0$; the reduced form (17) and the aggregation of Section 2.3 hold with the convention $0^0 = 1$.

## A.2 Proof of Proposition 1

**Regularity conditions.** Maintain the assumptions of the model and the integrability condition (A.3). When physical capital is present, also assume $\mu - 1 \geq \kappa_p/(1 + e)$.

**Proof.** We first establish that each proposed aggregate output determines a unique allocation satisfying firm optimality and labour-market clearing. We then show that exactly one proposed output also clears the goods and physical-capital markets.

Fix $Y > 0$. When physical capital is present, assign the candidate price

$$p_p(Y) = \left(\frac{\kappa_p}{\mu}\, Y\right)^{1/(1+e)}. \tag{A.12}$$

This price will clear its market if goods-market clearing holds. Automation capital is available at its competitive price $c_k$.

Optimising inputs and automation gives

$$r_i = Y^{\varrho_p}\, \Xi_i(\omega_i), \qquad I_i^d = Y^{\varrho_p}\, \Xi_i^I(\omega_i), \tag{A.13}$$

where

$$\varrho_p = \begin{cases} \dfrac{\mu - 1}{\mu - \kappa_f}, & \kappa_p = 0, \\ \dfrac{\mu - 1 - \kappa_p/(1+e)}{\mu - \kappa}, & \kappa_p > 0, \end{cases}$$

so that $\varrho_p$ equals $\varrho$ of Lemma 2 without physical capital, and $\Xi_i(\omega_i)$ is then the term $\Xi_i$ of (29). The assumptions imply $0 \le \varrho_p < 1$. The cost-and-demand characterisation of Lemma 2, with firm-specific installation costs and the physical-capital input included, gives positive, continuous, strictly decreasing functions $\Xi_i, \Xi_i^I$, with

$$\Xi_i^I(0+) = \infty, \qquad \Xi_i^I(\infty) = 0. \tag{A.14}$$

These properties hold with non-adoption (Appendix A.1).

*Conditional labour-market allocation.* Write $s_i \equiv s_{i|j}$. Worker sorting (1) and wage setting imply

$$w_i = \Phi_j^{1/\varsigma}\, (s_i/T_i)^{1/\varsigma}, \qquad \omega_i = \Phi_j^{1/\varsigma}\, f_i(s_i), \qquad f_i(s) \equiv \frac{(s/T_i)^{1/\varsigma}}{\delta(s)}. \tag{A.15}$$

Since $\delta$ is positive and decreasing, $f_i$ is continuous and strictly increasing.

First fix a market wage index $\Phi_j > 0$. Let $\Theta_j(\Phi_j)$ denote market employment divided by $Y^{\varrho_p}$. Firm labour-market clearing requires

$$\Xi_i^I\big(\Phi_j^{1/\varsigma} f_i(s_i)\big) = \Theta_j(\Phi_j)\, s_i, \qquad \sum_{i \in j} s_i = 1. \tag{A.16}$$

For each firm, $\Xi_i^I(\Phi_j^{1/\varsigma} f_i(s))/s$ decreases continuously from infinity as $s$ increases from zero. For $\Theta \geq \max_{i \in j} \Xi_i^I(\Phi_j^{1/\varsigma} f_i(1))$, the individual equations determine unique shares. Their sum is at least one at the lower bound and decreases continuously to zero as $\Theta \to \infty$. Thus (A.16) determines a unique $\Theta_j(\Phi_j)$ and share vector.

Increasing $\Phi_j$ reduces every share at a fixed $\Theta$, so $\Theta_j(\Phi_j)$ strictly decreases. Continuity, (A.14) and the share normalisation imply

$$\Theta_j(0+) = \infty, \qquad \Theta_j(\infty) = 0. \tag{A.17}$$

We now determine the market wage indices. By (1), market employment must equal both firms' demand and workers' supply, $Y^{\varrho_p}\Theta_j(\Phi_j) = L\,\Phi_j^{\eta/\varsigma}/\sum_h \Phi_h^{\eta/\varsigma}$. Equivalently,

$$\Theta_j(\Phi_j) = \mathfrak{q}\,\Phi_j^{\eta/\varsigma}, \qquad Y^{\varrho_p} \sum_j \Theta_j(\Phi_j) = L \tag{A.18}$$

for a common $\mathfrak{q} > 0$. Since $\Theta_j(\Phi_j)/\Phi_j^{\eta/\varsigma}$ decreases continuously from infinity to zero, the first equation uniquely determines $\Phi_j(\mathfrak{q})$, decreasing in $\mathfrak{q}$. Hence $\sum_j \Theta_j(\Phi_j(\mathfrak{q}))$ increases continuously from zero to infinity, and the second equation uniquely determines $\mathfrak{q}$.

This establishes a unique, continuous labour-market allocation for every $Y > 0$. The associated wage conditions give global firm optima by the argument of Lemma 2(v): capital-net revenue is strictly concave in employment and wage expenditure is strictly convex.

*Aggregate consistency.* An increase in $Y$ weakly lowers $\mathfrak{q}$ and weakly raises every $\Phi_j$. Every $\omega_i$ also weakly rises. Otherwise, (A.15) would imply a lower share $s_i$, making the right side of (A.16) fall while its left side weakly rises, a contradiction. Thus $\omega_i(Y)$ is nondecreasing in $Y$.

Define normalised excess revenue by

$$\mathcal{Z}(Y) \equiv \frac{\sum_i r_i(Y) - Y}{Y} = Y^{\varrho_p - 1} \sum_i \Xi_i\big(\omega_i(Y)\big) - 1. \tag{A.19}$$

Because $\varrho_p < 1$ and each $\Xi_i(\omega_i(Y))$ is positive and nonincreasing, $\mathcal{Z}$ is continuous and strictly

decreasing. Bounding the sum by its value at $Y = 1$, on either side of one, gives

$$\lim_{Y\to 0} \mathcal{Z}(Y) = \infty, \qquad \lim_{Y\to\infty} \mathcal{Z}(Y) = -1.$$

The intermediate value theorem therefore gives exactly one $Y^* > 0$ satisfying $\mathcal{Z}(Y^*) = 0$.

Goods-market clearing holds at $Y^*$. Physical-capital clearing follows because, along the candidate prices (A.12),

$$\frac{K^{p,d}(Y)}{K^{p,s}(Y)} = \frac{(\kappa_p/\mu)\sum_i r_i(Y)}{p_p(Y)^{1+e}} = 1 + \mathcal{Z}(Y).$$

Automation supply meets firms' demand at its fixed competitive price. All remaining quantities follow uniquely. Every equilibrium must satisfy the equations above, so the constructed equilibrium exists and is unique. □

## A.3 Proof of Proposition 2

**Proof.** Both firms are in the one market with the same amenity $T$, so by (1) the posted wage is $w_i = (s_i\Phi_j/T)^{1/\varsigma}$, which rises with the within-market share. By (23) and (A.1), the markdown $\delta_i = \delta(s_i)$ falls with it.

**Step 1: the share ranking.** Suppose $s_H \le s_L$. Then $w_H \le w_L$ and $\delta_H \ge \delta_L$, so $\omega_H = w_H/\delta_H \le \omega_L$. By Lemma 2(iii)–(iv), labour demand at the common $Y$ is strictly increasing in $\phi_h$ and strictly decreasing in $\omega$, so $l_H = l^d(\omega_H;\phi_{Hh}) > l^d(\omega_L;\phi_{Lh}) = l_L$. But $l_i = s_i L$ in the single market, so $s_H \le s_L$ gives $l_H \le l_L$: a contradiction. Hence $s_H > s_L$ and $l_H > l_L$.

**Step 2: labour market power.** With $s_H > s_L$, $w_H > w_L$ and $\delta_H < \delta_L$, hence $\omega_H > \omega_L$. This proves part (ii).

**Step 3: automation and revenue.** Both firms buy automation capital at the price $c_k$ and $\omega_H > \omega_L$, so Lemma 1(ii) gives $\tilde{x}_H > \tilde{x}_L$. Finally, by (16),

$$r_i = \frac{\omega_i l_i}{(\kappa_f/\mu)(1-\tilde{x}_i)}.$$

The numerator is strictly larger for firm $H$ and the denominator is smaller, so $r_H > r_L$. This proves parts (i) and (iii).

**Step 4: profit and revenue.** By (27), $\tilde{\pi}_i = m_i r_i$ with margin $m_i \equiv (1-\kappa_f/\mu)+(1-\delta_i)(\kappa_f/\mu)(1-\tilde{x}_i)$, local to this proof. Then $\tilde{\pi}_H/(\tilde{\pi}_H+\tilde{\pi}_L) > r_H/(r_H+r_L)$ if and only if $m_H > (m_H r_H + m_L r_L)/(r_H+r_L)$, that is, $m_H > m_L$, which is (34) after cancelling the common terms. This proves part (iv). □

## A.4 Comparative statics

**Maintained assumption.**

(A2) In the analytical economy, the equilibria along the family (36) exist for all sufficiently small $t \geq 0$ and are right-differentiable in $t$ at $t = 0$, and the oligopsonistic equilibrium is differentiable in $c_k$.

**The total differentials.** In equilibrium $l_i = l^d(\omega_i, c_k, Y; \phi_{ih})$ by Lemma 2(iv). Along any path on which $\phi_{ih}$ is fixed, write $z \equiv -d\log c_k$ for the fall in the automation price and $\mathcal{F}_i \equiv \mathcal{F}(\tilde{x}_i)$. Totally differentiating with the four elasticities (A.9),

$$d\log l_i = g_i\, z \;-\; \mathcal{F}_i\, d\log \omega_i \;+\; \varrho\, d\log Y: \tag{A.20}$$

$g_i z$ is the direct labour-demand shift from cheaper automation at a fixed marginal labour cost, $\mathcal{F}_i$ the absolute elasticity of labour demand with respect to that cost, and $\varrho\, d\log Y$ the common demand-expansion term. The firm's automation depends on the relative factor cost $\omega_i/c_k$ (Lemma 1), so write $a_i \equiv d\log\omega_i + z$ for its change. Using (A.11), (A.20) becomes

$$d\log l_i = \rho\varrho\, z \;-\; \mathcal{F}_i\, a_i \;+\; \varrho\, d\log Y, \tag{A.21}$$

in which only the middle term differs across firms. Likewise, by (A.8),

$$d\log r_i = \varrho\, d\log Y + \frac{\kappa_f}{\mu-\kappa_f}\big[z - (1-\tilde{x}_i)\, a_i\big], \tag{A.22}$$

in which the Shephard properties (A.6) do the envelope work of eliminating the induced movement of the threshold.

**Two equilibrium identities.** Propositions 3 and 4 are comparative statics of the analytical economy along a smooth path of equilibria ((A2)): in the wedge parameter $t$ at $t = 0$, or in the automation price $c_k$. All differentials are evaluated at the initial equilibrium, with $\phi_{ih}$ and $T$ fixed. With a single market, $\Phi = \Gamma^{\eta}\Phi_j^{\eta/\varsigma}$, so $s_j = 1$ identically: $l_i = s_i L$, $d\log l_i = d\log s_i$, and $ds_H + ds_L = 0$. Write $ds \equiv ds_H = -ds_L$ and $\varpi_i \equiv r_i/Y$ for the revenue weights, and

$$\Delta_Y \;\equiv\; g_H - g_L \;=\; \mathcal{F}_L - \mathcal{F}_H, \tag{A.23}$$

by (A.11). Incidence is concentrating exactly when $\Delta_Y > 0$ and equalising exactly when $\Delta_Y < 0$.

**Lemma 3** (Two equilibrium identities)**.** *Along either path,*

$$\frac{ds}{s_H s_L} \;=\; \mathcal{F}_L\, a_L - \mathcal{F}_H\, a_H, \tag{A.24}$$

$$A_H\, a_H + A_L\, a_L \;=\; \varrho\Big(\rho + \frac{\kappa_f}{1-\kappa_f}\Big) z, \qquad A_i \equiv s_i\,\mathcal{F}_i + \frac{\kappa_f\,\varrho}{1-\kappa_f}\,\varpi_i\,(1-\tilde{x}_i) \;>\; 0. \tag{A.25}$$

The first identity says that workers move towards the firm whose marginal labour cost rises by less relative to the price of automation capital, each firm's change weighted by its labour-demand elasticity. The second combines the two market-clearing conditions: a fall in the automation price raises the marginal labour cost relative to the price of automation capital on average across firms, with weights that add each firm's employment share to its share of labour cost.

**Proof.** For (A.24), subtract (A.21) for $L$ from that for $H$: the common terms cancel, and $d\log s_H - d\log s_L = ds/s_H + ds/s_L = ds/(s_H s_L)$.

For (A.25), labour clearing gives $s_H\, d\log s_H + s_L\, d\log s_L = ds_H + ds_L = 0$. Weighting (A.21) by $s_i$ and summing,

$$\varrho\; d\log Y = \sum_i s_i\,\mathcal{F}_i\, a_i - \rho\varrho\; z.$$

Goods clearing, Definition 1(iii), gives $d\log Y = \sum_i \varpi_i\, d\log r_i$; substituting (A.22), using $\sum_i \varpi_i = 1$

and $1-\varrho = (1-\kappa_f)/(\mu-\kappa_f)$,

$$d\log Y = \frac{\kappa_f}{1-\kappa_f}\Big[z - \sum_i \varpi_i\,(1-\tilde{x}_i)\,a_i\Big].$$

Multiplying by $\varrho$ and equating the two expressions for $\varrho\, d\log Y$ gives (A.25). Each $A_i$ is a sum of products of strictly positive terms. □

## A.5 Proof of Proposition 3

The family (36) nests the two equilibria. At every $t$, the firm's internal plan is given by Lemma 2 at $(\omega_i, \phi_{ih})$, employment satisfies $l^d(\omega_i;\phi_{ih}) = s_i L$, workers sort on posted wages by (1), and the goods market clears, Definition 1(iii); only the link between the posted wage and the marginal labour cost changes with $t$. At $t=1$ the rule is the wage-posting optimum (20)–(22), so the family ends at the oligopsonistic equilibrium. All differentials below are evaluated at $t=0$, with the automation price fixed, so $z=0$ and $a_i = d\log\omega_i$.

**Proof. Step 1: the ranking at the benchmark.** At $t=0$, $\omega_i = w_i = (s_i\Phi_j/T)^{1/\varsigma}$. Suppose $s_H \le s_L$. Then $\omega_H \le \omega_L$, and Lemma 2(iii)–(iv) gives $l_H > l_L$, contradicting $l_i = s_i L$. Hence $s_H > s_L$, and since $\delta(\cdot)$ is strictly decreasing,

$$\Delta\mathfrak{d} \;\equiv\; \delta(s_L) - \delta(s_H) \;>\; 0.$$

**Step 2: marginal labour costs and shares.** From (36), $\log\omega_i = \log w_i - \log\{1 - t\,[1-\delta(s_i)]\}$. At $t=0$ the differential of the last term is $[1-\delta(s_i)]\,dt$; the term in $ds_i$ carries the factor $t$ and vanishes. Sorting gives $d\log w_i = d\nu + \frac{1}{\varsigma}\,d\log s_i$ with $d\nu \equiv \frac{1}{\varsigma}\,d\log\Phi_j$ common to both firms. Subtracting,

$$a_H - a_L \;=\; \frac{ds}{\varsigma\, s_H s_L} + \Delta\mathfrak{d}\,dt. \tag{A.26}$$

Now suppose $a_H \le 0$. With $z=0$, (A.25) reads $A_H a_H + A_L a_L = 0$, so $a_L \ge 0$, and (A.24) then gives $ds \ge 0$. The left side of (A.26) is then nonpositive while the right side is strictly positive: a contradiction. Hence $a_H > 0$, so $a_L < 0$ by (A.25), and $ds < 0$ by (A.24). This proves part (ii).

**Step 3: wages and thresholds.** The wage premium moves with the shares, $d\log w_H - d\log w_L = ds/(\varsigma s_H s_L) < 0$, which with $ds < 0$ proves part (i). By (A.2) at fixed $c_k$, $d\tilde{x}_i = (1-\tilde{x}_i)\,h_i\,a_i$, so $d\tilde{x}_H > 0 > d\tilde{x}_L$ and the gap in automated shares widens, part (iii). The second form of (A.2) gives $d\log(1-\tilde{x}_i) = -h_i\,a_i$; subtracting the two firms and using $a_H > 0 > a_L$ gives (37), part (iv). □

## A.6 Proof of Proposition 4

Here the path is in $c_k$ with the wage-posting rule (22), $\omega_i = w_i/\delta(s_i)$, and $z > 0$.

**Proof. Step 1: both relative factor costs rise.** Sorting gives $d\log w_i = dv + \frac{1}{\varsigma}\,d\log s_i$, and the wage rule adds the markdown feedback, $d\log\omega_i = d\log w_i - [\delta'(s_i)/\delta_i]\,ds_i$. With $ds_L = -ds$,

$$a_H - a_L \;=\; C\,ds, \qquad C \;\equiv\; \frac{1}{\varsigma\, s_H s_L} - \frac{\delta'(s_H)}{\delta_H} - \frac{\delta'(s_L)}{\delta_L} \;>\; 0, \tag{A.27}$$

by (A.1). Substituting $ds = (a_H - a_L)/C$ into (A.24) and collecting,

$$\left(1 + C s_H s_L\,\mathcal{F}_H\right) a_H \;=\; \left(1 + C s_H s_L\,\mathcal{F}_L\right) a_L.$$

Both coefficients are positive, so $a_H$ and $a_L$ have the same sign. If both were nonpositive, the left side of (A.25) would be nonpositive while the right side is strictly positive for $z > 0$. Hence $a_H > 0$ and $a_L > 0$, and by (A.2) $d\tilde{x}_i = (1-\tilde{x}_i)\,h_i\,a_i > 0$ at both firms. This proves part (i).

**Step 2: reallocation.** Substituting $a_H = a_L + C\,ds$ into (A.24),

$$\left(\frac{1}{s_H s_L} + C\,\mathcal{F}_H\right) ds \;=\; (\mathcal{F}_L - \mathcal{F}_H)\,a_L \;=\; \Delta_\Upsilon\,a_L.$$

The coefficient on the left and $a_L$ are positive, so $\operatorname{sign}(ds) = \operatorname{sign}(\Delta_\Upsilon)$, and $ds = 0$ exactly when $\Delta_\Upsilon = 0$. The concentrating and equalising cases follow from the definitions after (A.23); one firm gains only if the other loses, since $ds_L = -ds$. This proves part (ii).

**Step 3: markdowns.** Since $\delta_i = \delta(s_i)$ and $ds_L = -ds$,

$$d\delta_H = \delta'(s_H)\,ds, \qquad d\delta_L = -\,\delta'(s_L)\,ds, \qquad d(\delta_L - \delta_H) = \left[|\delta'(s_L)| + |\delta'(s_H)|\right] ds,$$

with $\delta' < 0$ by (A.1). Thus $ds > 0$ lowers $\delta_H$, raises $\delta_L$, and widens the gap; all three movements reverse when $ds < 0$. Part (ii) supplies the incidence statements. This proves part (iii). □

## A.7 Proof of Proposition 5

**The adoption ordering.** The statement in Section 4.1 that firms with identical installation requirements retain the ranking of Proposition 2 follows from the adoption rule. Take two firms $i$ and $i'$ in the same market with the same amenity and $\delta_i < \delta_{i'}$. Since $\delta(\cdot)$ is strictly decreasing, $s_{i|j} > s_{i'|j}$. The posted wage rises with the share, $w_i = (s_{i|j}\Phi_j/T)^{1/\varsigma}$, and the markdown falls with it, so $\omega_i = w_i/\delta_i > \omega_{i'}$. If both firms draw the same requirement $b$ and firm $i'$ adopts, then $\omega_i > \omega_{i'} > c_k b R(0)$, so firm $i$ adopts, and by the interior branch of (42) $\tilde{x}_i > \tilde{x}_{i'}$.

**Setup.** Work in the analytical economy with $\phi_{Hh} > \phi_{Lh}$, a common amenity $T$, fixed total employment, no physical capital, a common installation requirement $b$, and the increasing task schedule of Assumption 1 with the intercept $R(0) > 0$. Wage-setting power is introduced along the family (36), with $c_k$, $b$, productivities, the amenity and the task schedule fixed. Assumption (A2) is used here only in a weaker form: equilibria exist for all sufficiently small $t \geq 0$ and converge to the equilibrium without labour markdowns as $t \to 0$.

**Proof. Step 1: neither firm adopts at $t = 0$.** Define the common adoption threshold $\bar{\omega} \equiv c_k b R(0)$; by hypothesis $\omega_H(0) = \bar{\omega}$. At $t = 0$, $\omega_i = w_i = (s_i\Phi_j/T)^{1/\varsigma}$. Suppose $s_H \leq s_L$. Then $\omega_H \leq \omega_L$, and Lemma 2(iii)–(iv), which hold at the corner by Appendix A.1, give $l_H > l_L$, contradicting $l_i = s_i L$. Hence $s_H(0) > s_L(0)$ and $\omega_L(0) < \omega_H(0) = \bar{\omega}$. The adoption rule (42) assigns no automated tasks at equality, so $d_H(0) = d_L(0) = 0$. Since $\delta(\cdot)$ is strictly decreasing,

$$\Delta\mathfrak{d} \;\equiv\; \delta(s_L(0)) - \delta(s_H(0)) \;>\; 0.$$

**Step 2: the economy without automation.** Consider the auxiliary economy in which both firms must use labour on every task, so $\tilde{x}_i \equiv 0$. It coincides with the actual economy at $t = 0$. Hats denote its allocations, $a_i \equiv d\log\widehat{\omega}_i/dt$ and $\dot{s} \equiv d\widehat{s}_H/dt$ at $t = 0$, and all weights and shares are evaluated at $t = 0$. With $\tilde{x}_i = 0$ and $h_i = g_i = 0$, (A.10) gives the same labour-demand elasticity

at both firms,

$$\bar{\mathcal{F}} \equiv \frac{\mu}{\mu - \kappa_f} > 0,$$

and the two identities of Lemma 3, whose derivation uses only labour demand, revenue and the two clearing conditions, hold with $z = 0$:

$$\frac{\dot{s}}{s_H s_L} = \bar{\mathcal{F}}(a_L - a_H), \qquad A_H a_H + A_L a_L = 0, \qquad A_i = s_i \bar{\mathcal{F}} + \frac{\kappa_f \varrho}{1 - \kappa_f} \varpi_i > 0. \tag{A.28}$$

**Step 3: wage-setting power raises $H$'s marginal labour cost in the auxiliary economy.** As in Step 2 of the proof of Proposition 3, the wage rule and sorting give (A.26), $a_H - a_L = \dot{s}/(\varsigma s_H s_L) + \Delta\mathfrak{d}$; the change in $\delta(s_i)$ inside (36) is multiplied by $t$ and has no first-order effect at $t = 0$. Substituting the first identity in (A.28),

$$\left(1 + \frac{\bar{\mathcal{F}}}{\varsigma}\right)(a_H - a_L) = \Delta\mathfrak{d} > 0,$$

and with the second identity,

$$a_H = \frac{A_L}{A_H + A_L} \frac{\Delta\mathfrak{d}}{1 + \bar{\mathcal{F}}/\varsigma} > 0, \qquad a_L = -\frac{A_H}{A_H + A_L} \frac{\Delta\mathfrak{d}}{1 + \bar{\mathcal{F}}/\varsigma} < 0. \tag{A.29}$$

The same equations give local uniqueness and smoothness of the auxiliary equilibrium. Set the forcing term $\Delta\mathfrak{d}$ to zero: the first identity and (A.26) then give $a_H = a_L$, the second gives $a_H = a_L = 0$, the first gives $\dot{s} = 0$, and goods-market clearing gives $d \log \widehat{Y} = 0$. The Jacobian of the auxiliary equilibrium system is therefore nonsingular. Its equations are smooth, since no threshold moves, so the implicit function theorem gives a unique differentiable auxiliary equilibrium near $t = 0$, which justifies (A.29). Hence, for every sufficiently small $t > 0$,

$$\widehat{\omega}_H(t) > \widehat{\omega}_H(0) = \bar{\omega}. \tag{A.30}$$

**Step 4: continued non-adoption is not an equilibrium.** In the actual economy, firm $L$ starts strictly below the threshold, so by continuity $\omega_L(t) < \bar{\omega}$ and $d_L(t) = 0$ for all sufficiently small $t > 0$. Suppose firm $H$ also did not adopt at such a $t$. The actual equilibrium would then

solve the equations of the auxiliary economy and, by their local uniqueness, coincide with the auxiliary equilibrium. By (A.30), $\omega_H(t)/(c_k b) > R(0)$. Since the task schedule is continuous and strictly increasing, automation capital is then strictly cheaper than labour on a positive interval of tasks, and (42) requires $\tilde{x}_H(t) > 0$, a contradiction. Hence

$$d_H(0) = d_L(0) = 0, \qquad d_H(t) = 1, \quad d_L(t) = 0 \quad \text{for all sufficiently small } t > 0. \qquad \square$$

## A.8 Proof of Proposition 6

**Maintained identification assumptions.** The distributional assumptions and normalisations of Section 5.3 hold, and the externally fixed parameters of Section 5.4 are known. Aggregate prices and labour-market indices are held fixed. The number of firms in each market is taken as given, with a positive share of single-firm markets. Parameter values are interior, with $\sigma_\phi > 0$, $\upsilon > 0$ and $|\mathcal{R}_{\phi T}| < 1$, and wages, employment, revenue and market membership are observed without error.

**Proof.** Write $u_i = \tilde{x}_i$ and $\tau_i = \log T_i$. Expectations are across firms, and aggregate equilibrium objects are finite, positive and constant across observations.

**Step 1: recover amenities and automation intensity.** Labour supply (1), the markdown schedule (23) and labour demand (S1.3) imply

$$\tau_i = \log s_{i|j} + \frac{\varsigma}{\eta} \log s_j - \varsigma \log w_i + K_T,$$
$$\omega_i = \frac{w_i}{\delta(s_{i|j})}, \qquad u_i = 1 - \frac{\mu}{\kappa_f} \frac{\omega_i l_i}{r_i}.$$

The normalisation $\mathbb{E}[\tau_i] = 0$ fixes the common constant $K_T$. The Pareto restriction then identifies

$$\iota = \mathrm{Var}(\tau_i)^{-1/2}, \qquad z_i = z(\tau_i) = F_N^{-1}\big(F_T(e^{\tau_i})\big),$$

where $F_T$ and $F_N$ are the amenity and standard normal distribution functions. Thus $(\tau_i, z_i, \omega_i, u_i)$ are recoverable from observables.

**Step 2: identify the task schedule.** Restrict attention to single-firm markets; exogeneity of

market sizes preserves the primitive distributions. Define the known constants

$$A = \mu - \kappa > 0, \qquad D = \mu - \kappa_p + \eta A, \qquad C = \frac{\eta A}{\varsigma} > 0.$$

In these markets, labour supply gives $\log l_i = \eta \log \omega_i + (\eta/\varsigma)\tau_i + K_l$. Combining this with production, factor demands and the automation condition (42) yields $H(u_i) = Z_i$ for adopters, where

$$H(u) = -\big[A + (D - \kappa_f)a\big] \log(1-u) + (\kappa_f a + D\beta_R)\, u - \frac{\kappa_f \beta_R}{2} u^2,$$
$$Z_i = \log \phi_{ih} - D \log b_i - C\tau_i - D \log c_k - K_0.$$

Here $K_l$ and $K_0$ are common constants. For non-adopters, $Z_i = D \log[\omega_i/(c_k b_i)] \le 0$. Moreover, $H(0) = 0$, $H(u) \to \infty$ as $u \to 1$, and

$$H'(u) = (D - \kappa_f u)\left(\beta_R + \frac{a}{1-u}\right) + \frac{A}{1-u} > 0,$$

because $D - \kappa_f = (1+\eta)A > 0$ and $\beta_R > -a$. Hence $u_i = 0$ when $Z_i \le 0$, and $u_i = H^{-1}(Z_i)$ otherwise.

The Gaussian copula (50) and independent lognormal installation costs imply

$$\log \phi_{ih} = \mu_\phi + \lambda z_i + \epsilon_{\phi i}, \qquad \log b_i = -\frac{\upsilon^2}{2} + \epsilon_{bi}, \qquad \lambda = \sigma_\phi \mathcal{R}_{\phi T},$$

where the residuals are independent centred normals, independent of amenities, with variances $\sigma_\phi^2(1 - \mathcal{R}_{\phi T}^2)$ and $\upsilon^2$. Consequently,

$$Z_i = m + \lambda z_i - C\tau_i + \epsilon_{\phi i} - D\epsilon_{bi}, \qquad V \equiv \mathrm{Var}(Z_i \mid \tau_i) = \sigma_\phi^2(1 - \mathcal{R}_{\phi T}^2) + D^2\upsilon^2,$$

for a common intercept $m$. The observable conditional distribution function therefore satisfies

$$F_N^{-1}\big[\Pr(u_i \le u \mid \tau_i = \tau)\big] = \frac{H(u) - m - \lambda z(\tau) + C\tau}{\sqrt{V}}, \qquad 0 \le u < 1. \tag{A.31}$$

At $u = 0$, (A.31) identifies the coefficients on 1, $z(\tau)$ and $\tau$. These functions are linearly independent because $z(\tau) \to -\infty$ at the finite lower endpoint $\tau = -1/\iota$. Since $C > 0$ is known,

the coefficient $C/\sqrt{V}$ identifies $V$; the other coefficients identify $m$ and $\lambda$.

Differencing (A.31) between $u > 0$ and $u = 0$ then identifies $H(u)$. Linear independence of $\log(1-u)$, $u$ and $u^2$ identifies $\beta_R$ from its quadratic coefficient and $a$ from its logarithmic coefficient. This includes $\beta_R = 0$.

**Step 3: separate installation costs from productivity heterogeneity.** For adopters, the now-observable quantity

$$q_i = \log \omega_i - \beta_R u_i + a \log(1-u_i) = \log c_k - \frac{\upsilon^2}{2} + \epsilon_{bi}$$

satisfies, by normal conditioning,

$$\mathbb{E}[q_i \mid u_i = u, \tau_i = \tau] = \log c_k - \frac{\upsilon^2}{2} - \frac{D\upsilon^2}{V}\big[H(u) - m - \lambda z(\tau) + C\tau\big], \qquad u > 0.$$

This follows from $\mathrm{Cov}(\epsilon_{bi}, \epsilon_{\phi i} - D\epsilon_{bi}) = -D\upsilon^2$. At fixed amenities, automation intensity has positive density throughout $(0,1)$. Variation in $H(u)$ therefore identifies the slope $-D\upsilon^2/V$, hence $\upsilon^2$; the level then identifies $c_k$. Conditioning on automation intensity accounts for selection into adoption.

Finally,

$$\sigma_\phi^2 = V - D^2\upsilon^2 + \lambda^2, \qquad \mathcal{R}_{\phi T} = \frac{\lambda}{\sigma_\phi}.$$

All seven parameters are uniquely recovered, so observationally equivalent interior parameter vectors coincide. □

# B Solution algorithm

This appendix gives the algorithm that solves the quantitative model of Section 4. The quantitative model extends the baseline model with heterogeneous installation costs, an extensive margin of automation, and physical capital.

**The iteration.** The equilibrium is computed by iterating on a state vector of employment shares and automation thresholds. The firm-level draws $(\phi_{ih}, T_i, b_i)$, and hence the firm-specific

capital costs $c_{k,i}$, are fixed before the iteration begins. Let $\chi_i \equiv s_{i|j}\, s_j$ denote the economy-wide employment share of firm $i$ — the unconditional choice probability of Supplemental Material S1.1 — so that $l_i = \chi_i L$, and let the state vector be

$$\mathbf{s} \equiv \left(\{\chi_i\},\, \{\tilde{x}_i\}\right). \tag{B.1}$$

Because the state carries the employment shares directly, labour-market clearing holds at every iteration.

**Step 0 (Initialise).** Guess $\mathbf{s}^{(0)} = \left(\{\chi_i^{(0)}\},\, \{\tilde{x}_i^{(0)}\}\right)$ with $\chi_i^{(0)} \in (0,1)$, $\sum_{i\in\mathcal{I}} \chi_i^{(0)} = 1$, and $\tilde{x}_i^{(0)} \in [0,1)$ for all $i$.

**Step 1 (Firm-level assignments).** Set $l_i = \chi_i^{(n)} L$ and define the gross-revenue shares

$$\sigma_p \equiv \frac{\kappa_p}{\mu}, \qquad \alpha_i \equiv \frac{\kappa_f}{\mu}\tilde{x}_i^{(n)}. \tag{B.2}$$

Let $\lambda_i \equiv \alpha_i + \sigma_p$ be the combined share of the two capital inputs concentrated out below, and let $\nu_i \equiv (\kappa_f/\mu)(1-\tilde{x}_i^{(n)})/(1-\lambda_i)$ be the remaining labour exponent.

**Step 2 (Aggregate output and the price of physical capital).** In Steps 2–4 the iteration superscript is dropped: $\tilde{x}_i$ denotes the current iterate $\tilde{x}_i^{(n)}$. Solve the goods-market condition

$$Y = \sum_{i\in\mathcal{I}} \Omega_i^{Q}(\tilde{x}_i; p_p)\, Y^{1/[\rho(1-\lambda_i)]}\, l_i^{\nu_i}, \qquad p_p = \left(\sigma_p\, Y\right)^{1/(1+e)}, \tag{B.3}$$

where

$$\Omega_i^{Q}(\tilde{x}_i; p_p) \equiv \left[\tilde{\phi}_i(\tilde{x}_i)^{\kappa_f/\mu}\left(\frac{\alpha_i}{c_{k,i}}\right)^{\alpha_i}\left(\frac{\sigma_p}{p_p}\right)^{\sigma_p}\right]^{1/(1-\lambda_i)}. \tag{B.4}$$

The second equation in (B.3) is the capital-market clearing condition (45). Substituting it into the first leaves one equation in $Y$, solved by one-dimensional root-finding. For a non-adopter, $\alpha_i = \tilde{x}_i = 0$, and both $(\alpha_i/c_{k,i})^{\alpha_i}$ and $\tilde{\phi}_i(0)$ are evaluated with the convention $0^0 = 1$. The equation follows by substituting the automation-capital and physical-capital demands into gross revenue and then imposing $\sum_i r_i = Y$.

**Step 3 (Firm-level objects).** With $Y$ and $p_p$ in hand, compute

$$r_i = \Omega_i^Q(\tilde{x}_i; p_p)\, Y^{1/[\rho(1-\lambda_i)]}\, l_i^{\nu_i}, \qquad r_i^{\mathrm{N}} = \frac{\tilde{\mu}}{\mu} r_i, \qquad k_i = \frac{\alpha_i r_i}{c_{k,i}}, \qquad k_i^p = \frac{\kappa_p r_i}{\mu\, p_p}. \tag{B.5}$$

**Step 4 (Wages).** Recover within-market shares $s_{i|j} = \chi_i^{(n)} / \sum_{i' \in j} \chi_{i'}^{(n)}$ and the markdown $\delta_i = \delta(s_{i|j})$. The marginal revenue product of labour and the wage are

$$\mathrm{MRPL}_i = \frac{\kappa_f}{\mu}(1-\tilde{x}_i)\frac{r_i}{l_i} = \frac{\kappa_f}{\tilde{\mu}}(1-\tilde{x}_i)\frac{r_i^{\mathrm{N}}}{l_i}, \qquad w_i = \delta_i \mathrm{MRPL}_i.$$

Set $\omega_i = w_i/\delta_i = \mathrm{MRPL}_i$.

**Step 5 (Model-implied state).** Given the wages $\{w_i\}$, compute the model-implied state $\hat{\mathbf{s}} = \big(\{\hat{\chi}_i\}, \{\hat{\tilde{x}}_i\}\big)$.

*Employment shares.* Form the wage indices and the implied share from the Fréchet labour supply (1):

$$\Phi_j = \sum_{i' \in j} T_{i'}\, w_{i'}^{\varsigma}, \qquad \Phi = \sum_{j'} \Gamma^{\eta}\, \Phi_{j'}^{\eta/\varsigma}, \qquad \hat{\chi}_i = \frac{T_i\, w_i^{\varsigma}}{\Phi_j} \cdot \frac{\Gamma^{\eta}\, \Phi_j^{\eta/\varsigma}}{\Phi}. \tag{B.6}$$

*Automation thresholds.* Using $\omega_i$ and $c_{k,i}$, the threshold condition (24) gives the update

$$\hat{\tilde{x}}_i \text{ solves } R\big(M\hat{\tilde{x}}_i\big) = \frac{\omega_i}{c_{k,i}} \text{ if } \frac{\omega_i}{c_{k,i}} > R(0), \qquad \hat{\tilde{x}}_i = 0 \text{ otherwise.} \tag{B.7}$$

Under the power–exponential schedule of Section 5.3, firms with $\omega_i \le c_{k,i}$ are placed at the corner. For adopters, strict monotonicity of $R$ guarantees a unique root in (B.7), which is computed by one-dimensional root-finding.

**Step 6 (Dampen and iterate).** Update with relaxation weight $\nu \in (0, 1]$:

$$\mathbf{s}^{(n+1)} = \nu\, \hat{\mathbf{s}} + (1-\nu)\, \mathbf{s}^{(n)}. \tag{B.8}$$

If $\|\hat{\mathbf{s}} - \mathbf{s}^{(n)}\|_\infty < 10^{-10}$, stop; otherwise return to Step 1. At convergence, compute firm profits as revenue net of the automation, labour, and physical-capital bills, the physical-capital stock $K^p = \sum_i k_i^p$, the upstream resource cost and profit from (44), owner consumption from (S1.6), and aggregate consumption from (S1.5).

# SUPPLEMENTAL MATERIAL – NOT FOR PUBLICATION

# LABOUR MARKET POWER, AUTOMATION AND WELFARE

Aseem Patel
*University of Essex*

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This Supplemental Material is organised as follows. Section S1 derives the labour supply curve, workers' welfare and social welfare, and the conditions of the model with physical capital. Section S2 sets out the normalisations. Section S3 describes the data, the measurement of automation capital and the sample, and estimates markdowns from production functions. Section S4 defines the targeted moments, checks local identification at the estimates and explains how the fall in the automation price is computed. Section S5 reports the sensitivity analysis of Section 6.3. Equations, propositions, sections, tables and figures without a prefix refer to the main paper; those with a letter prefix refer to the Online Appendix.

# S1 Derivations

## S1.1 Section 2: Model

**Choice probabilities and the labour supply curve.** This derivation follows Appendix A.1 of Azkarate-Askasua and Zerecero (2025), restated in the notation of Section 2.1. Worker $\ell$'s indirect utility from firm $i$ in market $j$ is $u_{\ell ij} = w_i\, z_{\ell i}\, v_{\ell j}$, with the shocks drawn independently from Fréchet distributions,

$$\mathbb{P}(z_{\ell i} \le z) = e^{-T_i\, z^{-\varsigma}}, \qquad \mathbb{P}(v_{\ell j} \le v) = e^{-v^{-\eta}},$$

independent across workers, firms, and markets, and of each other. The worker first observes the market shocks $\{v_{\ell j}\}$ and chooses a market; she then observes the firm shocks within the chosen market and chooses a firm. The unconditional probability that she works at firm $i$ of market $j$ is therefore

$$\chi_i = \underbrace{\mathbb{P}\Big(w_i z_{\ell i} \ge \max_{i' \ne i} w_{i'} z_{\ell i'}\Big)}_{\text{A. firm choice within } j} \times \underbrace{\mathbb{P}\Big[\mathbb{E}_j\big(\max_i w_i z_{\ell i}\big)\, v_{\ell j} \ge \max_{j' \ne j} \mathbb{E}_{j'}\big(\max_i w_i z_{\ell i}\big)\, v_{\ell j'}\Big]}_{\text{B. market choice}}.$$

**Step A: the firm choice.** The utility offer of firm $i$ has distribution

$$G_i(\vartheta) = \mathbb{P}\big[w_i z_{\ell i} \le \vartheta\big] = \mathbb{P}\Big[z_{\ell i} \le \frac{\vartheta}{w_i}\Big] = e^{-T_i w_i^{\varsigma} \vartheta^{-\varsigma}}, \qquad g_i(\vartheta) = T_i\, \varsigma\, w_i^{\varsigma}\, \vartheta^{-(\varsigma+1)}\, e^{-T_i w_i^{\varsigma} \vartheta^{-\varsigma}}.$$

By independence, the best rival offer has distribution

$$\mathbb{P}\Big(\max_{i' \ne i} w_{i'} z_{\ell i'} \le \vartheta\Big) = \prod_{i' \ne i} e^{-T_{i'} w_{i'}^{\varsigma} \vartheta^{-\varsigma}} = e^{-\Phi_j^{-i} \vartheta^{-\varsigma}}, \qquad \Phi_j^{-i} \equiv \sum_{i' \ne i} T_{i'} w_{i'}^{\varsigma},$$

and the best offer in the market, $G_j(\vartheta) = e^{-\Phi_j \vartheta^{-\varsigma}}$ with $\Phi_j = \sum_{i' \in j} T_{i'} w_{i'}^{\varsigma}$ and density $g_j(\vartheta) = \varsigma\, \Phi_j\, \vartheta^{-(\varsigma+1)} e^{-\Phi_j \vartheta^{-\varsigma}}$. Integrating over the own offer,

$$\mathbb{P}\Big(\max_{i' \ne i} w_{i'} z_{\ell i'} \le w_i z_{\ell i}\Big) = \int_0^{\infty} e^{-\Phi_j^{-i} \vartheta^{-\varsigma}}\, g_i(\vartheta)\, d\vartheta = \int_0^{\infty} T_i\, \varsigma\, w_i^{\varsigma}\, \vartheta^{-(\varsigma+1)}\, e^{-\Phi_j \vartheta^{-\varsigma}}\, d\vartheta$$

$$= \frac{T_i w_i^{\varsigma}}{\Phi_j} \int_0^{\infty} \varsigma\, \Phi_j\, \vartheta^{-(\varsigma+1)}\, e^{-\Phi_j \vartheta^{-\varsigma}}\, d\vartheta = \frac{T_i w_i^{\varsigma}}{\Phi_j} = s_{i|j},$$

the last integral being that of the density $g_j$, hence one.

**Step B: the market choice.** The expected best offer in market $j$ is

$$\mathbb{E}_j\big(\max_i w_i z_{\ell i}\big) = \int_0^{\infty} \vartheta\, g_j(\vartheta)\, d\vartheta = \int_0^{\infty} \varsigma\, \Phi_j\, \vartheta^{-\varsigma}\, e^{-\Phi_j \vartheta^{-\varsigma}}\, d\vartheta = \Phi_j^{1/\varsigma}\, \Gamma\Big(\frac{\varsigma-1}{\varsigma}\Big) = \Phi_j^{1/\varsigma}\, \Gamma,$$

by the change of variables $x = \Phi_j \vartheta^{-\varsigma}$, where $\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t}\, dt$ is the Gamma function and $\Gamma \equiv \Gamma(1 - 1/\varsigma)$, requiring $\varsigma > 1$. The worker therefore compares $\Phi_j^{1/\varsigma}\, \Gamma\, v_{\ell j}$ across markets; since $v_{\ell j}$ is Fréchet with shape $\eta$, the offer $\Phi_j^{1/\varsigma} \Gamma\, v_{\ell j}$ is Fréchet with scale $\Phi_j^{\eta/\varsigma} \Gamma^{\eta}$ and shape $\eta$, and the same integral as in Step A gives

$$\mathbb{P}\Big(\Phi_j^{1/\varsigma} \Gamma\, v_{\ell j} \geq \max_{j' \neq j} \Phi_{j'}^{1/\varsigma} \Gamma\, v_{\ell j'}\Big) = \frac{\Gamma^{\eta}\, \Phi_j^{\eta/\varsigma}}{\Phi} = s_j, \qquad \Phi \equiv \sum_{j'} \Gamma^{\eta}\, \Phi_{j'}^{\eta/\varsigma}.$$

Multiplying the two steps, $\chi_i = s_{i|j}\, s_j$, and with a mass $L$ of workers making i.i.d. choices, firm-level labour supply is $l_i^s(w_i) = s_{i|j}\, s_j\, L$, which is (1).

*The supply elasticity.* Holding rivals' wages and $\Phi$ fixed per (A1), while internalising the own-wage effect on $\Phi_j$,

$$\frac{d \log l_i^s}{d \log w_i} = \frac{d \log s_{i|j}}{d \log w_i} + \frac{d \log s_j}{d \log w_i} = \Big(\varsigma - \varsigma\, s_{i|j}\Big) + \frac{\eta}{\varsigma} \cdot \varsigma\, s_{i|j} = \varsigma\, (1 - s_{i|j}) + \eta\, s_{i|j},$$

using $d \log \Phi_j / d \log w_i = \varsigma\, s_{i|j}$, which is the elasticity (2).

**The workers' welfare index.** The representative-household formulation of Azkarate-Askasua and Zerecero (2025) supplies the foundation for the workers' welfare index. Consider a household that allocates its members across firms and markets at given positive wages. Let $\chi_i = l_i / L$, $s_j = \sum_{i \in j} \chi_i$, and $s_{i|j} = \chi_i / s_j$. With the amenities and wage indices of (1), define

$$H_j = -\sum_{i \in j} s_{i|j} \log \frac{s_{i|j}}{T_i}, \qquad H = -\sum_j s_j \log \frac{s_j}{\Gamma^{\eta}}.$$

These entropy terms value diversity in the allocation of employment, with amenities shifting the relative value of each job. The household solves

$$\max_{\{\chi_i \geq 0\}:\, \sum_i \chi_i = 1} \left\{ \sum_i \chi_i \log w_i + \frac{1}{\varsigma} \sum_j s_j H_j + \frac{1}{\eta} H \right\}, \tag{S1.1}$$

with the usual convention $0 \log 0 = 0$. Assume $\varsigma > 1$ and $0 < \eta < \varsigma$; the former makes $\Gamma = \Gamma(1 - 1/\varsigma)$ finite in the underlying two-step worker problem.

**Proposition S1.1.** *Problem* (S1.1) *has a unique solution, and this solution reproduces the labour supply* (1)*. Its maximised value is* $U^{\mathrm{RH}} = \eta^{-1} \log \Phi = \log \mathcal{W}$.

**Proof.** Fix a market share $s_j > 0$. The conditional shares maximise

$$\sum_{i \in j} s_{i|j} \log w_i - \frac{1}{\varsigma} \sum_{i \in j} s_{i|j} \log \frac{s_{i|j}}{T_i}, \qquad \sum_{i \in j} s_{i|j} = 1.$$

Strict concavity gives a unique interior optimum. The first-order conditions make $s_{i|j}$ proportional to $T_i w_i^{\varsigma}$, so adding up and substituting back gives

$$s_{i|j} = \frac{T_i w_i^{\varsigma}}{\Phi_j}, \qquad V_j = \frac{1}{\varsigma} \log \Phi_j, \qquad \Phi_j = \sum_{i \in j} T_i w_i^{\varsigma},$$

where $V_j$ is the optimised within-market value. The remaining problem is

$$\max_{\{s_j \geq 0\}:\, \sum_j s_j = 1} \left\{ \sum_j s_j V_j - \frac{1}{\eta} \sum_j s_j \log \frac{s_j}{\Gamma^{\eta}} \right\}.$$

Again the objective is strictly concave. Its first-order conditions make $s_j$ proportional to $\Gamma^{\eta} \exp(\eta V_j)$, yielding

$$s_j = \frac{\Gamma^{\eta} \Phi_j^{\eta/\varsigma}}{\Phi}, \qquad \Phi = \sum_j \Gamma^{\eta} \Phi_j^{\eta/\varsigma}.$$

Thus $I_i = L s_j s_{i|j}$ reproduces (1). At these shares, $\log(s_j/\Gamma^\eta) = \eta V_j - \log\Phi$, so substitution gives

$$U^{\mathrm{RH}} = \frac{1}{\eta}\log\Phi = \log\mathcal{W}. \tag{S1.2}$$

□

Note that this construction does not require $\eta > 1$. In the original two-step choice problem, the value of the chosen market before the firm-specific draw is realised is $\max_j\{\Gamma\Phi_j^{1/\varsigma} v_{\ell j}\}$. Its distribution function is $\exp(-\Phi u^{-\eta})$ for $u > 0$, so its quantile at probability $q \in (0,1)$ is $\mathcal{W}(-\log q)^{-1/\eta}$. All fixed quantiles, including the median, are proportional to $\mathcal{W}$. Its mean is finite only when $\eta > 1$; the representative-household value and the quantiles remain finite for every $\eta > 0$. Accordingly, $\mathcal{W}$ is a welfare index, not a claim that mean utility exists throughout the parameter space.

**The workers' consumption equivalent.** Scaling all wages by a common factor scales $\mathcal{W}$ and the wage bill $\mathcal{B}$ by that factor and leaves employment shares unchanged. Hence $\mathcal{N} = \mathcal{W}/\mathcal{B}$ is invariant to the wage level. Let $\mathcal{B}^{\mathrm{CE}}$ be the wage bill that, under a proportional scaling of the reference wage vector, gives the household its current utility. Since the reference employment index is $\bar{\mathcal{N}}$, (S1.2) requires

$$\log(\mathcal{B}^{\mathrm{CE}}\bar{\mathcal{N}}) = \log(\mathcal{B}\mathcal{N}), \qquad \mathcal{B}^{\mathrm{CE}} = \frac{\mathcal{B}\mathcal{N}}{\bar{\mathcal{N}}} = \frac{\sigma C\mathcal{N}}{\bar{\mathcal{N}}}.$$

This is the workers' term in (33). Multiplying household utility by the fixed worker mass $L$ leaves this conversion unchanged.

**The owners' consumption equivalent and social welfare.** Owners have linear utility, so their consumption-equivalent welfare is aggregate profit, $\mathcal{W}^f = \Pi = (1-\sigma)C$. Social welfare (33) adds the two consumption equivalents with equal weights per unit of reference consumption.

## S1.2 Section 4.3: Physical capital

Each firm spends a constant share of revenue on physical capital, and the task block receives the share $\kappa_f/\mu$ of gross revenue, as in the baseline:

$$p_p\, k_i^p \;=\; \frac{\kappa_p}{\mu}\, r_i, \qquad c_{k,i} k_i = \frac{\kappa_f}{\mu}\, \tilde{x}_i r_i, \qquad \omega_i l_i = \frac{\kappa_f}{\mu}\,(1-\tilde{x}_i) r_i. \tag{S1.3}$$

Automation changes the split of the task block between labour and automation capital, not the revenue share of physical capital. In terms of revenue net of the physical-capital bill, $r_i^{\mathrm{N}} \equiv r_i - p_p k_i^p = (\tilde{\mu}/\mu)\, r_i$ with

$$\tilde{\mu} \;\equiv\; \mu - \kappa_p, \tag{S1.4}$$

the firm-level conditions of the baseline carry over after replacing $(\mu, r_i)$ by $(\tilde{\mu}, r_i^{\mathrm{N}})$; $\tilde{\mu}$ is a curvature, not a markup, and the restriction $\kappa < \mu$ is equivalent to $\kappa_f < \tilde{\mu}$. The markdown schedule and the automation rule (42) are unchanged.

Summing the physical-capital shares over firms and using goods-market clearing, $p_p K^p = (\kappa_p/\mu)\, Y$, and the supply curve gives the price of physical capital as a function of aggregate output,

$$p_p \;=\; \left(\frac{\kappa_p}{\mu}\, Y\right)^{1/(1+e)},$$

which is (45) in the main text. Every solve imposes (45): the estimated economy, every candidate parameter vector in estimation, and every counterfactual allocation.

With $\mathcal{E} \equiv \sum_i c_{k,i} k_i$ the aggregate automation expenditure and $\Pi_p^U \equiv p_p K^p - G_p(K^p)$ the capital producer's profit, national income and consumption are

$$Y = \mathcal{B} + \mathcal{E} + p_p K^p + \Pi, \qquad C = \mathcal{B} + \Pi + \Pi_p^U = Y - \mathcal{E} - G_p. \tag{S1.5}$$

Owner $i$ consumes her operating profit and her dividend,

$$c_i^f \;\equiv\; \tilde{\pi}_i + \bar{s}_i^{\pi}\, \Pi_p^U, \tag{S1.6}$$

where the dividend share $\bar{s}_i^{\pi}$ is her profit share at the welfare reference allocation. The welfare

measure (33) keeps its form, with $\sigma = \mathcal{B}/C$, so owners' consumption-equivalent welfare is their income, $(1-\sigma)C = \Pi + \Pi_p^U$. Dividends enter no equilibrium condition.

## S2 Normalisations

The moments used in estimation describe differences across firms and patterns of automation adoption, but do not pin down the model's scale. The normalisations in Table 2 fix this scale without changing employment shares, markdowns, automation decisions or expenditure shares. Proposition S2.1 establishes these results.

**Proposition S2.1** (Normalisations). *Within the parametric family of Section 5.3, the following normalisations can be imposed jointly:*

(i) $M = 1$, $A_p = 1$, *and* $L$ *equal to sample employment, fixing task units, physical-capital units and the employment scale;*

(ii) $\mathbb{E}[\log T_i] = 0$, $\mathbb{E}[\phi_{ih}] = 1$, *and* $\mathbb{E}[b_i] = 1$, *implemented by* $x_{\min} = e^{-1/\iota}$, $\mu_\phi = -\sigma_\phi^2/2$, *and* $\mu_b = -\upsilon^2/2$;

(iii) *the common task-schedule scale* $\zeta = 1$ *and comparative-advantage intercept* $R(0) = 1$. *Allowing separate power exponents* $a_l$ *and* $a_k$ *in the task schedules does not enlarge the set of observable outcomes: the common exponent can be set to* $a_l = a_k \equiv a$.

*These normalisations leave the dispersions, the productivity–amenity dependence and the shape parameters* $(a, \beta_R)$ *unrestricted.*

**Proof.** Task units enter only through $\tau/M$ and $x_i/M$, so $M = 1$ is a relabelling. A common amenity multiplier cancels from both employment shares in (1) and from welfare ratios. Likewise, multiplying every $b_i$ by a constant and dividing $c_k$ by it leaves each installed-capital price $c_k b_i$ unchanged. The distributional means give the stated formulas for $x_{\min}$ and $\mu_b$; the copula leaves these marginals unchanged.

For the task schedules, write $u = \tau/M$ and allow

$$\phi_l(\tau) = \zeta_l u^{a_l} e^{\beta_R u}, \qquad \phi_k(\tau) = \zeta_k u^{a_l} (1-u)^{a_k}.$$

The common factor $u^{a_l}$ cancels from comparative advantage and contributes only $a_l \int_0^1 \log u \, du = -a_l$ to log composite productivity. Setting $a_l = a_k$ is therefore offset by a shift in $\mu_\phi$. Multiplying $\zeta_k$ and $c_k$ by the same constant, and dividing automation quantities by it, preserves effective inputs and expenditures; this permits $\zeta_k = \zeta_l$, hence $R(0) = 1$. The remaining common scale enters productivity only through $\mu_\phi/\kappa_f + \log \zeta$ and can be set to one.

To verify the remaining level conventions, restore the physical-capital supply scale $p_p = A_p (K^p)^{1/e}$. For any $h > 0$, changing units as

$$(k_i^p, p_p, A_p, \phi_{ih}) \longmapsto (k_i^p/h,\ h p_p,\ h^{1+1/e} A_p,\ h^{\kappa_p} \phi_{ih})$$

preserves output, expenditures and resource costs. Separately, multiplying every $\phi_{ih}$ by $q > 0$ scales output, revenues, wages, both capital prices and $A_p$ by $q$, leaving input quantities and all targeted moments unchanged. Together they scale $\mathbb{E}[\phi_{ih}]$ by $q h^{\kappa_p}$ and $A_p$ by $q h^{1+1/e}$; because $\kappa_p < 1 < 1 + 1/e$, a unique pair $(h, q)$ sets $A_p = 1$ and $\mathbb{E}[\phi_{ih}] = 1$. Finally, multiplying $L$ and all input quantities by $\lambda > 0$, all wages and input prices by $\lambda^{\kappa-1}$, and $A_p$ by $\lambda^{\kappa-1-1/e}$ preserves shares and automation choices. The preceding transformations restore the two level conventions. Thus $L$ can match sample employment without changing the targeted moments. □

# S3 Data

The main text measures labour market power with the model's markdown. This section describes the data and estimates markdowns from production functions, a measure that does not rely on the model's labour supply curve.

## S3.1 Data, sample construction, and sample overview

### S3.1.1 Datasets

I construct a firm-year panel of French manufacturing firms for 2009–2019 from four administrative datasets. They share a firm identifier, the SIREN number, so each firm's production,

workforce, and imports can be followed in one panel. The panel starts in 2009, when the product-level production data become available.

**Balance sheets.** The *Fichier approché des résultats d'Esane* (FARE) provides balance sheets for French firms. I use its revenue, value added, physical capital, intermediate inputs, employment, labour costs, location, and industry.

**Production survey.** The *Enquête Annuelle de Production* (EAP) reports revenue and quantities by product for manufacturing firms. It covers every firm with at least 20 employees or turnover above €5 million, and a representative sample of smaller firms. Because it reports prices and quantities separately, it gives firm-level output prices (Section S3.2.3).

**Workforce records.** The *Déclaration Annuelle des Données Sociales* (DADS Postes) reports, for every job, the worker's wage, hours, age, occupation, and commuting zone. I aggregate the jobs to the firm-year. The records describe each firm's workforce in each year; they do not follow workers across years.

**Customs records.** French Customs report each firm's imports by product, country of origin, and year from 1996 to 2019. They measure automation capital (Section S3.1.2). Because they begin well before the panel, every firm-year in the panel has a complete history of automation imports.

### S3.1.2 Automation capital

**Classification.** Following Acemoglu and Restrepo (2022), an import is automation capital when its product belongs to a list of 141 six-digit products of the 1992 Harmonised System (HS92). The list has eight categories: industrial robots (1 code), numerically controlled machines (12), automatic machine tools (45), automatic welding equipment (2), weaving machinery (13), dedicated textile machinery (25), conveyors and transfer equipment (4), and regulating and control instruments (39). It excludes hand-operated machine tools, machine tools classified as

not numerically controlled, household appliances, and information-technology equipment. The codes are listed at the end of this subsection.

**From Customs products to the classification.** Customs record products in the eight-digit Combined Nomenclature (CN8), which changes from year to year. For each year from 1996 to 2019, I map that year's CN8 codes to HS92 through the correspondence tables between the CN8 nomenclature in force and the HS revision of that year. I keep only codes that map to exactly one HS92 product, so no import is counted twice. I keep imports of non-French origin only.[1] The measure records the automation equipment a firm imports itself; equipment bought from domestic producers or through domestic intermediaries is not observed.

**The automation stock.** Let $e_{it} \geq 0$ denote firm $i$'s imports of automation capital in year $t$, deflated to 2015 euros with the French national-accounts investment deflator for machinery and equipment. Customs declarations cover every importing firm, so a sample firm with no automation import recorded in a year has $e_{it} = 0$. I accumulate the flows into the stock (46) of Section 5.1, with a straight-line service life of eight years:

$$K_{it}^{A} = \sum_{s=0}^{7} \left(1 - \frac{s}{8}\right) e_{i,t-s}.$$

Each purchase enters the stock at its full value, loses one eighth of that value each year, and leaves after eight years.[2]

**Adoption and intensity.** A firm is an adopter in year $t$ when it holds a positive stock:

$$d_{it} = 1\{K_{it}^{A} > 0\}. \tag{S3.1}$$

[1] Intra-EU records are restricted to ordinary import regimes, and special-use CN8 codes, such as grouped or simplified declarations, are excluded.

[2] I use a stock rather than the annual flow because automation projects can involve several purchases spread across fiscal years, and because equipment installed in one year remains in use for several more. Eight years is a standard accounting depreciation period for industrial machinery.

Adoption is not an absorbing state: a firm stops counting as an adopter once all its purchases have aged out of the eight-year window. Among adopters, automation intensity is the stock relative to value added in the previous year, $K^A_{it}/\text{VA}_{i,t-1}$.

**The 141 products.** The six-digit HS92 codes, by category, are as follows.
*Industrial robots:* 847989.
*Numerically controlled machines:* 845811, 845891, 845921, 845931, 845951, 845961, 846011, 846021, 846031, 846221, 846231, 846241.
*Automatic machine tools:* 845610, 845620, 845630, 845690, 845910, 845940, 845970, 846040, 846090, 846110, 846120, 846130, 846140, 846150, 846190, 846210, 846291, 846299, 846310, 846320, 846330, 846390, 846410, 846420, 846490, 846510, 846591, 846592, 846593, 846594, 846595, 846596, 846599, 846610, 846620, 846630, 846691, 846692, 846693, 846694, 846820, 846880, 846890, 851511, 851519.
*Automatic welding equipment:* 851521, 851531.
*Weaving machinery:* 844610, 844621, 844629, 844630, 844711, 844712, 844720, 844790, 844841, 844842, 844849, 844851, 844859.
*Dedicated textile machinery:* 844400, 844511, 844512, 844513, 844519, 844520, 844530, 844540, 844590, 844811, 844819, 844820, 844831, 844832, 844833, 844839, 844900, 845221, 845230, 845240, 845290, 845310, 845320, 845380, 845390.
*Conveyors and transfer equipment:* 842832, 842833, 842839, 845730.
*Regulating and control instruments:* 902511, 902519, 902580, 902590, 902610, 902620, 902680, 902690, 902710, 902720, 902730, 902750, 902780, 902790, 902810, 902820, 902830, 902890, 902910, 902920, 902990, 903010, 903020, 903031, 903039, 903040, 903081, 903089, 903090, 903110, 903120, 903140, 903180, 903190, 903210, 903220, 903281, 903289, 903290.

### S3.1.3 Sample construction

I prepare each dataset separately, merge the balance sheets, the production survey, and the workforce records by firm and year, and apply the sample restrictions. I then append each firm's Customs history to compute the automation capital stock.

**Balance sheets.** I keep manufacturing firms, with two-digit industry codes from 10 to 33, from 2009 to 2019. I remove duplicate observations on the identifier, industry, revenue, value added, capital, investment, and labour costs.

**Production survey.** I construct firm-level output prices and quantities following De Ridder et al. (2026) (Section S3.2.3).

**Workforce records.** I drop jobs with a missing or invalid identifier, including empty strings and "000000000", workers outside ages 16–65, non-positive gross wages, and missing commuting zones, and aggregate hours and gross wages to the firm-year.

**Customs records.** I standardise identifiers to nine-digit strings and remove administrative placeholders, such as repeated-digit patterns and codes shorter than six digits. I sum the automation imports of Section S3.1.2 to the firm-year.

**Merge and restrictions.** I merge the balance sheets with the production survey and keep firms present in both, which limits the sample to firms that meet the survey's reporting thresholds. I then merge the workforce records and commuting zones by identifier and year, and apply the following restrictions in order.

- *Industries.* I drop food and beverages (10 and 11), coke and refined petroleum (19), and pharmaceuticals (21), which have very few observations, and installation and repair of machinery (33), whose firms may be intermediaries for automation equipment rather than its users. Tobacco (12) has no firms in the panel, so eighteen industries remain.
- *Missing data.* I drop observations with missing revenue, physical output, capital, hours, employment, intermediate inputs, output prices, labour costs, wages, or commuting zone.
- *Consecutive years.* Firms may enter and leave the sample, but each firm must be observed in consecutive years. For a firm with gaps, I keep its longest run of consecutive years, and I drop firms whose longest run is shorter than three years.

- *Size.* I keep firms with at least five employees on average over their time in the sample.
- *Outliers.* I winsorise revenue, output prices and quantities, capital, intermediate inputs, hours, employment, gross wages, wages, and the material and wage shares at the 1st and 99th percentiles.

I fix the sample at this point and append each firm's 1996–2019 automation imports. All monetary variables are in 2015 euros, using the deflators of Section S3.2.2. The production-function estimation of Section S3.3 uses the twelve two-digit industries with the largest shares of automation imports, shown in bold in Table S3.1; I call them the production-function sample.

### S3.1.4 Sample overview

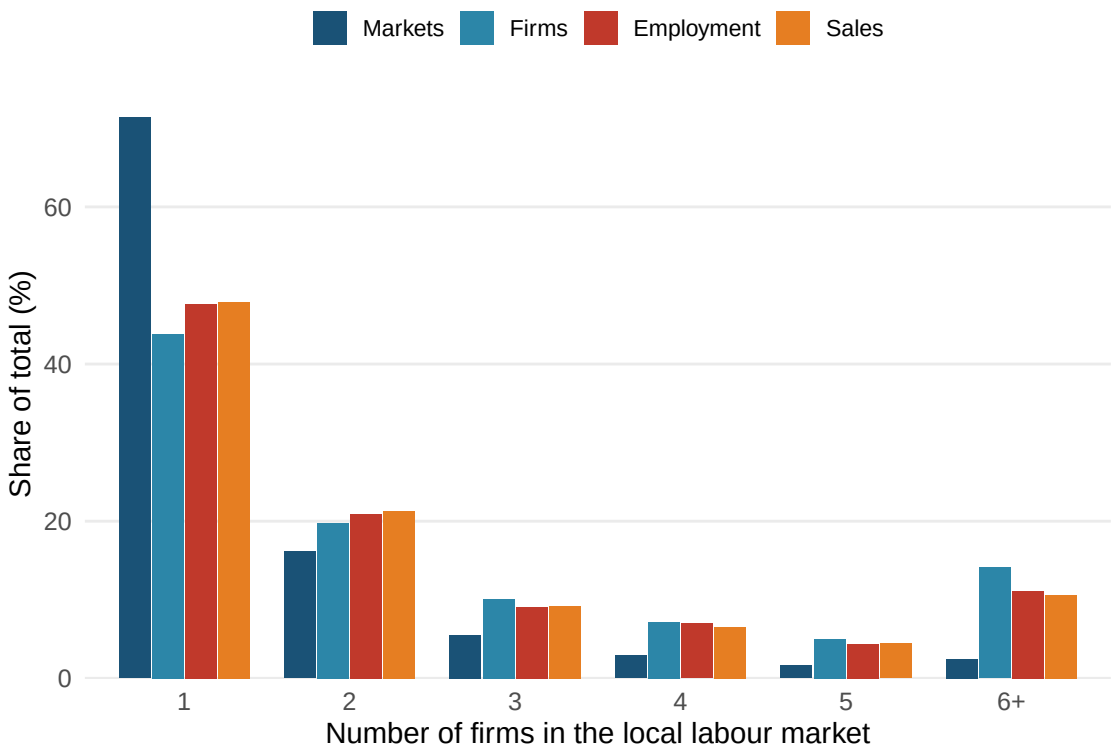
The panel has 21,918 firms; the production-function sample has 134,762 firm-years for 16,525 firms. Table S3.1 lists every two-digit manufacturing industry in order of its share of manufacturing automation imports. The twelve selected industries account for 97.8 per cent of automation imports, 86.0 per cent of revenue, and 84.0 per cent of hours worked in the eighteen manufacturing industries of the panel.

Automation is common but lumpy. In 2019, 43.7 per cent of the panel's firms hold a positive automation stock, while in a given year about three firm-years in ten in the production-function sample record automation imports.

Table S3.2 describes the distribution of firm characteristics in the production-function sample. The median firm has revenue of €4.31 million, 25 workers, and a wage bill of €0.87 million. At the ninetieth percentile the figures are €41.39 million, 168 workers, and €6.88 million.

Figure S3.1 shows how markets, firms, employment and sales are distributed across local labour markets by the number of firms they contain; Section 5.4 summarises it.

Figure S3.1: DISTRIBUTION OF FIRMS ACROSS LOCAL LABOUR MARKETS




*Notes:* The figure groups local labour markets (five-digit industry × commuting zone) by the number of firms they contain and plots, for each group, its share of all markets, of all firms, of total employment (worker headcounts), and of total sales. Shares within each series sum to one.

## S3.2 Data sources and variable definitions

### S3.2.1 Variables

The four datasets are linked through the SIREN identifier. The variables below enter the production-function estimation, the automation measure, and the moments of the quantification.

1. **FARE**

   - **Revenue:** total revenue, `REDI_R310`.
   - **Value added:** value added as reported in FARE, the denominator of automation intensity.

Table S3.1: ALL 2-DIGIT MANUFACTURING SECTORS

| Code | Sector | # of Firms | Revenue Share | Hours Share | Automation Expenditure Share |
|---|---|---|---|---|---|
| **28** | **Machinery** | **2,362** | **12.54%** | **12.78%** | **31.89%** |
| **22** | **Rubber and plastic products** | **2,346** | **9.75%** | **10.66%** | **16.29%** |
| **29** | **Motor vehicles** | **856** | **7.54%** | **6.97%** | **13.94%** |
| **26** | **Computer and optical products** | **905** | **5.76%** | **6.02%** | **6.81%** |
| **23** | **Other non-metallic mineral products** | **1,642** | **7.26%** | **7.00%** | **5.68%** |
| **27** | **Electrical equipment** | **1,033** | **6.91%** | **6.85%** | **5.61%** |
| **30** | **Other transport equipment** | **252** | **2.74%** | **2.42%** | **4.83%** |
| **25** | **Fabricated metal products** | **4,445** | **9.40%** | **12.02%** | **4.73%** |
| **20** | **Chemicals** | **1,182** | **13.59%** | **9.11%** | **3.20%** |
| **13** | **Textiles** | **923** | **2.00%** | **2.27%** | **2.01%** |
| **32** | **Other manufacturing** | **626** | **2.41%** | **2.87%** | **1.45%** |
| **24** | **Basic metals** | **601** | **6.13%** | **5.00%** | **1.41%** |
| 17 | Paper | 908 | 5.77% | 5.00% | 0.85% |
| 31 | Furniture | 1,182 | 2.11% | 3.07% | 0.52% |
| 16 | Wood products | 1,268 | 2.34% | 2.79% | 0.40% |
| 14 | Wearing apparel | 710 | 1.88% | 2.40% | 0.31% |
| 18 | Printing | 1,349 | 0.82% | 1.32% | 0.06% |
| 15 | Leather | 309 | 1.05% | 1.45% | 0.02% |
| - | **Selected Industries** | **16,525** | **86.02%** | **83.97%** | **97.83%** |

*Notes:* Shares are of the eighteen-industry panel after the restrictions of Section S3.1.3, before the industry selection. Totals are computed from unrounded shares. Hours are from the workforce records. A firm that changes two-digit industry during the panel is counted in each industry it appears in, so the industry counts sum to more than the 16,525 selected firms. Industries in **bold** form the production-function sample. The automation shares use an earlier version of the product list of Section S3.1.2, on which the twelve industries were selected.

- **Capital:** total fixed physical assets, `IMMO_CORP`.
- **Intermediate inputs:** materials bought for production (`REDI_212`) and goods bought for resale (`REDI_210`), net of the changes in their stocks (`REDI_213` and `REDI_211`): (`REDI_210` + `REDI_212`) − (`REDI_211` + `REDI_213`).
- **Employment:** the firm's number of workers. It defines employment shares in local labour markets and enters the moments.
- **Labour cost:** gross wages and salaries, including base pay, overtime, and bonuses, plus employer social security contributions: `REDI_R216` + `REDI_R217`.

Table S3.2: SUMMARY STATISTICS OF OBSERVED FIRM CHARACTERISTICS

| | Revenue (million €) | Fixed Assets (million €) | Wage Bill (DADS gross) (million €) | Employment | Log Price | Log Price (5-digit) |
|---|---|---|---|---|---|---|
| 5th percentile | 0.65 | 0.11 | 0.17 | 6.00 | -3.04 | -2.21 |
| 10th percentile | 0.91 | 0.18 | 0.22 | 7.00 | -1.89 | -1.42 |
| 25th percentile | 1.74 | 0.43 | 0.39 | 12.00 | -0.44 | -0.39 |
| Median | 4.31 | 1.30 | 0.87 | 25.00 | 0.43 | 0.08 |
| Mean | 19.37 | 9.90 | 3.12 | 73.81 | 0.00 | 0.00 |
| 75th percentile | 12.64 | 4.78 | 2.27 | 60.00 | 0.74 | 0.53 |
| 90th percentile | 41.39 | 19.60 | 6.88 | 168.00 | 1.12 | 1.27 |
| 95th percentile | 82.83 | 47.89 | 13.37 | 312.00 | 1.65 | 1.80 |
| Std. deviation | 49.85 | 29.64 | 7.27 | 150.70 | 1.37 | 1.19 |

*Notes:* The table covers the firm-years of the production-function sample. The first "Log Price" column reports log output prices after winsorisation and mean normalisation, across all firms in the sample. The second also demeans them by five-digit-industry by year fixed effects.

- **Firm age:** from the creation date, `ENT_CRE_DAAAAMMJJ`.
- **Location:** the five-digit commune code of the registered head office, `DEPCOM`, whose first two digits give the department. It is the parent company's location, not that of every establishment.
- **Industry:** the five-digit APE code of the firm's main activity, in revision 2 of the French NAF classification. Manufacturing is two-digit codes 10 to 33.

2. **EAP**

- **Output prices:** built from the ten-digit product code (`PRODFRA`), the quantity and revenue indicators (`C_UNITE_VAR`), and the values in quantity and revenue (`VAL_REF`), as described in Section S3.2.3.

3. **DADS Postes**

- **Hours:** annual hours of each job, `NBHEUR`.

- **Wages:** gross annual earnings of each job, `S_BRUT`, excluding employer social security contributions.
- **Commuting zone:** `ZEMPT`. A local labour market is a five-digit industry by commuting zone.

4. **Customs.** I clean the Customs data with the code of Bergounhon et al. (2018).
   - **Product:** the eight-digit Combined Nomenclature code, `NC8`, mapped to HS92 (Section S3.1.2).
   - **Origin:** the country of origin, `PYOD`.
   - **Import value:** `VALSTAT`.

### S3.2.2 Deflators

I convert nominal values to 2015 euros with industry deflators from EU KLEMS: the gross-output deflator (`GO_CP`) for revenue, the intermediate-input deflator (`II_CP`) for intermediate inputs, the value-added price index for value added, and the investment-price deflator (`IP_GFCF`) for capital. Wages, labour costs, and output prices are deflated with the consumer price index. Automation imports are deflated with the French national-accounts investment deflator for machinery and equipment.

### S3.2.3 Firm-level output prices

Following De Ridder et al. (2026) and Wong (2026), I build a firm-level output price from the EAP in three steps. Let $\mathcal{G}_{it}$ be the set of products firm $i$ sells in year $t$, $\mathcal{I}_{nt} \subseteq \mathcal{I}$ the set of firms that sell product $n$, and $P_{int}$ and $\text{Quantity}_{int}$ the price and quantity of product $n$ at firm $i$.

**Step 1.** The price of each product at each firm is its revenue divided by its quantity, $P_{int} = \text{Revenue}_{int}/\text{Quantity}_{int}$.

**Step 2.** Each price is divided by the revenue-weighted average price of the product across the

firms that sell it,

$$\overline{P}_{nt} = \sum_{i' \in \mathcal{I}_{nt}} \frac{\text{Revenue}_{i'nt}}{\sum_{i'' \in \mathcal{I}_{nt}} \text{Revenue}_{i''nt}} P_{i'nt}.$$

**Step 3.** The firm's price is the revenue-weighted average of its relative prices,

$$P_{it} = \sum_{n \in \mathcal{G}_{it}} s^R_{int} \frac{P_{int}}{\overline{P}_{nt}}, \qquad s^R_{int} = \frac{\text{Revenue}_{int}}{\sum_{n' \in \mathcal{G}_{it}} \text{Revenue}_{in't}}.$$

A firm whose products sell above their market averages has a high price, with each product weighted by its share of the firm's revenue. The log output price $p_{it}$ is the logarithm of $P_{it}$ deflated by the consumer price index, and physical output is $q_{it} = r_{it} - p_{it}$.

## S3.3 Markdown estimation

### S3.3.1 Markdowns from production functions

The markdown is the ratio of the wage to the marginal revenue product of labour, the object $\delta_i = w_i/\omega_i$ of the model. Following Yeh et al. (2022), it can be written in terms of observed shares and two output elasticities:

$$\delta_{it} = \frac{\mu_{it}}{\theta^L_{it}} s^L_{it}, \qquad \mu_{it} = \theta^M_{it} \frac{\text{Revenue}_{it}}{\text{Intermediate-input expenditure}_{it}}, \tag{S3.2}$$

where $s^L_{it}$ is the wage bill divided by revenue, with revenue as observed rather than corrected for the measurement error removed in Step 1 below, $\mu_{it}$ is the price-cost markup, and $\theta^L_{it}$ and $\theta^M_{it}$ are the output elasticities of labour and intermediate inputs. A markdown below one means the firm pays workers less than their marginal revenue product. The wage bill is DADS gross earnings (`S_BRUT`), which exclude employer social security contributions; the same wage bill gives the average hourly wage behind efficiency hours and the wage moments of the estimation.

Separating the markdown from the markup requires three assumptions, which I maintain throughout: firms take the prices of intermediate inputs as given; capital is predetermined and dynamic; and labour and intermediate inputs are static. Revenue, the wage bill, and

intermediate-input expenditure are observed. The two output elasticities are not, and I estimate them from production functions.

### S3.3.2 Production-function estimation

**Specification.** For each two-digit industry, output of firm $i$ in year $t$ is

$$y_{it} = f(k_{it}, l_{it}, m_{it}; \beta) + \varphi_{it}, \tag{S3.3}$$

where lower-case letters are logarithms of output $y$, capital $k$, labour $l$, and intermediate inputs $m$, $f$ is translog, and $\varphi_{it}$ is log Hicks-neutral productivity. The output elasticities are $\theta^L_{it} = \partial y_{it}/\partial l_{it}$ and $\theta^M_{it} = \partial y_{it}/\partial m_{it}$. Productivity follows a first-order Markov process,

$$\varphi_{it} = \mathcal{M}(\varphi_{i,t-1}) + \varepsilon^{\varphi}_{it}. \tag{S3.4}$$

A tilde denotes log expenditure, the sum of the log price and the log quantity, as in $\widetilde{k}_{it} = k_{it} + p_{k,it}$.

**Labour in efficiency hours.** Raw hours mix the quantity of labour with its quality, so a change in skill mix would show up as a change in productivity (Chan et al., 2026). I therefore measure labour in efficiency hours. Let $\bar{w}_{it}$ be the firm's average hourly wage, its wage bill divided by its hours. I regress its logarithm on five-digit-industry by year and commuting-zone by year fixed effects,

$$\log \bar{w}_{it} = \Lambda_{n(i)t} + \Xi_{c(i)t} + \varepsilon^{w}_{it},$$

where $n(i)$ is the firm's five-digit industry and $c(i)$ its commuting zone, and scale hours by the exponential of the residual:

$$L^{\text{eff}}_{it} = L^{\text{hours}}_{it} \exp(\hat{\varepsilon}^{w}_{it}).$$

A firm that pays more per hour than its industry and location predict is counted as employing more efficiency units per hour. Because the residual also contains within-market wage differences from productivity, amenities and markdowns, efficiency hours count part of these differences as differences in labour quality.

**Three biases.** Three variables the econometrician does not observe bias the estimated elasticities. Writing revenue in terms of what is observed makes them explicit:

$$\begin{aligned} r_{it} &= p_{it} + q_{it}, \qquad q_{it} \equiv y_{it} + \varepsilon^{q}_{it} \\ &= p_{it} + f(k_{it}, l_{it}, m_{it}; \beta) + \varphi_{it} + \varepsilon^{q}_{it} \\ &= p_{it} + f(\widetilde{k}_{it}, l_{it}, \widetilde{m}_{it}; \beta) + B(\boldsymbol{p}_{xit}, \widetilde{\boldsymbol{x}}_{it}, l_{it}; \beta, \boldsymbol{\vartheta}) + \varphi_{it} + \varepsilon^{q}_{it}, \end{aligned} \tag{S3.5}$$

where $p_{it}$ is the log output price, $q_{it}$ is measured output, $\varepsilon^{q}_{it}$ is measurement error, $\widetilde{\boldsymbol{x}}_{it} = \{\widetilde{k}_{it}, \widetilde{m}_{it}\}$ collects input expenditures, with capital entering as the value of fixed assets, $\boldsymbol{p}_{xit} = \{p_{k,it}, p_{m,it}\}$ collects input prices, and $B(\cdot)$, with its own parameters $\boldsymbol{\vartheta}$, absorbs the input-price terms (De Loecker et al., 2016).

- *Productivity.* More productive firms produce more and use more inputs, which biases the elasticities upwards.
- *Output prices.* Firms that charge more sell less for the same revenue and use fewer inputs, which biases the elasticities downwards (Bond et al., 2021).
- *Input prices.* Capital and intermediate inputs are observed only as expenditures, so differences in input prices across firms, from market access, input quality, or bargaining, contaminate the elasticities.

**Corrections.** A control function addresses the first bias: the demand for intermediate inputs reveals productivity and can be inverted to recover it (Olley and Pakes, 1996; Ackerberg et al., 2015). The EAP addresses the second: it reports prices and quantities separately, so I deflate revenue by the firm's output price and estimate a quantity production function (De Ridder et al., 2026). For the third, sector and location fixed effects absorb systematic differences in input prices, under the assumption that firms take input prices as given. A second control function, following De Loecker et al. (2016) and Wong (2026), uses the output price as a proxy for input quality: under a broad class of demand systems, firms that use better inputs sell at higher prices. Input prices are then a function of observables, $\boldsymbol{p}_{xit} = \boldsymbol{p}_x(p_{it}, \boldsymbol{Z}_{it})$, where $\boldsymbol{Z}_{it}$

contains sector-by-location fixed effects and controls for markups and markdowns: the firm's revenue share in its five-digit industry and its employment share in its local labour market.

**Estimating equation.** Substituting the input-price function into (S3.5), and writing $B(\cdot)$ as a flexible basis in $(p_{it}, \boldsymbol{Z}_{it})$ and their interactions with $\{1, \widetilde{\boldsymbol{x}}_{it}, l_{it}\}$ (De Loecker et al., 2016), gives the equation taken to the data:

$$q_{it} = f(\widetilde{k}_{it}, l_{it}, \widetilde{m}_{it}; \beta) + B\left[(p_{it}, \boldsymbol{Z}_{it}) \times \{1, \widetilde{\boldsymbol{x}}_{it}, l_{it}\}; \beta, \boldsymbol{\vartheta}\right] + \varphi_{it} + \varepsilon^{q}_{it}. \tag{S3.6}$$

I estimate it in two steps, following Ackerberg et al. (2015) and De Ridder et al. (2026). The first step separates measurement error from productivity. The second estimates $(\beta, \boldsymbol{\vartheta})$ together with the law of motion of productivity.

**Step 1: measurement error.** Productivity is recovered by inverting the demand for intermediate inputs,

$$\varphi_{it} = m^{-1}(\widetilde{k}_{it}, l_{it}, \widetilde{m}_{it}, p_{it}, s^{Y}_{it}, \boldsymbol{Z}_{it}). \tag{S3.7}$$

The inversion requires that productivity is the only unobserved variable in the demand for intermediate inputs, and that this demand rises with productivity at given controls (Olley and Pakes, 1996). Every argument of $m^{-1}$ must therefore be observed. The EAP supplies output prices. Markups are not observed, so I control for them with the firm's revenue share in its five-digit industry and year, $s^{Y}_{it}$ (De Ridder et al., 2026), since a firm's relative size determines its markup (Edmond et al., 2023). Year fixed effects absorb common movements in the prices of intermediate inputs. Substituting (S3.7) into (S3.6) gives

$$q_{it} = \mathcal{P}(\widetilde{k}_{it}, l_{it}, \widetilde{m}_{it}, p_{it}, s^{Y}_{it}, \boldsymbol{Z}_{it}) + \varepsilon^{q}_{it}, \tag{S3.8}$$

which I estimate by ordinary least squares. Here $\mathcal{P}$ is a polynomial in log capital, log intermediate-input expenditure, log efficiency hours, the log output price, the revenue share in the five-digit industry, and the employment share in the local labour market, with year and industry-by-location fixed effects entering linearly. The local labour market in this step is the five-digit

industry by commuting zone. The polynomial is of second order in industries 20, 27, 29, and 30, and of third order in industries 13, 22, 23, 24, 25, 26, 28, and 32. Industry 29 uses five-digit-industry by region by year fixed effects in the first step, and controls for firm age in both steps.

**Step 2: elasticities.** With $\hat{q}_{it}$ the fitted output from the first step, productivity at any candidate $(\beta, \boldsymbol{\vartheta})$ is

$$\varphi_{it}(\beta, \boldsymbol{\vartheta}) = \hat{q}_{it} - f(\widetilde{k}_{it}, l_{it}, \widetilde{m}_{it}; \beta) - B\big[(p_{it}, \boldsymbol{Z}_{it}) \times \{1, \widetilde{\boldsymbol{x}}_{it}, l_{it}\}; \beta, \boldsymbol{\vartheta}\big]. \tag{S3.9}$$

The law of motion (S3.4) is linear in every industry, and its innovation is uncorrelated with predetermined variables:

$$\mathbb{E}\big[\varepsilon^{\varphi}_{it}(\beta, \boldsymbol{\vartheta})\, \mathbf{X}_{it}\big] = 0.$$

The translog $f$ contains the first- and second-order terms in $(\widetilde{k}_{it}, l_{it}, \widetilde{m}_{it})$ and their three-way interaction, so $\beta$ has ten parameters. The basis $B(\cdot)$ is of first order in every industry: it interacts the output price and the two share controls with $\{1, \widetilde{k}_{it}, \widetilde{m}_{it}, l_{it}\}$, so $\boldsymbol{\vartheta}$ has twelve parameters (sixteen in industry 29, where firm age is a further control). The share controls enter this step in logs. The fixed effects in $\boldsymbol{Z}_{it}$ are removed by a within-transformation and do not enter these interactions. The instruments $\mathbf{X}_{it}$ are current fixed assets; the lagged inputs with their squares, two-way interactions and three-way interaction; and the lagged basis of $B(\cdot)$, with lagged export status and firm age added to the share controls. This gives 31 instruments for 22 parameters (26 in industry 29). The elasticities $\theta^{L}_{it}$ and $\theta^{M}_{it}$ in (S3.2) are the derivatives of $f$ plus those of $B(\cdot)$ with respect to $l_{it}$ and $\widetilde{m}_{it}$, evaluated at the estimates. For each candidate $(\beta, \boldsymbol{\vartheta})$, I compute productivity, recover the innovation from the law of motion, and evaluate the moment conditions; the estimates minimise their quadratic form.

### S3.3.3 Results

**Levels and dispersion.** Table S3.3 reports the markdowns by industry. The median markdown is 0.68 and the mean is 0.78: at the median firm, a worker is paid 68 cents per euro of marginal revenue product. The median markdown is below one in eleven of the twelve industries, and

Table S3.3: ESTIMATES OF MARKDOWNS BY INDUSTRY

| Code | Industry | Mean | Median | IQR |
|---|---|---|---|---|
| 13 | Textiles | 0.57 | 0.47 | 0.28 |
| 20 | Chemicals | 0.74 | 0.75 | 0.42 |
| 22 | Rubber and plastic products | 0.54 | 0.50 | 0.23 |
| 23 | Other non-metallic mineral products | 0.81 | 0.73 | 0.39 |
| 24 | Basic metals | 0.70 | 0.66 | 0.37 |
| 25 | Fabricated metal products | 0.83 | 0.78 | 0.35 |
| 26 | Computer and optical products | 1.06 | 0.87 | 0.61 |
| 27 | Electrical equipment | 0.59 | 0.56 | 0.22 |
| 28 | Machinery | 0.72 | 0.71 | 0.23 |
| 29 | Motor vehicles | 0.51 | 0.45 | 0.25 |
| 30 | Other transport equipment | 1.31 | 0.88 | 0.88 |
| 32 | Other manufacturing | 2.19 | 1.73 | 1.21 |
| - | **Aggregate** | 0.78 | 0.68 | 0.39 |

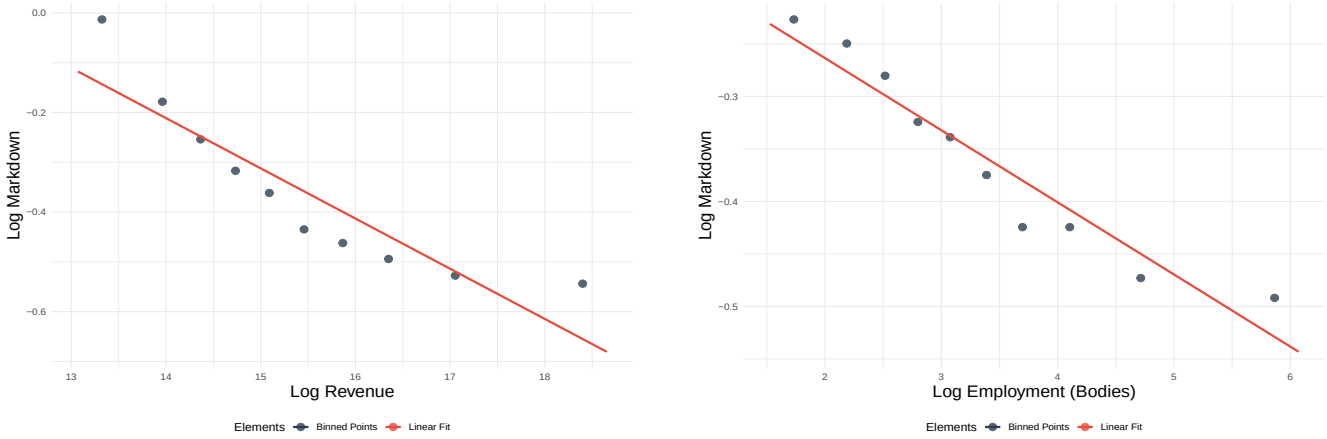
*Notes:* I estimate a translog production function separately for each two-digit manufacturing industry listed in the table, using a two-step procedure: (i) isolate measurement error; (ii) recover the key translog parameters. Additional details are provided in Section S3.3.2. Markdowns are winsorised at the 1st and 99th percentiles within each industry; the statistics are over the firm-years of the production-function sample.

ranges from 0.45 in motor vehicles to 0.88 in other transport equipment; the exception is other manufacturing. Markdowns also differ within industries, with an aggregate interquartile range of 0.39.

**Firm size.** Figure S3.2 shows that labour market power rises with firm size: log markdowns fall with log revenue and with log employment. Controlling for local labour market by year fixed effects, a 10 per cent larger revenue goes with a 1.1 per cent lower markdown (elasticity −0.11, standard error 0.0016), and a 10 per cent larger workforce with a 0.6 per cent lower markdown (−0.06, standard error 0.0018).[3]

[3]With local labour markets defined as three-digit industry by commuting zone, the elasticities are −0.10 for revenue and −0.07 for employment.

Figure S3.2: SIZE-BASED LABOUR MARKET POWER




(a) Log of Revenue

(b) Log of Employment (headcounts)

*Notes:* The figures show binscatter plots with data divided into 10 quantile-based bins. Panel (a) has log of revenues on the x-axis and Panel (b) has log of employment on the x-axis. Log of markdowns is on the y-axis. Each bin displays mean values, and linear regression fits are overlaid to illustrate the relationship between markdowns and firm size. The results are robust to using alternative measures when hours are used instead of headcounts.

**Automation.** I sort firm-years from 2010 to 2019 into deciles of the production-function markdown, from the least labour market power in bin 1, the highest markdowns, to the most in bin 10, and estimate regression (47) with these deciles. The bin boundaries are computed once on the pooled sample. The sample keeps market-year cells with at least two firms and variation in markdowns within the cell, and standard errors are clustered by firm. For the extensive margin the outcome is the adoption indicator (S3.1). For the intensive margin the sample is restricted to adopters and the outcome is the log of automation intensity, $\log(K^A_{it}/\text{VA}_{i,t-1})$.

Panel (a) of Figure 4 shows that, with the production-function markdown, the probability of holding automation capital rises with labour market power within a local labour market. In the bottom decile, 40 per cent of firm-years hold a positive stock. Relative to that decile, firms in the fifth decile are 15.2 percentage points more likely to hold automation capital (standard error 1.8), the gap peaks at 37.0 points in the ninth decile (standard error 2.2), and firms in the top decile are 31.3 points more likely to adopt (standard error 2.3).

Panel (b) shows that, among adopters, those with more labour market power also hold more automation capital per euro of value added. Relative to adopters in the bottom decile of

the same market and year, intensity is higher by 0.59 log points in the fifth decile (95 per cent confidence interval 0.13 to 1.04) and by 1.27 log points in the top decile (0.74 to 1.80), so the top decile holds about three and a half times as much automation capital per euro of lagged value added. The differences are significant from the fourth decile onwards.

Both margins therefore show the pattern of Section 5.2, with a measure of labour market power that does not use the model's labour supply curve.

# S4 Estimation

**Simulation and bootstrap.** The standard shocks behind $(\phi_{ih}, T_i, b_i)$ are drawn once and transformed at every candidate $\theta$, so the objective (51) is a deterministic function of the parameters. Each evaluation uses one set of draws ($S = 1$). To measure the simulation noise this leaves in $\mathbf{m}(\hat{\theta})$, I re-solve the model at $\hat{\theta}$ under 100 independent sets of standard shocks; the covariance of the resulting moments is $\hat{\mathbf{C}}_{\text{sim}}$ in (53). Relative to the data standard error, the simulation standard deviation is about 0.5 for (M1), 0.8 for (M2), 1.3 for (M3), between 0.9 and 1.0 for (M4), and between 1.1 and 1.5 for (M5), similar under the three schedules. The draw also moves the point estimates. To first order, fitting the moments averaged over the 100 draws would lower $\hat{o}$ and $\hat{c}_k$ by about one standard error of Table 2 and $\hat{\mathcal{R}}_{\phi T}$ by about 0.8 of one, and would leave $\hat{\sigma}_\phi$ and $\hat{\iota}$ essentially unchanged. Each bootstrap replication draws $J$ local labour markets with replacement from the $J$ observed markets, keeps every firm in a drawn market, and recomputes the full moment vector. Replications in which any moment is undefined are discarded and redrawn until 1,000 are obtained.

## S4.1 Local identification at the estimates

Let $\mathbf{D}$ be the Jacobian of the five model moments with respect to the five estimated parameters at $\hat{\theta}$, computed by central finite differences with the simulation draws held fixed. Table S4.1 reports two diagnostics under each task schedule.

The first is the matrix $\mathbf{W}^{1/2}\mathbf{D}\,\mathrm{diag}(s)$, in which each moment is measured in units of its standard error and each parameter in units of its scale $s_r$ (the estimate itself, and 0.1 for $\mathcal{R}_{\phi T}$).

Table S4.1: LOCAL IDENTIFICATION

| Moment | $\sigma_\phi$ | $\iota$ | $\mathcal{R}_{\phi T}$ | $\upsilon$ | $c_k$ |
|---|---|---|---|---|---|
| *Falling hazard* | | | | | |
| (M1) | 39 | 48 | 6 | 2 | 4 |
| (M2) | 57 | 37 | 3 | 2 | 1 |
| (M3) | 51 | 43 | 1 | 2 | 3 |
| (M4) | 32 | 4 | 4 | 59 | 1 |
| (M5) | 9 | 3 | 0 | 26 | 62 |
| Weakest direction | −0.01 | 0.02 | 1.00 | −0.06 | −0.02 |
| Singular values: 138.6, 99.5, 56.8, 21.9, 3.2 | | | | | |
| *Constant hazard* | | | | | |
| (M1) | 40 | 49 | 6 | 1 | 3 |
| (M2) | 58 | 36 | 3 | 1 | 1 |
| (M3) | 52 | 44 | 1 | 1 | 2 |
| (M4) | 21 | 8 | 3 | 57 | 11 |
| (M5) | 10 | 2 | 0 | 29 | 59 |
| Weakest direction | −0.01 | 0.02 | 1.00 | −0.05 | −0.02 |
| Singular values: 139.8, 100.1, 56.9, 20.2, 3.3 | | | | | |
| *Rising hazard* | | | | | |
| (M1) | 41 | 52 | 6 | 1 | 0 |
| (M2) | 62 | 34 | 3 | 1 | 0 |
| (M3) | 54 | 44 | 1 | 1 | 0 |
| (M4) | 12 | 11 | 2 | 59 | 16 |
| (M5) | 3 | 1 | 0 | 41 | 54 |
| Weakest direction | −0.01 | 0.02 | 1.00 | −0.06 | −0.04 |
| Singular values: 131.1, 104.1, 57.0, 9.2, 3.4 | | | | | |

*Notes:* Each row gives the share, in per cent, of each parameter in the response of the moment, $100\,|D_{kr}|\,s_r / \sum_{r'} |D_{kr'}|\,s_{r'}$, where $D_{kr}$ is the derivative of moment $k$ with respect to parameter $r$ at the estimate and $s_r$ is the parameter's scale (the estimate, and 0.1 for $\mathcal{R}_{\phi T}$). Rows sum to 100. The Jacobian has rank 5 under each schedule, and the condition number (largest over smallest singular value) is 43, 43 and 39. Singular values are those of $\mathbf{W}^{1/2}\mathbf{D}\,\mathrm{diag}(s)$, so each is a change in the moments, in standard errors, per unit-length move in scaled parameter space; they are listed from largest to smallest. The weakest direction is the right singular vector of the smallest singular value, with one entry per parameter in scale units and unit length. Moment labels as in Table 3.

Its singular values are bounded away from zero and it has full rank, 5, under every schedule, so $\mathbf{D}'\mathbf{W}\mathbf{D}$ is invertible and the estimates are locally identified.

The smallest singular value, between 3.2 and 3.4, says that a move of one scale unit in the worst-determined direction of the parameter space still changes the moments by more than three standard errors. That direction is almost entirely $\mathcal{R}_{\phi T}$ under every schedule: its $\mathcal{R}_{\phi T}$ entry is 1.00 and no other entry exceeds 0.06 in absolute value (row "Weakest direction" in Table S4.1).

The second diagnostic is the share of each parameter in the response of each moment: the entry in row $k$ and column $r$ is $100\,|D_{kr}|\,s_r / \sum_{r'} |D_{kr'}|\,s_{r'}$. The dispersion moments (M1)–(M3) respond mostly to $\sigma_\phi$ and $\iota$, the adoption slope (M4) mostly to $\upsilon$, and the adoption rate (M5) mostly to $c_k$, under all three schedules.

The shares of $\mathcal{R}_{\phi T}$ are small because its scale is small, not because it is weakly identified: it is the only parameter that moves the two variances in opposite directions, with $\partial(\text{M1})/\partial\mathcal{R}_{\phi T} < 0$, $\partial(\text{M2})/\partial\mathcal{R}_{\phi T} > 0$ and $\partial(\text{M3})/\partial\mathcal{R}_{\phi T}$ small under every schedule.

The sensitivity of the estimate to the data, $\partial\hat{\theta}/\partial\mathbf{m} = (\mathbf{D}'\mathbf{W}\mathbf{D})^{-1}\mathbf{D}'\mathbf{W}$ (Andrews et al., 2017), confirms it. Measured in standard errors of both the moment and the estimate, a one-standard-error rise in the wage variance (M1) lowers $\hat{\mathcal{R}}_{\phi T}$ by 1.0 standard error, a rise in the employment variance (M2) raises it by 0.5, the covariance (M3) moves it by 0.1, and the two adoption moments move it by less than 0.05, with the same values under all three schedules.

## S4.2 Moment construction

This subsection records the measurement conventions behind the moments in Table 3. The moments use the firms of the panel of Section S3.1.3 in 2019. The estimation targets the five moments (M1)–(M5).

**Variables.** Employment $l_i^{\text{data}}$ is the firm's number of workers. It is measured in worker bodies rather than in the efficiency hours used for the production-function estimation of Section S3.3, because the model's $l_i$ is a measure of workers. The employment peg $L$ of Section 5.4 is the corresponding 2019 total. The wage bill $\mathcal{B}_i^{\text{data}}$ is deflated. The average wage is $w_i^{\text{data}} =$

$\mathcal{B}_i^{\text{data}}/l_i^{\text{data}}$, the markdown $\delta_i^{\text{data}} = \delta(s_{i|j}^{\text{data}})$ evaluates the model's markdown schedule at the within-market employment share defined in Section 2.1, and the marginal labour cost is $\omega_i^{\text{data}} = w_i^{\text{data}}/\delta_i^{\text{data}}$.

**Adoption state.** The adoption indicator $d_i^{\text{data}}$ equals one when firm $i$ holds a positive stock of imported automation capital in 2019, where the stock (46) accumulates deflated automation imports under an eight-year straight-line service life (Section S3.1.2). Adoption is not an absorbing state: a firm ceases to count as an adopter once its acquisitions have fully aged out.

**Regression moment.** Regression (52) is estimated by within-market ordinary least squares on the 2019 firms with a defined adoption state and a finite marginal labour cost, restricted to markets containing at least two such firms and exhibiting within-market variation in $\log \omega_i^{\text{data}}$. The regression is reproduced in each simulated economy.

**Sample and weighting.** All moments are unweighted across firms, and no observation is trimmed or winsorised. Let $\mathcal{P}$ be the firms in markets containing at least two valid observations, $\mathcal{E}_d$ the eligible firms with a defined adoption state, and $u_i$ and $v_i$ the deviations of $\log w_i$ and $\log l_i$ from their means over the firms of $\mathcal{P}$ in firm $i$'s market. Then

$$\begin{gathered}(\text{M1}) = \frac{\sum_{i\in\mathcal{P}} u_i^2}{|\mathcal{P}|-1}, \qquad (\text{M2}) = \frac{\sum_{i\in\mathcal{P}} v_i^2}{|\mathcal{P}|-1}, \qquad (\text{M3}) = \frac{\sum_{i\in\mathcal{P}} u_i v_i}{|\mathcal{P}|-1}, \\ (\text{M4}) = \hat{\beta}_4, \qquad (\text{M5}) = \frac{1}{|\mathcal{E}_d|}\sum_{i\in\mathcal{E}_d} d_i,\end{gathered} \tag{S4.1}$$

with $\hat{\beta}_4$ from (52). The simulated counterparts use the identical formulas, so the convention is common to both sides of (51).

Moment (M5) uses all eligible 2019 firms with a defined adoption state. Computing the aggregate adoption rate requires neither multiple firms per market nor variation within markets, so firms in single-firm markets remain in its sample. Every firm in the 2019 sample is assigned to a local labour market, so (M5), the simulated market structure and the bootstrap use the same 15,189 firms in 9,326 markets.

Table S4.2: FRANCE'S OPERATIONAL STOCK OF INDUSTRIAL ROBOTS

| | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
|---|---|---|---|---|---|---|
| Stock at year end | 42,054 | 44,817 | 49,914 | 55,244 | 58,572 | 60,480 |
| Annual growth (%) | | 6.6 | 11.4 | 10.7 | 6.0 | 3.3 |

*Notes:* Operational industrial robots at the end of each year, from International Federation of Robotics (2025). The 2019 and 2020 figures are from the October 2023 edition, the later figures from the October 2025 edition; the two editions differ by one robot in 2022, the only revised year they share.

## S4.3 The cut in the automation price

In each estimated oligopsonistic economy, I lower the automation price to increase aggregate automation capital by 80 per cent, matching the projected growth of France's industrial robot stock between 2019 and 2029.

Table S4.2 reports France's operational robot stock from International Federation of Robotics (2025). I project the stock to 2029 using its average annual growth rate over 2022–2024:

$$g = \left(\frac{S_{2024}}{S_{2022}}\right)^{1/2} - 1 = 4.63\%, \qquad q = \frac{S_{2024}\,(1+g)^5}{S_{2019}} \approx 1.80,$$

where $S_t$ denotes the stock at the end of year $t$.

Let $K^0$ denote aggregate automation capital in the estimated 2019 economy and $K(\Delta)$ its equilibrium value after reducing $c_k$ by a proportion $\Delta$. Holding all other parameters, firm draws and market structure fixed, I solve for

$$\frac{K(\Delta^*)}{K^0} = q.$$

Each task schedule implies a different price reduction, reported in Table 5. I apply the same proportional price reduction to the corresponding economy without labour markdowns.

# S5 Sensitivity analysis

This section reports the specifications behind Section 6.3. Each specification is estimated by (51) with the moments, weighting matrix and externally fixed parameters of Section 5, and both experiments of Section 6 are repeated on the estimated economy. Welfare uses the estimated 2019 economy of each specification as reference.

## S5.1 Task schedule

The baseline schedules share $a = 0.5$. Here $a$ is set to 0.1 and to 0.9, and under each value $\beta_R$ is chosen so that the displacement hazard $h(\tilde{x}) = [a + \beta_R(1 - \tilde{x})]^{-1}$ falls, stays constant or rises, as in the baseline. The values are $\beta_R = -a/2$, 0 and $2a$, which keep the ratio $h(0)/h(1) = a/(a + \beta_R)$ at its baseline values of 2, 1 and $1/3$, so that $a$ moves only the level of the hazard, $h(1) = 1/a$. Table S5.1 reports the estimates, Table S5.2 the fit, and Table S5.3 the two experiments.

Table S5.1: ESTIMATES UNDER ALTERNATIVE TASK SCHEDULES

| | $a = 0.1$ | | | $a = 0.9$ | | |
|---|---|---|---|---|---|---|
| | Falling | Constant | Rising | Falling | Constant | Rising |
| *Fixed schedule parameters* | | | | | | |
| $a$ | 0.1 | 0.1 | 0.1 | 0.9 | 0.9 | 0.9 |
| $\beta_R$ | −0.05 | 0 | 0.2 | −0.45 | 0 | 1.8 |
| *Estimated* | | | | | | |
| $\sigma_\phi$ | 0.637 (0.013) | 0.627 (0.013) | 0.594 (0.013) | 0.553 (0.009) | 0.546 (0.009) | 0.534 (0.009) |
| $\iota$ | 0.759 (0.020) | 0.758 (0.020) | 0.757 (0.020) | 0.754 (0.020) | 0.754 (0.020) | 0.753 (0.020) |
| $\mathcal{R}_{\phi T}$ | 0.247 (0.037) | 0.258 (0.037) | 0.299 (0.034) | 0.393 (0.027) | 0.402 (0.027) | 0.416 (0.027) |
| $\upsilon$ | 0.351 (0.011) | 0.362 (0.011) | 0.416 (0.018) | 0.591 (0.046) | 0.659 (0.059) | 0.817 (0.081) |
| $c_k$ | 1.349 (0.024) | 1.370 (0.024) | 1.492 (0.031) | 0.657 (0.030) | 0.811 (0.048) | 1.811 (0.164) |

*Notes:* Each column is estimated by (51) on the five moments of Table 3, with the externally fixed parameters and normalisations of Table 2. Standard errors from (53) in parentheses.

Table S5.2: MODEL FIT UNDER ALTERNATIVE TASK SCHEDULES

| | | $a = 0.1$ | | | $a = 0.9$ | | |
|---|---|---|---|---|---|---|---|
| | Data | Falling | Constant | Rising | Falling | Constant | Rising |
| (M1) Variance of log wages | 0.048 | 0.048 | 0.048 | 0.048 | 0.048 | 0.048 | 0.048 |
| (M2) Variance of log employment | 0.765 | 0.765 | 0.765 | 0.765 | 0.765 | 0.765 | 0.765 |
| (M3) Covariance | 0.020 | 0.020 | 0.020 | 0.020 | 0.020 | 0.020 | 0.020 |
| (M4) Adoption slope | 0.438 | 0.438 | 0.438 | 0.438 | 0.438 | 0.438 | 0.438 |
| (M5) Adoption rate | 0.437 | 0.437 | 0.437 | 0.437 | 0.437 | 0.437 | 0.437 |

*Notes:* Moments as in Table 3; (M1)–(M3) are within-market moments and (M4) is the within-market slope of adoption on the log marginal labour cost. The Data column repeats Table 3. The other columns report the simulated moments at the estimates of Table S5.1.

Table S5.3: COUNTERFACTUALS UNDER ALTERNATIVE TASK SCHEDULES

| | $a = 0.1$ | | | $a = 0.9$ | | |
|---|---|---|---|---|---|---|
| | Falling | Constant | Rising | Falling | Constant | Rising |
| **Panel A. Welfare cost of labour market power (% of 2019 $C$)** | | | | | | |
| Endogenous automation, $G$ | 3.69 | 3.99 | 5.23 | 7.70 | 8.52 | 10.42 |
| No automation, $G^0$ | 1.91 | 2.18 | 3.43 | 6.61 | 7.59 | 10.05 |
| Difference (% of $G$) | 48.3 | 45.3 | 34.4 | 14.2 | 10.9 | 3.6 |
| **Panel B. Cheaper automation (welfare cost in % of 2019 $C$)** | | | | | | |
| Automation-price reduction (%) | 11.3 | 11.3 | 11.6 | 13.9 | 14.7 | 18.0 |
| Welfare cost, 2019 price | 3.69 | 3.99 | 5.23 | 7.70 | 8.52 | 10.42 |
| Welfare cost, lower price | 3.86 | 4.18 | 5.50 | 7.92 | 8.74 | 10.52 |
| Increase (% of cost at 2019 price) | 4.6 | 4.8 | 5.2 | 2.8 | 2.6 | 1.0 |
| *Gain from cheaper automation in the oligopsonistic economy (% of 2019 $C$)* | | | | | | |
| Social welfare $\mathcal{S}$ | 40.81 | 39.37 | 33.42 | 22.17 | 18.13 | 8.30 |
| Workers' welfare $\widetilde{\mathcal{W}}$ | −0.04 | −0.02 | 0.08 | 0.58 | 0.63 | 0.54 |
| Owners' welfare $\mathcal{W}^f$ | 40.85 | 39.40 | 33.34 | 21.58 | 17.50 | 7.77 |

*Notes:* Panel A repeats the experiment of Table 4: $G$ and $G^0$ are the welfare gains from removing labour markdowns with endogenous automation and with every automated share held at zero, and the difference is $(G - G^0)/G$. Panel B repeats the experiment of Table 5, with the price cut set for each column to deliver the same growth in automation capital as in the baseline. Workers' and owners' gains sum to the social welfare gain up to rounding.

## S5.2 Market definition

The baseline defines a local labour market as a five-digit industry within a commuting zone. Here a market is a three-digit industry within a commuting zone. The market structure $J$, $\{N_j\}$ and the within-market shares change with the definition, and so do the markdowns $\delta(s_{i|j})$, the marginal labour costs and the within-market moments (M1)–(M4), which are recomputed on the 2019 cross-section; the adoption rate (M5) does not depend on the definition. The task schedules are the three baseline schedules, with $a = 0.5$. Table S5.4 reports the estimates and fit, and Table S5.5 the two experiments.

Table S5.4: ESTIMATES AND FIT WITH THREE-DIGIT MARKETS

| | | Task schedule | | |
|---|---|---|---|---|
| | | Falling hazard | Constant hazard | Rising hazard |
| **Panel A. Estimates** | | | | |
| $\sigma_\phi$ | | 0.567 (0.008) | 0.559 (0.008) | 0.540 (0.007) |
| $\iota$ | | 0.741 (0.017) | 0.741 (0.017) | 0.741 (0.017) |
| $\mathcal{R}_{\phi T}$ | | 0.324 (0.027) | 0.338 (0.027) | 0.369 (0.026) |
| $\upsilon$ | | 0.486 (0.024) | 0.535 (0.025) | 0.692 (0.050) |
| $c_k$ | | 0.972 (0.024) | 1.089 (0.029) | 1.755 (0.094) |
| **Panel B. Moments** | | | | |
| | Data | Model | | |
| (M1) Variance of log wages | 0.055 (0.003) | 0.055 | 0.055 | 0.055 |
| (M2) Variance of log employment | 0.880 (0.018) | 0.880 | 0.880 | 0.880 |
| (M3) Covariance | 0.023 (0.003) | 0.023 | 0.023 | 0.023 |
| (M4) Adoption slope | 0.437 (0.021) | 0.437 | 0.437 | 0.437 |
| (M5) Adoption rate | 0.437 (0.006) | 0.437 | 0.437 | 0.437 |

*Notes:* A local labour market is a three-digit industry within a commuting zone. The task schedules have $a = 0.5$ and $\beta_R = -0.25$, 0 and 1. Moments as in Table 3, recomputed with the three-digit definition; data standard errors, in parentheses, from the cluster bootstrap of Section 5.5, clustered by three-digit market. Parameter standard errors from (53) in parentheses.

Table S5.5: COUNTERFACTUALS WITH THREE-DIGIT MARKETS

| | Falling hazard | Constant hazard | Rising hazard |
|---|---|---|---|
| **Panel A. Welfare cost of labour market power (% of 2019 $C$)** | | | |
| Endogenous automation, $G$ | 7.36 | 8.06 | 9.94 |
| No automation, $G^0$ | 5.90 | 6.73 | 9.11 |
| Difference (% of $G$) | 19.8 | 16.5 | 8.4 |
| **Panel B. Cheaper automation (welfare cost in % of 2019 $C$)** | | | |
| Automation-price reduction (%) | 12.6 | 13.1 | 15.0 |
| Welfare cost, 2019 price | 7.36 | 8.06 | 9.94 |
| Welfare cost, lower price | 7.66 | 8.38 | 10.22 |
| Increase (% of cost at 2019 price) | 4.0 | 3.9 | 2.8 |
| *Gain from cheaper automation in the oligopsonistic economy (% of 2019 $C$)* | | | |
| Social welfare $\mathcal{S}$ | 25.57 | 22.40 | 13.54 |
| Workers' welfare $\widetilde{\mathcal{W}}$ | 0.43 | 0.52 | 0.65 |
| Owners' welfare $\mathcal{W}^f$ | 25.14 | 21.88 | 12.89 |

*Notes:* As Table S5.3, for the economies of Table S5.4.

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